

# **Academic Regulations**

## **Course Structure & Detailed Syllabus**

**CHOICE BASED CREDIT SYSTEM**

### **R22**

## **MECHANICAL ENGINEERING**

### **Bachelor of Technology (B.Tech)**

**For the batches admitted from the A.Y. 2022-23**



**MARRI  
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# **MLR Institute of Technology**

**(Autonomous)**

**Laxman Reddy Avenue, Dundigal  
Hyderabad – 500043, Telangana State**

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**CHOICE BASED CREDIT SYSTEM**

**R22**

**MECHANICAL ENGINEERING**

**Bachelor of Technology (B.Tech.)**

**B. Tech. - Regular Four Year Degree Programme  
(For batches admitted from the academic year 2022 - 2023)  
&**

**B. Tech. - Lateral Entry Scheme  
(For batches admitted from the academic year 2023 - 2024)**



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## JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

*(Established by State Act No. 30 of 2008)*

Kukatpally, Hyderabad, Telangana (India).

### **ACADEMIC REGULATIONS (R22) FOR B.TECH REGULAR STUDENTS**

#### **WITH EFFECT FROM THE ACADEMIC YEAR 2022-23**

#### **1.0 Under-Graduate Degree Programme in Engineering & Technology (UGP in E&T)**

Jawaharlal Nehru Technological University Hyderabad (JNTUH) offers a 4-year (8 semesters) **Bachelor of Technology (B.Tech.)** degree programme, under Choice Based Credit System (CBCS) at its non-autonomous constituent and affiliated colleges with effect from the academic year **2022-23**.

#### **2.0 Eligibility for Admission**

- 2.1** Admission to the undergraduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government (EAMCET) or the University or on the basis of any other order of merit approved by the University, subject to reservations as prescribed by the government from time to time.
- 2.2** The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

#### **3.0 B.Tech. Programme Structure**

- 3.1** A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA  $\geq 5$ ) required for the completion of the undergraduate programme and award of the B.Tech. degree.
- 3.2** **UGC/ AICTE** specified definitions/ descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms, which are listed below.
  - 3.2.1 Semester Scheme**

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks ( $\geq 90$  instructional days) each and in each semester - ‘Continuous Internal Evaluation (CIE)’ and ‘Semester End Examination (SEE)’ under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

### 3.2.2 Credit Courses

All subjects/ courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/ course in an L: T: P: C (lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/ week/ semester for Theory/ Lecture (L) courses or Tutorials.
- One credit for two hours/ week/ semester for Laboratory/ Practical (P) courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

### 3.2.3 Subject Course Classification

All subjects/ courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The University has followed almost all the guidelines issued by AICTE/UGC.

| S. No. | Broad Course Classification | Course Group/ Category              | Course Description                                                                                                                                   |
|--------|-----------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Foundation Courses (FnC)    | BS – Basic Sciences                 | Includes Mathematics, Physics and Chemistry subjects                                                                                                 |
| 2      |                             | ES - Engineering Sciences           | Includes Fundamental Engineering Subjects                                                                                                            |
| 3      |                             | HS – Humanities and Social Sciences | Includes subjects related to Humanities, Social Sciences and Management                                                                              |
| 4      | Core Courses (CoC)          | PC – Professional Core              | Includes core subjects related to the parent discipline/ department/ branch of Engineering.                                                          |
| 5      | Elective Courses (ElC)      | PE – Professional Electives         | Includes elective subjects related to the parent discipline/ department/ branch of Engineering.                                                      |
| 6      |                             | OE – Open Electives                 | Elective subjects which include inter-disciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering. |
| 7      | Core Courses                | Project Work                        | B.Tech. Project or UG Project or UG Major Project or Project Stage I & II                                                                            |



|    |                        |                                                                                                         |                                                                                                             |
|----|------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 8  |                        | Industry Training/ Internship/ Industry Oriented Mini-project/ Mini- Project/ Skill Development Courses | Industry Training/ Internship/ Industry Oriented Mini-Project/ Mini-Project/ Skill Development Courses      |
| 9  |                        | Seminar                                                                                                 | Seminar/ Colloquium based on core contents related to parent discipline/ department/ branch of Engineering. |
| 10 | Minor Courses          | -                                                                                                       | 1 or 2 Credit Courses (subset of HS)                                                                        |
| 11 | Mandatory Courses (MC) | -                                                                                                       | Mandatory Courses (non-credit)                                                                              |

#### 4.0 Course Registration

- 4.1** A ‘faculty advisor or counselor’ shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/ courses, based on their competence, progress, pre-requisites and interest.
- 4.2** The academic section of the college invites ‘registration forms’ from students before the beginning of the semester through ‘on-line registration’, ensuring ‘date and time stamping’. The online registration requests for any ‘current semester’ shall be **completed before the commencement of SEEs (Semester End Examinations) of the ‘preceding semester’**.
- 4.3** A student can apply for **on-line** registration, **only after** obtaining the ‘**written approval**’ from faculty advisor/counselor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/ Counselor and the student.
- 4.4** A student may be permitted to register for all the subjects/ courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits (any 2 elective subjects), based on **progress** and SGPA/ CGPA, and completion of the ‘**pre-requisites**’ as indicated for various subjects/ courses, in the department course structure and syllabus contents.
- 4.5** Choice for ‘**additional subjects/courses**’, not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.
- 4.6** If the student submits ambiguous choices or multiple options or erroneous entries during **online** registration for the subject(s) / course(s) under a given/ specified course group/ category as listed in the course structure, only the first mentioned subject/ course in that category will be taken into consideration.
- 4.7** Subject/ course options exercised through **on-line** registration are final and **cannot** be changed or inter-changed; further, alternate choices also will not be considered.

However, if the subject/ course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternate choice either for a new subject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within a **week** after the commencement of class-work for that semester.

**4.8** Dropping of subjects/ courses may be permitted, only after obtaining prior approval from the faculty advisor/ counselor ‘within a period of 15 days’ from the beginning of the current semester.

**4.9 Open Electives:** The students have to choose three Open Electives (OE-I, II & III) from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered by his own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.

**4.10 Professional Electives:** The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.

## **5.0 Subjects/ courses to be offered**

**5.1** A subject/ course may be offered to the students, **only if** a minimum of 15 students opt for it.

**5.2** More than **one faculty member** may offer the **same subject** (lab/ practical may be included with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on - ‘**first come first serve** basis and CGPA criterion’ (i.e. the first focus shall be on early **on-line entry** from the student for registration in that semester, and the second focus, if needed, will be on CGPA of the student).

**5.3** If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide, whether or not to offer such a subject/ course for **two (or multiple) sections**.

**5.4** In case of options coming from students of other departments/ branches/ disciplines (not considering **open electives**), first **priority** shall be given to the student of the ‘**parent department**’.

## **6.0 Attendance requirements:**

**6.1** A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/ courses

(including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject. **This attendance should also be included in the attendance uploaded every fortnight in the University Website.**

- 6.2 Shortage of attendance in aggregate upto 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.
- 6.3 A stipulated fee shall be payable for condoning of shortage of attendance.
- 6.4 Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.
- 6.5 **Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examinations of that semester. They get detained and their registration for that semester shall stand cancelled,** including all academic credentials (internal marks etc.) of that semester. **They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking re-admission into that semester as and when offered; if there are any professional electives and/ or open electives, the same may also be re-registered if offered. However, if those electives are not offered in later semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.
- 6.6 A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

## 7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in Item No. 6.

- 7.1 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course, if student secures not less than 35% (14 marks out of 40 marks) in the Continuous Internal Evaluation (CIE), not less than 35% (21 marks out of 60 marks) in the semester end examinations (SEE), and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of letter grades, this implies securing 'C' grade or above in that subject/ course.
- 7.2 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research

Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such 'one reappearance' evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

### 7.3 Promotion Rules

| S. No. | Promotion                                                 | Conditions to be fulfilled                                                                                                                                                                                                                                                                             |
|--------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | First year first semester to first year second semester   | Regular course of study of first year first semester.                                                                                                                                                                                                                                                  |
| 2      | First year second semester to Second year first semester  | (i) Regular course of study of first year second semester.<br><br>(ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.   |
| 3.     | Second year first semester to Second year second semester | Regular course of study of second year first semester.                                                                                                                                                                                                                                                 |
| 4      | Second year second semester to Third year first semester  | (i) Regular course of study of second year second semester.<br><br>(ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not. |
| 5      | Third year first semester to Third year second semester   | Regular course of study of third year first semester.                                                                                                                                                                                                                                                  |
| 6      | Third year second semester to Fourth year first semester  | (i) Regular course of study of third year second semester.<br><br>(ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.  |
| 7      | Fourth year first semester to                             | Regular course of study of fourth year first                                                                                                                                                                                                                                                           |

|  |                                    |                  |
|--|------------------------------------|------------------|
|  | <b>Fourth year second semester</b> | <b>semester.</b> |
|--|------------------------------------|------------------|

- 7.4** A student (i) shall register for all courses/subjects covering 160 credits as specified and listed in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing SGPA  $\geq 5.0$  (in each semester), and—CGPA  $\geq 5$  (at the end of 8 semesters), (iv) **passes all the mandatory courses**, to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA (**at the end of undergraduate programme**), and shall be indicated in the grade card / marks memo of IV-year II semester.
- 7.5** If a student registers for ‘**extra subjects**’ (in the parent department or other departments/branches of Engg.) other than those listed subjects totaling to 160 credits as specified in the course structure of his department, the performances in those ‘**extra subjects**’ (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such ‘**extra subjects**’ registered, percentage of marks and letter grade alone will be indicated in the grade card / marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items 6 and 7.1 – 7.4 above.
- 7.6** A student eligible to appear in the semester end examination for any subject/ course, but absent from it or failed (thereby failing to secure ‘**C**’ grade or above) may reappear for that subject/ course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/ course will be carried over, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in that subject.
- 7.7** A student **detained in a semester due to shortage of attendance may be re-admitted in the same semester in the next academic year for fulfillment of academic requirements**. The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/ CGPA calculations will be done for the entire semester in which the student has been detained.
- 7.8** A student detained **due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits**. The academic regulations under which the student has been readmitted shall be applicable to him.
- 8.0 Evaluation - Distribution and Weightage of Marks**
- 8.1** The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).

**8.2** In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours as follows:

1. Mid Term Examination for 30 marks:
  - a. Part - A : Objective/quiz paper for 10 marks.
  - b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the concerned subject for 5 marks.

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination. The average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

*The student is eligible to write Semester End Examination of the concerned subject, if the student scores  $\geq 35\%$  (14 marks) of 40 Continuous Internal Examination (CIE) marks.*

*In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.*

There is NO Computer Based Test (CBT) for R22 regulations.

The details of the end semester question paper pattern are as follows:

**8.2.1** The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an “either” “or” choice, which means that there will be two questions from each unit and the student should answer either of the two questions.
- The duration of Semester End Examination is 3 hours.

**8.2.2** For the subject, **Computer Aided Engineering Graphics**, the Continuous Internal Evaluation (CIE) and Semester End Examinations (SEE) evaluation pattern is same as for other theory subjects.

**8.3** For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation:

1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
2. **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.
4. The remaining 10 marks are for Laboratory Report/Project and Presentation, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the cluster / other colleges which will be decided by the examination branch of the University.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
  2. 15 for experiment/program
  3. 15 for evaluation of results
  4. 10 marks for presentation on another experiment/program in the same laboratory course and
  5. 10 marks for viva-voce on concerned laboratory course.
- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all

40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

*The student is eligible to write Semester End Examination of the concerned subject, if the student scores  $\geq 35\%$  (14 marks) of 40 Continuous Internal Examination (CIE) marks.*

*In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.*

**8.4** The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I Semester course (ex., ***Elements of CE/ME/EEE/ECE/CSE etc***): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

**For CSE/IT and allied branches and Mining Engineering**, the Continuous Internal Evaluation (CIE) will be for 50 marks. Each Mid-Term examination consists of two parts i) **Part – A** for 20 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours.

**Part A:** Objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 20 marks. **Part B:** Descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks.

The remaining 10 marks of Continuous Internal Evaluation are for Assignment (5 marks) and Subject Viva-Voce/PPT/Poster Presentation/ Case Study (5 marks) and the evaluation pattern will remain same as for other theory subjects.

**For all other branches**, the Continuous Internal Evaluation (CIE) will be for 50 marks. Out of the 50 marks for internal evaluation:

- a) A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
- b) **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
- c) Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 15 marks.
- d) The remaining 15 marks are for Laboratory Report/Project and Presentation, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be



evaluated after completion of laboratory course and before semester end practical examination.

2. II Year II Semester *Real-Time (or) Field-based Research Project* course: The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (ii) secures less than 40% marks in this course.
- 
- 8.5 There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.
  - 8.6 The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his project work.
  - 8.7 UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage – II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEE Theory examinations.
  - 8.8 For Project Stage – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a

report on Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

- 8.9** For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/ Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of project, University selects an external examiner from the list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

- 8.10** A student can re-register for subjects in a semester:

- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva-voce/PPT/ Poster presentation/ Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

They may seek re-registration for all those subjects registered in that semester in which the student is failed. The student has to re-appear for CIE and SEE as and when offered.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the classwork in next academic year. His Continuous Internal Evaluation marks for 40 obtained in the previous attempt stand cancelled. The student has to obtain fresh set of marks for 40 allotted for CIE (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva-voce/PPT/ Poster presentation/ Case Study on a topic in the concerned subject). Head of the Dept. will take care of this.

- 8.11** For mandatory courses of Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization lab, a student has to secure 40 marks out of 100 marks (i.e. 40% of the 100 marks allotted) in the Continuous Internal Evaluation

for passing the subject/course. These marks should also be uploaded along with the internal marks of other subjects.

- 8.12** No marks or letter grades shall be allotted for mandatory/non-credit courses. Only Pass/Fail shall be indicated in Grade Card.

## **9.0 Grading Procedure**

- 9.1** Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/ Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.
- 9.2** As a measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

| <b>% of Marks Secured in a Subject/Course<br/>(Class Intervals)</b> | <b>Letter Grade<br/>(UGC Guidelines)</b> | <b>Grade Points</b> |
|---------------------------------------------------------------------|------------------------------------------|---------------------|
| <b>Greater than or equal to 90%</b>                                 | <b>O<br/>(Outstanding)</b>               | <b>10</b>           |
| <b>80 and less than 90%</b>                                         | <b>A<sup>+</sup><br/>(Excellent)</b>     | <b>9</b>            |
| <b>70 and less than 80%</b>                                         | <b>A<br/>(Very Good)</b>                 | <b>8</b>            |
| <b>60 and less than 70%</b>                                         | <b>B<sup>+</sup><br/>(Good)</b>          | <b>7</b>            |
| <b>50 and less than 60%</b>                                         | <b>B<br/>(Average)</b>                   | <b>6</b>            |
| <b>40 and less than 50%</b>                                         | <b>C<br/>(Pass)</b>                      | <b>5</b>            |
| <b>Below 40%</b>                                                    | <b>F<br/>(FAIL)</b>                      | <b>0</b>            |
| <b>Absent</b>                                                       | <b>Ab</b>                                | <b>0</b>            |

- 9.3** A student who has obtained an ‘F’ grade in any subject shall be deemed to have ‘**failed**’ and is required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.
- 9.4** To a student who has not appeared for an examination in any subject, ‘**Ab**’ grade will be allocated in that subject, and he is deemed to have ‘**Failed**’. A student will be required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.

- 9.5 A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.
- 9.6 A student earns Grade Point (GP) in each subject/ course, on the basis of the letter grade secured in that subject/ course. The corresponding ‘Credit Points’ (CP) are computed by multiplying the grade point with credits for that particular subject/ course.

**Credit Points (CP) = Grade Point (GP) x Credits .... For a course**

- 9.7 A student passes the subject/ course only when **GP ≥ 5 (‘C’ grade or above)**
- 9.8 The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points ( $\Sigma CP$ ) secured from all subjects/ courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA is thus computed as

$$SGPA = \{ \sum_{i=1}^N C_i G_i \} / \{ \sum_{i=1}^N C_i \} \dots \text{For each semester,}$$

where ‘i’ is the subject indicator index (considering all subjects in a semester), ‘N’ is the no. of subjects **‘registered’** for the semester (as specifically required and listed under the course structure of the parent department),  $C_i$  is the no. of credits allotted to the  $i^{th}$  subject, and  $G_i$  represents the grade points (GP) corresponding to the letter grade awarded for that  $i^{th}$  subject.

- 9.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all** semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to **two** decimal places. CGPA is thus computed from the I year II semester onwards at the end of each semester as per the formula

$$CGPA = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S semesters registered}$$

**(i.e., up to and inclusive of S semesters,  $S \geq 2$ ),**

where ‘M’ is the **total** no. of subjects (as specifically required and listed under the course structure of the parent department) the student has **‘registered’** i.e., from the 1<sup>st</sup> semester onwards up to and inclusive of the 8<sup>th</sup> semester, ‘j’ is the subject indicator index (takes into account all subjects from 1 to 8 semesters),  $C_j$  is the no. of credits allotted to the  $j^{th}$  subject, and  $G_j$  represents the grade points (GP) corresponding to the letter grade awarded for that  $j^{th}$  subject. After registration and completion of I year I semester, the SGPA of that semester itself may be taken as the CGPA, as there are no cumulative effects.

**Illustration of calculation of SGPA:**

| Course/Subject | Credits | Letter Grade | Grade Points | Credit Points      |
|----------------|---------|--------------|--------------|--------------------|
| Course 1       | 4       | A            | 8            | $4 \times 8 = 32$  |
| Course 2       | 4       | O            | 10           | $4 \times 10 = 40$ |

|          |    |    |   |                   |
|----------|----|----|---|-------------------|
| Course 3 | 4  | C  | 5 | $4 \times 5 = 20$ |
| Course 4 | 3  | B  | 6 | $3 \times 6 = 18$ |
| Course 5 | 3  | A+ | 9 | $3 \times 9 = 27$ |
| Course 6 | 3  | C  | 5 | $3 \times 5 = 15$ |
|          | 21 |    |   | 152               |

$$\text{SGPA} = 152/21 = 7.24$$

**Illustration of Calculation of CGPA up to 3<sup>rd</sup> Semester:**

| Semester | Course/<br>Subject Title | Credits<br>Allotted | Letter<br>Grade<br>Secured | Corresponding<br>Grade Point<br>(GP) | Credit<br>Points<br>(CP) |
|----------|--------------------------|---------------------|----------------------------|--------------------------------------|--------------------------|
| I        | Course 1                 | 3                   | A                          | 8                                    | 24                       |
| I        | Course 2                 | 3                   | O                          | 10                                   | 30                       |
| I        | Course 3                 | 3                   | B                          | 6                                    | 18                       |
| I        | Course 4                 | 4                   | A                          | 8                                    | 32                       |
| I        | Course 5                 | 3                   | A+                         | 9                                    | 27                       |
| I        | Course 6                 | 4                   | C                          | 5                                    | 20                       |
| II       | Course 7                 | 4                   | B                          | 6                                    | 24                       |
| II       | Course 8                 | 4                   | A                          | 8                                    | 32                       |
| II       | Course 9                 | 3                   | C                          | 5                                    | 15                       |
| II       | Course 10                | 3                   | O                          | 10                                   | 30                       |
| II       | Course 11                | 3                   | B+                         | 7                                    | 21                       |
| II       | Course 12                | 4                   | B                          | 6                                    | 24                       |
| II       | Course 13                | 4                   | A                          | 8                                    | 32                       |
| II       | Course 14                | 3                   | O                          | 10                                   | 30                       |
| III      | Course 15                | 2                   | A                          | 8                                    | 16                       |
| III      | Course 16                | 1                   | C                          | 5                                    | 5                        |
| III      | Course 17                | 4                   | O                          | 10                                   | 40                       |
| III      | Course 18                | 3                   | B+                         | 7                                    | 21                       |
| III      | Course 19                | 4                   | B                          | 6                                    | 24                       |
| III      | Course 20                | 4                   | A                          | 8                                    | 32                       |
| III      | Course 21                | 3                   | B+                         | 7                                    | 21                       |
|          | <b>Total<br/>Credits</b> | <b>69</b>           |                            | <b>Total Credit<br/>Points</b>       | <b>518</b>               |

$$\text{CGPA} = 518/69 = 7.51$$

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8<sup>th</sup> semester. The CGPA obtained at the end of 8<sup>th</sup> semester will become the final CGPA secured for entire B.Tech. programme.

**9.10** For merit ranking or comparison purposes or any other listing, **only** the ‘**rounded off**’ values of the CGPAs will be used.

**9.11** SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

## **10.0 Passing Standards**

**10.1** A student shall be declared successful or ‘passed’ in a semester, if he secures a GP  $\geq 5$  (‘C’ grade or above) in every subject/course in that semester (i.e. when the student gets an SGPA  $\geq 5.0$  at the end of that particular semester); and he shall be declared successful or ‘passed’ in the entire undergraduate programme, only when gets a CGPA  $\geq 5.00$  (‘C’ grade or above) for the award of the degree as required.

**10.2** After the completion of each semester, a grade card or grade sheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

## **11.0 Declaration of results**

**11.1** Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.

**11.2** For final percentage of marks equivalent to the computed final CGPA, the following formula may be used.

$$\% \text{ of Marks} = (\text{final CGPA} - 0.5) \times 10$$

## **12.0 Award of Degree**

**12.1** A student who registers for all the specified subjects/ courses as listed in the course structure and secures the required number of 160 credits (with CGPA  $\geq 5.0$ ), within 8 academic years from the date of commencement of the first academic year, shall be declared to have ‘**qualified**’ for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.

**12.2** A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.

**12.3** A student with final CGPA (at the end of the undergraduate programme)  $> 8.00$ , and fulfilling the following conditions - shall be placed in ‘**First Class with Distinction**’. However, he

- (i) Should have passed all the subjects/courses in ‘**First Appearance**’ within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.

- (ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.

A student not fulfilling any of the above conditions with final CGPA  $> 8$  shall be placed in '**First Class**'.

**12.4** Students with final CGPA (at the end of the undergraduate programme)  $\geq 7.0$  but  $< 8.00$  shall be placed in '**First Class**'.

**12.5** Students with final CGPA (at the end of the undergraduate programme)  $\geq 6.00$  but  $< 7.00$ , shall be placed in '**Second Class**'.

**12.6** All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme)  $\geq 5.00$  but  $< 6$ , shall be placed in '**pass class**'.

**12.7** A student with final CGPA (at the end of the undergraduate programme)  $< 5.00$  will not be eligible for the award of the degree.

**12.8** Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of '**Gold Medal**'.

**12.9 Award of 2-Year B.Tech. Diploma Certificate**

1. A student is awarded 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (within 4 years from the date of admission) upto B.Tech. II Year II Semester, if the student want to exit the 4-Year B.Tech. program and *requests for the 2 -Year B. Tech. (UG) Diploma Certificate*.
2. The student **once opted and awarded 2-Year UG Diploma Certificate, the student will be permitted to join** in B. Tech. III Year I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree ONLY in the next academic year along with next batch students. *However, if any student wishes to continue the study after opting for exit, he/she should register for the subjects/courses in III Year I Semester before commencement of classwork for that semester.*
3. *The students, who exit the 4-Year B. Tech. program after II Year of study and wish to re-join the B.Tech. program, must submit the 2 -Year B. Tech. (UG) Diploma Certificate awarded to him, subject to the eligibility for completion of Course/Degree.*
4. A student may be permitted to take one year break after completion of II Year II Semester or B. Tech. III Year II Semester (with university permission through the principal of the college well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).

### 13.0 Withholding of results

- 13.1** If the student has not paid the fees to the University at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

### 14.0 Transitory Regulations

A. For students detained due to shortage of attendance:

1. A student who has been detained in any semester of I, II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of R22 Regulations and is required to complete the study of B.Tech. within the stipulated period of eight academic years from the date of first admission in I Year. The R22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory Regulations.

B. For students detained due to shortage of credits:

1. A student of R18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of R22 Regulations only after acquiring the required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both R18 & R22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The R22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations.

C. For readmitted students in R22 Regulations:

1. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
2. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including R22 Regulations. **There is NO exemption of credits in any case.**
3. If a student is readmitted to R22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in R22 Regulations will be substituted by another subject to be suggested by the University.

Note: If a student readmitted to R22 Regulations and has not studied any subjects/topics in his/her earlier regulations of study which is prerequisite for further subjects in R22



Regulations, the College Principals concerned shall conduct remedial classes to cover those subjects/topics for the benefit of the students.

#### **15.0 Student Transfers**

- 15.1** There shall be no branch transfers after the completion of admission process.
- 15.2** There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.
- 15.3** The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.
- 15.4** The transferred students from other Universities/Institutions to JNTUH affiliated colleges who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter issued by the University.
- 15.5** The autonomous affiliated colleges have to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

#### **16.0 Scope**

- 16.1** The academic regulations should be read as a whole, for the purpose of any interpretation.
- 16.2** In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.
- 16.3** The University may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the University authorities.
- 16.4** Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

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## **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**

*(Established by State Act No. 30 of 2008)*

Kukatpally, Hyderabad, Telangana (India).

### **ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY SCHEME) FROM THE AY 2023-24**

**1. Eligibility for the award of B.Tech Degree (LES)**

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The student shall register for 120 credits and secure 120 credits with CGPA  $\geq 5$  from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.

3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.

4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).

**5. Promotion rule**

| S. No | Promotion                                                 | Conditions to be fulfilled                                                                                                                                                                                                                                                                             |
|-------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Second year first semester to second year second semester | Regular course of study of second year first semester.                                                                                                                                                                                                                                                 |
| 2     | Second year second semester to third year first semester  | (i) Regular course of study of second year second semester.<br><br>(ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not. |
| 3     | Third year first semester to third year second semester   | Regular course of study of third year first semester.                                                                                                                                                                                                                                                  |
| 4     | Third year second semester to fourth year first semester  | (i) Regular course of study of third year second semester.<br><br>(ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to                                                                                                                                                 |

|          |                                                                  |                                                                                                                                                      |
|----------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                  | <b>third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.</b> |
| <b>5</b> | <b>Fourth year first semester to fourth year second semester</b> | <b>Regular course of study of fourth year first semester.</b>                                                                                        |

- 6. All the other regulations as applicable to B. Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).**
- 7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.**

**\*\*\*\***

## Malpractices Rules

### Disciplinary Action For / Improper Conduct in Examinations

|        | Nature of Malpractices/Improper conduct                                                                                                                                                                                                                                                                                                                                                                                                 | Punishment                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | If the student:                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1. (a) | Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination) | Expulsion from the examination hall and cancellation of the performance in that subject only.                                                                                                                                                                                                                                                                                                             |
| (b)    | Gives assistance or guidance or receives it from any other student orally or by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.                                                                                                                                                                                                   | Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.                                                                                                                                                                             |
| 2.     | Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.                                                                                                                                                                                                     | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year.<br><br>The hall ticket of the student is to be cancelled and sent to the University.         |
| 3.     | Impersonates any other student in connection with the examination.                                                                                                                                                                                                                                                                                                                                                                      | The student who has impersonated shall be expelled from examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | semester/year. The student is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.                                                                                                                                                                          |
| 4. | Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. |
| 5. | Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cancellation of the performance in that subject.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6. | Refuses to obey the orders of the chief superintendent/assistant – superintendent / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of | In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.                                                    |

|     |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7.  | Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.                                                                                                               | Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all University examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. |
| 8.  | Possesses any lethal weapon or firearm in the examination hall.                                                                                                                                                                                                | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.                                                                                                                                                                                     |
| 9.  | If student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.                                                        | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.<br><br>Person(s) who do not belong to the college will be handed over to the police and, a police case will be registered against them.                                             |
| 10. | Comes in a drunken condition to the examination hall.                                                                                                                                                                                                          | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations                                                                                                                                                                                                                                                                                                                                                       |

|     |                                                                                                                                                                         |                                                                                                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                                                                                                                                         | and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.                                                                         |
| 11. | Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.                                                               | Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations and project work of that semester/year examinations. |
| 12. | If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the University for further action to award a suitable punishment. |                                                                                                                                                                                           |

#### **Malpractices identified by squad or special invigilators**

1. Punishments to the students as per the above guidelines.
2. Punishment for Institutions: (if the squad reports that the college is also involved in encouraging malpractices)
  - a. A show-cause notice shall be issued to the college.
  - b. Impose a suitable fine on the college.
  - c. Shifting the examination center from one college to another college for a specific period of not less than one year.

\* \* \* \* \*

# **COURSE STRUCTURE**



**DEPARTMENT OF MECHANICAL ENGINEERING**  
**R22 COURSE STRUCTURE**

| <b>I B.Tech.- I SEMESTER</b> |                                |             |                |          |           |           |                                     |                |            |
|------------------------------|--------------------------------|-------------|----------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
| Course Code                  | Course Title                   | Course Area | Hours per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|                              |                                |             | L              | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A6BS01                       | Linear Algebra and Calculus    | BSC         | 3              | 1        | 0         | 4         | 40                                  | 60             | 100        |
| A6BS07                       | Applied Physics                | BSC         | 3              | 1        | 0         | 4         | 40                                  | 60             | 100        |
| A6BS09                       | Engineering Chemistry          | BSC         | 3              | 1        | 0         | 4         | 40                                  | 60             | 100        |
| A6ME01                       | Engineering Graphics           | ESC         | 1              | 0        | 4         | 3         | 40                                  | 60             | 100        |
| A6BS08                       | Applied Physics Lab            | BSC         | 0              | 0        | 3         | 1.5       | 40                                  | 60             | 100        |
| A6BS10                       | Engineering Chemistry Lab      | BSC         | 0              | 0        | 3         | 1.5       | 40                                  | 60             | 100        |
| A6ME03                       | Engineering Workshop Practices | ESC         | 0              | 1        | 2         | 2         | 40                                  | 60             | 100        |
| A6BS11                       | Environmental Science          | MC          | 3              | 0        | 0         | 0         | 50                                  | -              | 50         |
| <b>TOTAL</b>                 |                                |             | <b>13</b>      | <b>4</b> | <b>12</b> | <b>20</b> | <b>330</b>                          | <b>420</b>     | <b>750</b> |

| <b>I B.Tech.- II SEMESTER</b> |                                               |             |                |          |           |           |                                     |                |            |
|-------------------------------|-----------------------------------------------|-------------|----------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
| Course Code                   | Course Title                                  | Course Area | Hours per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|                               |                                               |             | L              | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A6BS02                        | Numerical Methods and Integral Transforms     | BSC         | 3              | 1        | 0         | 4         | 40                                  | 60             | 100        |
| A6HS01                        | English for Skill Enhancement                 | HSMC        | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME05                        | Engineering Mechanics                         | ESC         | 3              | 1        | 0         | 4         | 40                                  | 60             | 100        |
| A6CS02                        | Programming for Problem Solving               | ESC         | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6HS02                        | English Language and Communication Skills Lab | HSMC        | 0              | 0        | 3         | 1.5       | 40                                  | 60             | 100        |
| A6CS03                        | Programming for Problem Solving Lab           | ESC         | 0              | 0        | 3         | 1.5       | 40                                  | 60             | 100        |
| A6ME06                        | Elements of Mechanical Engineering Design     | ESC         | 1              | 0        | 2         | 2         | 50                                  | -              | 50         |
| A6HS04                        | Seminar                                       | HSMC        | 0              | 0        | 2         | 1         | 50                                  | -              | 50         |
| <b>TOTAL</b>                  |                                               |             | <b>13</b>      | <b>2</b> | <b>10</b> | <b>20</b> | <b>340</b>                          | <b>360</b>     | <b>700</b> |

**II B.Tech.- I SEMESTER**

| Course Code  | Course Title                                                  | Course Area | Hours per Week |          |          | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|---------------------------------------------------------------|-------------|----------------|----------|----------|-----------|-------------------------------------|----------------|------------|
|              |                                                               |             | L              | T        | P        |           | Internal (CIE)                      | External (SEE) | Total      |
| A6BS05       | Probability, Statistics and Complex Analysis                  | BSC         | 3              | 1        | 0        | 4         | 40                                  | 60             | 100        |
| A6ME09       | Strength of Materials                                         | PCC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME10       | Material Science and Metallurgy                               | PCC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME12       | Manufacturing Processes                                       | PCC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME14       | Thermodynamics                                                | PCC         | 3              | 1        | 0        | 4         | 40                                  | 60             | 100        |
| A6ME13       | Manufacturing Processes Lab                                   | PCC         | 0              | 0        | 2        | 1         | 40                                  | 60             | 100        |
| A6ME11       | Strength of Materials and Material Science and Metallurgy lab | PCC         | 0              | 0        | 2        | 1         | 40                                  | 60             | 100        |
| A6ME15       | Python Lab for Mechanical Applications                        | ESC         | 0              | 0        | 2        | 1         | 40                                  | 60             | 100        |
| A6HS06       | Constitution of India                                         | MC          | 2              | 0        | 0        | 0         | 50                                  | -              | 50         |
| <b>TOTAL</b> |                                                               |             | <b>17</b>      | <b>2</b> | <b>6</b> | <b>20</b> | <b>370</b>                          | <b>480</b>     | <b>850</b> |

**II B.Tech.- II SEMESTER**

| Code         | Course                                           | Course Area | Periods per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|--------------------------------------------------|-------------|------------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
|              |                                                  |             | L                | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A6EE60       | Basic Electrical and Electronics Engineering     | ESC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME16       | Fluid Mechanics and Hydraulic Machines           | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME18       | Thermal Engineering - I                          | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME19       | Theory of Machines - I                           | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME20       | Design of Machine Elements                       | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6EE61       | Basic Electrical and Electronics Engineering Lab | ESC         | 0                | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME17       | Fluid Mechanics and Hydraulic Machines Lab       | PCC         | 0                | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME21       | Computer Aided Machine Drawing Lab               | PCC         | 0                | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME22       | Real Time Project / Field based Project          | PWC         | 0                | 0        | 4         | 2         | 50                                  | -              | 50         |
| A6HS05       | Gender Sensitization                             | MC          | 0                | 0        | 2         | 0         | 50                                  | -              | 50         |
| <b>TOTAL</b> |                                                  |             | <b>15</b>        | <b>0</b> | <b>12</b> | <b>20</b> | <b>420</b>                          | <b>480</b>     | <b>900</b> |

**III B.Tech.- I SEMESTER**

| Course Code  | Course Title                              | Course Area | Hours per Week |          |          | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|-------------------------------------------|-------------|----------------|----------|----------|-----------|-------------------------------------|----------------|------------|
|              |                                           |             | L              | T        | P        |           | Internal (CIE)                      | External (SEE) | Total      |
| A6HS08       | Business Economics and Financial Analysis | HSMC        | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME23       | Machine Design                            | PCC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME24       | Theory of Machines - II                   | PCC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME25       | CAD/CAM                                   | PCC         | 2              | 0        | 0        | 2         | 40                                  | 60             | 100        |
| A6ME27       | Thermal Engineering - II                  | PCC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
|              | PROFESSIONAL ELECTIVE - I                 | PEC         | 3              | 0        | 0        | 3         | 40                                  | 60             | 100        |
| A6ME26       | CAD Lab                                   | PCC         | 0              | 0        | 2        | 1         | 40                                  | 60             | 100        |
| A6ME28       | Thermal Engineering Lab                   | PCC         | 0              | 0        | 2        | 1         | 40                                  | 60             | 100        |
| A6ME30       | Independent Study / Moocs                 | PWC         | 0              | 0        | 2        | 1         | -                                   | 100            | 100        |
| A6HS10       | Human Values and Professional Ethics      | MC          | 2              | 0        | 0        | 0         | 50                                  | -              | 50         |
| <b>TOTAL</b> |                                           |             | <b>19</b>      | <b>0</b> | <b>6</b> | <b>20</b> | <b>370</b>                          | <b>580</b>     | <b>950</b> |

**III B.Tech.- II SEMESTER**

| Code                                 | Course                                                       | Course Area | Periods per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------------------------------|--------------------------------------------------------------|-------------|------------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
|                                      |                                                              |             | L                | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A6ME31                               | Heat Transfer                                                | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME33                               | Machine Tools                                                | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME35                               | Finite Element Analysis                                      | PCC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
|                                      | OPEN ELECTIVE-I                                              | OEC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
|                                      | PROFESSIONAL ELECTIVE - II                                   | PEC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME32                               | Heat Transfer Lab                                            | PCC         | 0                | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME34                               | Machine Tools Lab                                            | PCC         | 0                | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME36                               | Production Drawing Practice & Computer Aided Engineering Lab | PCC         | 0                | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME40                               | Industry Oriented Mini Project / Internship                  | PWC         | 0                | 0        | 4         | 2         | -                                   | 100            | 100        |
| <b>TOTAL</b>                         |                                                              |             | <b>15</b>        | <b>0</b> | <b>10</b> | <b>20</b> | <b>320</b>                          | <b>580</b>     | <b>900</b> |
| <b>Mandatory Course (Non-Credit)</b> |                                                              |             |                  |          |           |           |                                     |                |            |
| A6BS11                               | Environmental Science (LE Students)                          | MC          | 2                | 0        | 0         | 0         | 50                                  | -              | 50         |

**IV B.Tech.- I SEMESTER**

| Course Code  | Course Title                                  | Course Area | Hours per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|-----------------------------------------------|-------------|----------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
|              |                                               |             | L              | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A6ME41       | Composite Materials                           | PCC         | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME43       | Engineering Metrology and Instrumentation     | PCC         | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
|              | OPEN ELECTIVE-II                              | OEC         | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
|              | PROFESSIONAL ELECTIVE - III                   | PEC         | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
|              | PROFESSIONAL ELECTIVE - IV                    | PEC         | 3              | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME42       | Composite Materials Lab                       | PCC         | 0              | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME44       | Engineering Metrology and Instrumentation Lab | PCC         | 0              | 0        | 2         | 1         | 40                                  | 60             | 100        |
| A6ME45       | Research Project Phase - I                    | PWC         | 0              | 0        | 6         | 3         | 100                                 | -              | 100        |
| <b>TOTAL</b> |                                               |             | <b>15</b>      | <b>0</b> | <b>10</b> | <b>20</b> | <b>380</b>                          | <b>420</b>     | <b>800</b> |

**IV B.Tech.- II SEMESTER**

| Code         | Course                      | Course Area | Periods per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|-----------------------------|-------------|------------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
|              |                             |             | L                | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
|              | OPEN ELECTIVE-III           | OEC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
|              | PROFESSIONAL ELECTIVE - V   | PEC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
|              | PROFESSIONAL ELECTIVE - VI  | PEC         | 3                | 0        | 0         | 3         | 40                                  | 60             | 100        |
| A6ME50       | Research Project Phase - II | PWC         | 0                | 0        | 22        | 11        | 40                                  | 60             | 100        |
| <b>TOTAL</b> |                             |             | <b>9</b>         | <b>0</b> | <b>22</b> | <b>20</b> | <b>160</b>                          | <b>240</b>     | <b>400</b> |

**List of Open Electives from Mechanical Department:**

Open Elective (OE – I)

1. A6ME55 – Basics of Mechanical Engineering
2. A6ME60 - Fundamentals of Engineering Materials

Open Elective (OE – II)

1. A6ME65 - Fundamentals of Operations Research
2. A6ME70 –Introduction to Industrial Engineering

Open Elective (OE – III)

1. A6ME75 - Renewable Energy Sources
2. A6ME80 - Elements of Electric and Hybrid vehicles

| PROFESSIONAL ELECTIVE COURSES (PEC) |                                                   |        |                                       |
|-------------------------------------|---------------------------------------------------|--------|---------------------------------------|
| PE-I                                |                                                   | PE-II  |                                       |
| A6ME51                              | Artificial Intelligence in Mechanical Engineering | A6ME56 | Sustainable Engineering               |
| A6ME52                              | Mechatronics                                      | A6ME57 | Machine Dynamics & Vibrations         |
| A6ME53                              | Advanced Welding Techniques & NDT                 | A6ME58 | Design for Manufacturing and Assembly |
| A6ME54                              | Non-Conventional Energy Sources                   | A6ME59 | Computational Fluid Dynamics          |
| PE-III                              |                                                   | PE-IV  |                                       |
| A6ME61                              | Automation in Manufacturing                       | A6ME66 | Automobile Engineering                |
| A6ME62                              | Unconventional Machining Processes                | A6ME67 | Power Plant Engineering               |
| A6ME63                              | Additive Manufacturing                            | A6ME68 | Refrigeration and Air Conditioning    |
| A6ME64                              | Production Planning and Control                   | A6ME69 | Electric and Hybrid Vehicles          |
| PE-V                                |                                                   | PE-VI  |                                       |
| A6ME71                              | Industry 4.0 and IOT                              | A6ME76 | Industrial Engineering and Management |
| A6ME72                              | Nanotechnology                                    | A6ME77 | Operations Research                   |
| A6ME73                              | Robotics Engineering                              | A6ME78 | Total Quality Management              |
| A6ME74                              | Industrial Safety and Maintenance Engineering     | A6ME79 | Product Life Cycle Management         |

# **B.TECH I SEMESTER SYLLABUS**

## LINEAR ALGEBRA & CALCULUS

| I Year I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                        |              |   |   |         |               |     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Category                                               | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6BS01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | BSC                                                    | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                        | 3            | 1 | - | 4       | 40            | 60  | 100   |
| <b>Course Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |              |   |   |         |               |     |       |
| To learn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |              |   |   |         |               |     |       |
| 1. Concept of Rank of a matrix, Consistency and solving system of linear equations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                        |              |   |   |         |               |     |       |
| 2. Concept of eigen values, eigen vectors and diagonalization of the matrix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                        |              |   |   |         |               |     |       |
| 3. The concept of differential equations and solve them using appropriate methods.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                        |              |   |   |         |               |     |       |
| 4. Evaluate multiple integrals and improper integrals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                        |              |   |   |         |               |     |       |
| 5. The partial derivatives of several variable functions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                        |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MATRICES AND THEIR APPLICATIONS                        |              |   |   |         |               |     |       |
| Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew – Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m x n and n x n linear system of equations by Gauss Elimination, Gauss siedel Method |                                                        |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | EIGEN VALUES, EIGEN VECTORS                            |              |   |   |         |               |     |       |
| Eigen values and Eigen vectors and its properties (without proof), Cayley - Hamilton theorem (Statement and verification) - Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices. Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to Canonical forms by Orthogonal Transformation.                                                                                                                                                                                                                   |                                                        |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS |              |   |   |         |               |     |       |
| Introduction- Exact and reducible to Exact differential equations-Newton’s Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$ , $\text{Sin}ax$ , $\text{Cos}ax$ , $e^{ax} v(x)$ , $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.                                                                                                                                                                                                  |                                                        |              |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS</b> |
| <p>Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integral.</p> <p>Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.</p>                                                                                                                                                     |                                                     |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>CALCULUS OF SEVERAL VARIABLES</b>                |
| <p>Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians - functional dependence &amp; independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints).</p>                                                                                                                                                                                                                                                |                                                     |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |
| <ol style="list-style-type: none"> <li>1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley &amp; Sons, 2006.</li> <li>2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.</li> </ol>                                                                                                                                                                                                                                                                                                     |                                                     |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                     |
| <ol style="list-style-type: none"> <li>1. G.B.Thomas, calculus and analytical geometry, 9th Edition, Pearson Reprint 2006.</li> <li>2. N.P Bali and Manish Goyal, A Text of Engineering Mathematics, Laxmi publications, 2008.</li> <li>3. E.L.Ince, Ordinary differential Equations, Dover publications, 1958.</li> </ol>                                                                                                                                                                                                                          |                                                     |
| <p><b>Course Outcomes</b></p> <p><b>At the end of the course, student will be able to:</b></p> <ol style="list-style-type: none"> <li>1. Solve the system of linear equations using rank of the matrices.</li> <li>2. Find the Eigen values and Eigen vectors of a matrix</li> <li>3. Identify the different types of differential equations and solve them using appropriate methods.</li> <li>4. Evaluate the improper integrals using beta and gamma functions.</li> <li>5. Find the Maxima and Minima of several variable functions.</li> </ol> |                                                     |



## APPLIED PHYSICS

**I Year I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)**

| Course Code: | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6BS07       | BSC      | L            | T | P | C       | CIE           | SEE | Total |
|              |          | 3            | 1 | 0 | 4       | 40            | 60  | 100   |

### Course Objectives:

The course should enable the students to:

1. Learn the basic principles of quantum physics and its applications
2. Understand the formation of energy bands and atomic structure in solids for material classification
3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices
4. Learn the basic principles of laser and optical fiber in information technology
5. Identify the importance of nanoscale and various fabrication and characterization techniques and quantum computations in engineering applications

### Course Outcomes:

At the end of the course students will be able to:

1. **Analyze** the microscopic properties of materials using principles of quantum physics for engineering applications
2. **Explain** the behaviour of different electronic materials based on the concepts of band theory
3. **Apply** the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation
4. **Gain** the knowledge of production of laser and usage of fibers in fiber optic communication technology
5. **Comprehend** the knowledge of quantum physics in quantum computation for secure information technology

|                                                                                                                                                                                                                                                                                                                                                                                          |                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>Quantum Physics</b>                                                |
| Black body radiation, Stefan-Boltzmann's law, Planck's radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box. |                                                                       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                           | <b>Electronic properties of Materials &amp; Band theory of solids</b> |
| <b>Electronic properties of Materials:</b> Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig - Penny model (Qualitative treatment), E-k diagram, effective mass of electron.                                                                                                                                     |                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Band theory of solids:</b> Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                         |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Semi-conductors &amp; Semiconductor Devices</b>      |
| <p><b>Semiconductors:</b> Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications.</p> <p><b>Semiconductor Devices:</b> Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.</p>                                                                                                                                                                                                                                                               |                                                         |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Laser &amp; Fiber Optics</b>                         |
| <p><b>Laser:</b> Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO<sub>2</sub> laser, Applications of lasers in different fields.</p> <p><b>Fiber Optics:</b> Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.</p> |                                                         |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Physics of Quantum computing &amp; Quantum gates</b> |
| <p><b>Physics of Quantum computing:</b> Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits.</p> <p><b>Quantum gates</b>-Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates - CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.</p>                                                                                                                           |                                                         |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                         |
| <ol style="list-style-type: none"> <li>1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, 4th Edition.</li> <li>2. Nielsen M. A., I. L Chung, Quantum Computation &amp; Quantum Information, Cambridge Univ. Press.</li> <li>3. M. N. Avadhanulu, P.G. Kshirsagar &amp; TVS Arun Murthy” A Text book of Engineering Physics”- S. Chand Publications, 11th Edition 2019.</li> </ol>                                                                                                                                                                                                                      |                                                         |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |
| <ol style="list-style-type: none"> <li>1. Quantum Physics, H.C.Verma, TBS Publiication, 2<sup>nd</sup> Edition 2012</li> <li>2. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley &amp; Sons, 11<sup>th</sup> Edition, 2018.</li> <li>3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication,2019</li> <li>4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2<sup>nd</sup> Edition, 2022</li> </ol>                                                                                                                                                    |                                                         |

| ENGINEERING CHEMISTRY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                                        |   |   |         |               |     |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------|---|---|---------|---------------|-----|-------|
| I Year I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                                        |   |   |         |               |     |       |
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Category | Hours/Week                             |   |   | Credits | Maximum Marks |     |       |
| A6BS09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | BSC      | L                                      | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          | 3                                      | 1 | 0 | 4       | 40            | 60  | 100   |
| <b>COURSE OBJECTIVES:</b> <ol style="list-style-type: none"><li>1. To bring adaptability to new developments in Engineering Chemistry and to acquire the skills required to become a perfect engineer.</li><li>2. To include the importance of water in industrial usage, fundamental aspects of battery chemistry, significance of corrosion it's control to protect the structures.</li><li>3. To imbibe the basic concepts of petroleum and its products.</li><li>4. To acquire required knowledge about engineering materials like cement, smart materials and Lubricants.</li></ol>                                                                                                                                                                                                                                                                           |          |                                        |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course student will be able to:</b> <ol style="list-style-type: none"><li>1. Understand the basic properties of water and its usage in domestic and industrial purpose.</li><li>2. Acquire the basic knowledge of electrochemical procedures related to corrosion</li><li>3. Learn the fundamentals and general properties of polymers and other engineering materials.</li><li>4. Predict potential applications of chemistry and practical utility in order to become good engineers and entrepreneurs.</li></ol>                                                                                                                                                                                                                                                                                                |          |                                        |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          | <b>WATER AND ITS TREATMENT</b>         |   |   |         |               |     |       |
| <p>Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break - point chlorination. Defluoridation - Determination of F- ion by ion - selective electrode method.</p> <p>Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water - Calgon conditioning - Phosphate conditioning - Colloidal conditioning, External treatment methods -Softening of water by ion- exchange processes. Desalination of water – Reverse osmosis.</p>                                                                                                                                                                  |          |                                        |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          | <b>BATTERY CHEMISTRY AND CORROSION</b> |   |   |         |               |     |       |
| <p>Introduction - Classification of batteries- primary, secondary and reserve batteries with examples. Basic requirements for commercial batteries. Construction, working and applications of: Zn-air and Lithium ion battery, Applications of Li-ion battery to electrical vehicles. Fuel Cells- Differences between battery and a fuel cell, Construction and applications of Methanol Oxygen fuel cell and Solid oxide fuel cell. Solar cells - Introduction and applications of Solar cells.</p> <p><b>Corrosion:</b> Causes and effects of corrosion – theories of chemical and electrochemical corrosion –mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection – Sacrificial anode and impressed current methods.</p> |          |                                        |   |   |         |               |     |       |

| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | POLYMERIC MATERIALS   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <p>Definition – Classification of polymers with examples – Types of polymerization – addition (free radical addition) and condensation polymerization with examples – Nylon 6:6, Terylene</p> <p><b>Plastics:</b> Definition and characteristics- thermoplastic and thermosetting plastics, Preparation, Properties and engineering applications of PVC and Bakelite, Teflon, Fiber reinforced plastics (FRP). <b>Rubbers:</b> Natural rubber and its vulcanization. <b>Elastomers:</b> Characteristics –preparation – properties and applications of Buna-S, Butyl and Thiokol rubber.</p> <p><b>Conducting polymers:</b> Characteristics and Classification with examples-mechanism of conduction in trans-polyacetylene and applications of conducting polymers.</p> <p><b>Biodegradable polymers:</b> Concept and advantages - Polylactic acid and poly vinyl alcohol and their applications.</p> |                       |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ENERGY SOURCES        |
| <p>Introduction, Calorific value of fuel – HCV, LCV- Dulong's formula. Classification- solid fuels: coal –analysis of coal – proximate and ultimate analysis and their significance. Liquid fuels – petroleum and its refining, cracking types – moving bed catalytic cracking. Knocking – octane and cetane rating, synthetic petrol - Fischer-Tropsch's process; Gaseous fuels – composition and uses of natural gas, LPG and CNG, Biodiesel – Transesterification, advantages.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |                       |
| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ENGINEERING MATERIALS |
| <p><b>Cement:</b> Portland cement, its composition, setting and hardening. Smart materials and their engineering applications, Shape memory materials- Poly L- Lactic acid. Thermo response materials- Polyacryl amides, Poly vinylamides</p> <p><b>Lubricants:</b> Classification of lubricants with examples-characteristics of a good lubricants – mechanism of lubrication (thick film, thin film and extreme pressure)- properties of lubricants: viscosity, cloud point, pour point, flash point and fire point.</p>                                                                                                                                                                                                                                                                                                                                                                            |                       |
| <b>TEXTBOOKS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |
| <ol style="list-style-type: none"> <li>1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010</li> <li>2. Engineering Chemistry by Rama Devi, Venkata Ramana Reddy and Rath, Cengage learning, 2016</li> <li>3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K.Shashikala, Pearson Publications, 2021.</li> <li>4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |
| <b>REFERENCEBOOKS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                       |
| <ol style="list-style-type: none"> <li>1. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi (2015)</li> <li>2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi (2011)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |

## ENGINEERING GRAPHICS

| I Year I Semester: MECHANICAL ENGINEERING, AERONAUTICAL ENGINEERING & EEE                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                  |              |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Category                                         | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ESC                                              | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                  | 1            | 0 | 4 | 3       | 40            | 60  | 100   |
| <b>COURSE OUTCOMES:</b><br>At the end of the course the student should be able to:<br>1. Explain various commands and create drawing in AutoCAD.<br>2. Sketch the various curves used in engineering applications<br>3. Prepare orthographic projections of lines, planes by visualizing them in different positions.<br>4. Solve the problem of projections of solids and development of surfaces for industrial needs.<br>5. Construct the isometric view into orthographic views and vice versa. |                                                  |              |   |   |         |               |     |       |
| UNIT- I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | INTRODUCTION                                     |              |   |   |         |               |     |       |
| Introduction to Engineering Graphics: Principles and their significance,<br>INTRODUCTION TO COMPUTER AIDED DRAFTING: Initial Setup Commands, Draw Commands, Modify Commands, 2D Drawings-Simple Exercises.                                                                                                                                                                                                                                                                                          |                                                  |              |   |   |         |               |     |       |
| UNIT- II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ENGINEERING CURVES                               |              |   |   |         |               |     |       |
| Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only).<br>Special curves: Cycloids, Involute.                                                                                                                                                                                                                                                                                                                                                                                  |                                                  |              |   |   |         |               |     |       |
| UNIT- III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ORTHOGRAPHIC PROJECTION                          |              |   |   |         |               |     |       |
| Principles of Orthographic Projections – conventions – first and third angle projections. Projections of points-<br>Projection of lines inclined to both the planes. (First angle projection only)<br>PROJECTIONS OF PLANES: Projections of regular planes inclined to both planes.                                                                                                                                                                                                                 |                                                  |              |   |   |         |               |     |       |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PROJECTION OF SOLIDS AND DEVELOPMENT OF SURFACES |              |   |   |         |               |     |       |
| PROJECTION OF SOLIDS: Regular Solids inclined to both planes<br>DEVELOPMENT OF SURFACE/SOLIDS: Theory of development, development of lateral surface with base.                                                                                                                                                                                                                                                                                                                                     |                                                  |              |   |   |         |               |     |       |
| UNIT- V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ISOMETRIC DRAWINGS                               |              |   |   |         |               |     |       |
| Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections. Drawing<br>Isometric circles, Dimensioning Isometric objects.<br>Conversion of Isometric view to Orthographic views and Orthographic views to isometric view.                                                                                                                                                                                                                             |                                                  |              |   |   |         |               |     |       |
| Text Books:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                  |              |   |   |         |               |     |       |
| 1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House.<br>2. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication                                                                                                                                                                                                                                                                                                                    |                                                  |              |   |   |         |               |     |       |
| Reference Books:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                  |              |   |   |         |               |     |       |
| 1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications, 2013.<br>2. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers.<br>3. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education.                                                                                                                                                                                            |                                                  |              |   |   |         |               |     |       |

## APPLIED PHYSICS LAB

|                                                                                                                                                                                                                                                                                                                     |          |              |   |   |         |               |     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|---|---|---------|---------------|-----|-------|
| I Year I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)                                                                                                                                                                                                                                           |          |              |   |   |         |               |     |       |
| Course Code:                                                                                                                                                                                                                                                                                                        | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6BS08                                                                                                                                                                                                                                                                                                              | BSC      | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                     |          | 0            | 0 | 3 | 1.5     | 40            | 60  | 100   |
| Course Objectives:                                                                                                                                                                                                                                                                                                  |          |              |   |   |         |               |     |       |
| The course should enable the students to:                                                                                                                                                                                                                                                                           |          |              |   |   |         |               |     |       |
| 1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications.                                                                                                                                                                                   |          |              |   |   |         |               |     |       |
| 2. To know the Laser and Fibre optic technologies and its applications in real time scenario.                                                                                                                                                                                                                       |          |              |   |   |         |               |     |       |
| 3. Understand the electromagnetic properties using experimental knowledge.                                                                                                                                                                                                                                          |          |              |   |   |         |               |     |       |
| 4. Understand the method of least squares fitting.                                                                                                                                                                                                                                                                  |          |              |   |   |         |               |     |       |
| Course Outcomes:                                                                                                                                                                                                                                                                                                    |          |              |   |   |         |               |     |       |
| By the end of the course students will be able to:                                                                                                                                                                                                                                                                  |          |              |   |   |         |               |     |       |
| 1. <b>Analyze</b> the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photo diodes, solar cell, and temperature effect on resistance using thermistor. |          |              |   |   |         |               |     |       |
| 2. <b>Identify</b> the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank’s constant using a light source and interference by using Newton’s rings   |          |              |   |   |         |               |     |       |
| 3. <b>Analyze</b> the electromagnetic properties of a current carrying coil by using Stewart Gee’s experiment                                                                                                                                                                                                       |          |              |   |   |         |               |     |       |
| 4. <b>Analyze</b> the least squares fitting method for data analysis using experimental data of Tensional Pendulum                                                                                                                                                                                                  |          |              |   |   |         |               |     |       |

**APPLIED PHYSICS LABORATORY****LIST OF EXPERIMENTS**

|                      |                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Experiment-1</b>  | <b>ENERGY GAP OF P-N JUNCTION DIODE:</b> To determine the energy gap of a given semiconductor diode                                          |
| <b>Experiment-2</b>  | <b>SOLAR CELL:</b> To study the V-I and V-P characteristics and determine the fill factor of solar cell                                      |
| <b>Experiment-3</b>  | <b>LIGHT EMITTING DIODE:</b> To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs       |
| <b>Experiment-4</b>  | <b>HALL EFFECT:</b> To determine Hall co-efficient and charge carrier concentration of a given semiconductor                                 |
| <b>Experiment-5</b>  | <b>PIN PHOTO DIODE:</b> To study the V-I Characteristics of Photo Diode with respect to intensity of light                                   |
| <b>Experiment-6</b>  | <b>OPTICAL FIBRE:</b> To determine the numerical aperture and acceptance angle of an optical fiber                                           |
| <b>Experiment-7</b>  | <b>LASER:</b> To determine the wavelength of a given laser source by using diffraction grating method                                        |
| <b>Experiment-8</b>  | <b>NEWTON'S RINGS:</b> To determine the radius of curvature of a given Plano convex lens by forming Newton's rings                           |
| <b>Experiment-9</b>  | <b>THERMISTOR:</b> To study the variation of resistance with respect to temperature using thermistor                                         |
| <b>Experiment-10</b> | Understanding the method of least squares - Torsional pendulum as an example                                                                 |
| <b>Experiment-11</b> | <b>PLANCK'S CONSTANT:</b> To determine value of planck's constant by measuring radiation in fixed spectral range                             |
| <b>Experiment-12</b> | <b>STEWART GEE'S EXPERIMENT:</b> To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux |

**Note: Students have to perform any 8 experiments**

**Reference Books:**

1. "Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel
2. S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.

## ENGINEERING CHEMISTRY LABORATORY

| I Year I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                 |              |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Category                                                                        | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6BS10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | BSC                                                                             | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                 | 0            | 0 | 3 | 1.5     | 40            | 60  | 100   |
| <b>Course Outcomes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                 |              |   |   |         |               |     |       |
| The experiments will make the student gain skills on:                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                 |              |   |   |         |               |     |       |
| <ul style="list-style-type: none"><li>• Determination of parameters like hardness of water and rate of corrosion of mild steel in various conditions.</li><li>• Able to perform methods such as conductometry, potentiometry and pH metry in order to find out the concentrations or equivalence points of acids and bases.</li><li>• Students are able to prepare polymers like bakelite and nylon - 6.</li><li>• Estimations saponification value, surface tension and viscosity of lubricant oils.</li></ul> |                                                                                 |              |   |   |         |               |     |       |
| <b>LIST OF EXPERIMENTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                 |              |   |   |         |               |     |       |
| Experiment-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Determination of total hardness of water by complexometric method using EDTA    |              |   |   |         |               |     |       |
| Experiment-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Determination of Alkalinity of given water sample                               |              |   |   |         |               |     |       |
| Experiment-3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Estimation of Chloride content of water by Argentometry.                        |              |   |   |         |               |     |       |
| Experiment-4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Estimation of the concentration of an acid by Conductometry                     |              |   |   |         |               |     |       |
| Experiment-5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Estimation of amount of ferrous ion by potentiometry using potassium dichromate |              |   |   |         |               |     |       |
| Experiment-6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Estimation of HCl by potentiometry                                              |              |   |   |         |               |     |       |
| Experiment-7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Preparation of Bakelite                                                         |              |   |   |         |               |     |       |
| Experiment-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Preparation of Thiokol Rubber                                                   |              |   |   |         |               |     |       |
| Experiment-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Estimation of acid value of given lubricant oil                                 |              |   |   |         |               |     |       |
| Experiment-10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Estimation of Viscosity of lubricant oil using Ostwald’s Viscometer.            |              |   |   |         |               |     |       |
| Experiment-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Determination of rate of corrosion of mild steel in the presence and absence of |              |   |   |         |               |     |       |
| Experiment-12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Estimation of Surface tension of a given liquids using Stalagmometer            |              |   |   |         |               |     |       |
| <b>VIRTUAL LABS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                 |              |   |   |         |               |     |       |
| Experiment-13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Construction of Fuel cell and its working                                       |              |   |   |         |               |     |       |
| Experiment-14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Smart materials for Biomedical applications                                     |              |   |   |         |               |     |       |
| Experiment-15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Batteries for electrical vehicles                                               |              |   |   |         |               |     |       |
| Experiment-16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Functioning of solar cell and its applications                                  |              |   |   |         |               |     |       |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                 |              |   |   |         |               |     |       |
| <ol style="list-style-type: none"><li>1. Lab manual for Engineering chemistry by B. Ramadevi and P. Aparna, S Chand Publications, New Delhi (2022)</li><li>2. Vogel’s textbook of practical organic chemistry 5th edition</li><li>3. Inorganic Quantitative analysis by A.I. Vogel, ELBS Publications.</li><li>4. College Practical Chemistry by V.K. Ahluwalia, Narosa Publications Ltd. New Delhi (2007)</li></ol>                                                                                            |                                                                                 |              |   |   |         |               |     |       |



## ENGINEERING WORKSHOP PRACTICES

| I Year I Semester: MECHANICAL ENGINEERING, AERONAUTICAL ENGINEERING & EEE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |              |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ESC      | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          | 0            | 1 | 2 | 2       | 40            | 60  | 100   |
| <b>COURSE OBJECTIVES:</b><br>1. To Study of different hand operated tools, uses and their demonstration.<br>2. To gain a good basic working knowledge required for the production of various engineering products.<br>3. To provide hands on experience about use of different engineering materials, tools, equipment's and processes those are common in the engineering field.<br>4. It explains the construction, function, use and application of different working tools, equipment and machines.<br>5. To study commonly used carpentry joints<br>6. Identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances.<br>7. To develop a right attitude, team working, precision and safety at work place |          |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>Student will be able to:</b><br>1. Assemble different components<br>2. Identify and apply suitable tools for different trades of Engineering processes.<br>3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, House wiring, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.<br>4. Apply basic electrical engineering knowledge for house wiring practice.<br>5. Learn the safety precautions for various operations in basic trade operations.                                                                                                                                                                                                 |          |              |   |   |         |               |     |       |

| <b>WEEKS</b>  | <b>TRADES FOR EXERCISES</b>        |
|---------------|------------------------------------|
|               | <b>Fitting</b>                     |
| <b>Week 1</b> | Filing Four Sides of MS Work piece |
| <b>Week 2</b> | L Fit                              |
|               | <b>Carpentry</b>                   |
| <b>Week 3</b> | Half Lap Joint                     |
| <b>Week 4</b> | Dove Tail Joint                    |
|               | <b>Tin Smithy</b>                  |

|                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Week 5</b>                                                                                                                                                                                                                                                                                                                                                                               | Prepare a Rectangular Tray                                                                                               |
| <b>Week 6</b>                                                                                                                                                                                                                                                                                                                                                                               | Prepare A Square Tin                                                                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                             | <b>Electrical</b>                                                                                                        |
| <b>Week 7</b>                                                                                                                                                                                                                                                                                                                                                                               | Series and Parallel Connection One Way Switch                                                                            |
| <b>Week 8</b>                                                                                                                                                                                                                                                                                                                                                                               | Two Way Switch Connection Stair Case Wiring                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                             | <b>Electronics</b>                                                                                                       |
| <b>Week 9</b>                                                                                                                                                                                                                                                                                                                                                                               | Soldering - Series Connection & Parallel Connection                                                                      |
| <b>Week 10</b>                                                                                                                                                                                                                                                                                                                                                                              | Desoldering & Construction of Wheat stone bridge                                                                         |
| <b>WEEK</b>                                                                                                                                                                                                                                                                                                                                                                                 | <b>TRADES FOR DEMONSTRATION &amp; EXPOSURE</b>                                                                           |
| <b>Week 11</b>                                                                                                                                                                                                                                                                                                                                                                              | Introduction to Black smithy , Plumbing , Lathe Machine , Injection Moulding , Welding , Milling , Drilling and foundry. |
| <b>TEXT BOOKS</b>                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                          |
| 1. Work shop Manual - P. Kannaiah/ K.L. Narayana/ Scitech.<br>2. Workshop Practice /B. L. Juneja / Cengage<br>3. Rao P.N., "ManufacturingTechnology", Tata Mc Graw Hill House, Vol.I and Vol.II.<br>4. Workshop Manual / K. Venugopal / Anuradha Publications.                                                                                                                              |                                                                                                                          |
| <b>REFERENCE BOOKS:</b>                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                          |
| 1. Hajra Choudhury S.K., Hajra Choudhury A.K.and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.<br>2. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4 <sup>th</sup> edition, 2002.<br>3. Workshop Manual / Venkat Reddy/ BSP |                                                                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                          |

| ENVIRONMENTAL SCIENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |            |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------|---|---|---------|---------------|-----|-------|
| I B.TECH-I/II SEMESTER: COMMON TO ALL BRANCHES                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                         |            |   |   |         |               |     |       |
| Course Code:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Category                                                | Hours/Week |   |   | Credits | Maximum Marks |     |       |
| A6BS11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MC                                                      | L          | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                         | 3          | 0 | 0 | 0       | 50            | -   | 50    |
| <b>COURSEOUTCOMES:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                         |            |   |   |         |               |     |       |
| <b>On Successful completion of this course, Students will be able to</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                         |            |   |   |         |               |     |       |
| <div>1. Identify the consequences of human actions on the web of life, global economy and quality of human life.</div> <div>2. Evaluate the strategies for scientific, social, economic and legal environmental protection.</div> <div>3. Study the impact of conservation of biodiversity.</div> <div>4. Analyze the reasons for environmental pollution.</div> <div>5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.</div> |                                                         |            |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ECOSYSTEMS</b>                                       |            |   |   |         |               |     |       |
| <b>Ecosystems:</b> Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids.                                                                                                                                                                                                                                                                                                                                              |                                                         |            |   |   |         |               |     |       |
| <b>Flow of Energy:</b><br>Biogeochemical cycles, Bioaccumulation, Biomagnification, ecosystem value, services and carrying capacity.                                                                                                                                                                                                                                                                                                                                                                                  |                                                         |            |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>NATURAL RESOURCES</b>                                |            |   |   |         |               |     |       |
| <b>Natural Resources:</b> Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems - case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources.                                                                                                                                           |                                                         |            |   |   |         |               |     |       |
| <b>Energy resources:</b> Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.                                                                                                                                                                                                                                                                                                                                                                              |                                                         |            |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>BIODIVERSITY AND BIOTIC RESOURCES</b>                |            |   |   |         |               |     |       |
| <b>Biodiversity and Biotic Resources:</b> Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values.                                                                                                                                                                                                                                                                                                 |                                                         |            |   |   |         |               |     |       |
| <b>India as a mega diversity nation:</b> Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man - wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity Act - Case studies.                                                                                                                                                                                                                                                  |                                                         |            |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES</b> |            |   |   |         |               |     |       |
| <b>Environmental Pollution and Control Technologies:</b> Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation.                                                                                                                                                                                                                  |                                                         |            |   |   |         |               |     |       |
| <b>Global Environmental Problems and Global Efforts:</b> Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions /Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-GoI Initiatives, COP 24, COP25.                                                                                                                                                                                                                                              |                                                         |            |   |   |         |               |     |       |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ENVIRONMENTAL POLICY, LEGISLATION & EIA |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <p><b>Environmental Policy, Legislation &amp; Environmental Impact Assessment (EIA):</b> Environmental Protection act, Legal aspects Air Act 1981, Water Act, Forest Act, Wildlife Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules.</p> <p><b>Environmental Impact Assessment:</b> EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.</p> |                                         |
| <p><b>TEXT BOOKS:</b></p> <ol style="list-style-type: none"><li>1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.</li><li>2. Environmental Studies by R.Rajagopalan, Oxford University Press.</li></ol>                                                                                                                                                                                                                                                                              |                                         |

# **B.TECH II SEMESTER SYLLABUS**

## NUMERICAL METHODS AND INTEGRAL TRANSFORMS

| I Year II Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                        |              |   |   |         |               |     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Category                               | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6BS02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | BSC                                    | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                        | 3            | 1 | 0 | 4       | 40            | 60  | 100   |
| <b>Course Objectives</b><br>To learn<br>1. Curve fitting and Interpolation techniques.<br>2. Numerical techniques.<br>3. Fourier series for periodic function<br>4. Laplace transforms<br>5. Concept and application of Fourier Transforms and Vector differentiation.                                                                                                                                                                                                                                                                                                                                |                                        |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>INTERPOLATION AND CURVE FITTING</b> |              |   |   |         |               |     |       |
| <b>INTERPOLATION:</b> Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton’s forward interpolation, Newton’s backward interpolation, Gauss’s forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange’s interpolation.<br><br><b>CURVE FITTING:</b> Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{bx}$ , $y = a x^b$ , $y = a b^x$ by the method of least squares. |                                        |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>NUMERICAL TECHNIQUES</b>            |              |   |   |         |               |     |       |
| Root Finding Techniques:<br><br>Bisection method Regula falsi method, Iteration method and Newton Raphson method.<br><br>Numerical Integration :<br><br>Trapezoidal rule - Simpson’s one-third rule - Simpson’s three-eighth rule.<br><br>Numerical Solution of Ordinary Differential Equations:<br><br>Taylor’s series method – Euler’s - modified Euler’s Method – Runge - Kutta method.                                                                                                                                                                                                            |                                        |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>FOURIER SERIES</b>                  |              |   |   |         |               |     |       |
| Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.                                                                                                                                                                                                                                                                                                                                                                                                                     |                                        |              |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>LAPLACE TRANSFORMS</b>                            |
| Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by $t^n$ - Division by $t$ – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations. |                                                      |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>FOURIER TRANSFORMS AND VECTOR DIFFERENTIATION</b> |
| Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms.<br><br>Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.                                                                                                                                                                                                                                |                                                      |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                      |
| <ol style="list-style-type: none"> <li>1. Ervin Kreyszig, Advanced Engineering Mathematics, 9<sup>th</sup> Edition, John Wiley &amp; Sons, 2006.</li> <li>2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.</li> </ol>                                                                                                                                                                                                                                                            |                                                      |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                      |
| <ol style="list-style-type: none"> <li>1. G.B.Thomas, calculus and analytical geometry, 9<sup>th</sup> Edition, Pearson Reprint 2006.</li> <li>2. N.P Bali and Manish Goyal, A Text of Engineering Mathematics, Laxmi publications, 2008.</li> <li>3. E.L.Ince, Ordinary differential Equations, Dover publications, 1958.</li> </ol>                                                                                                                                                                                 |                                                      |
| <b>Course Outcomes:</b><br><b>At the end of the course, student will be able to:</b> <ol style="list-style-type: none"> <li>1. Apply Curve fitting and Interpolation techniques.</li> <li>2. Apply various numerical techniques</li> <li>3. Find the Fourier series of the periodic functions.</li> <li>4. Obtain the Laplace transforms of functions</li> <li>5. Find Fourier transforms and apply vector differentiation techniques.</li> </ol>                                                                     |                                                      |

## ENGLISH FOR SKILL ENHANCEMENT

### **I B.TECH II Semester: Common to All Branches**

| Course Code   | Category    | Hours / Week |          |          | Credits  | Maximum Marks |            |              |
|---------------|-------------|--------------|----------|----------|----------|---------------|------------|--------------|
| <b>A6HS01</b> | <b>HSMC</b> | <b>L</b>     | <b>T</b> | <b>P</b> | <b>C</b> | <b>CIE</b>    | <b>SEE</b> | <b>Total</b> |
|               |             | 3            | 0        | 0        | 3        | 40            | 60         | 100          |

### **COURSE OBJECTIVES:**

**The course will enable the students to:**

1. Develop language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

### **UNIT-I**

#### **‘Toasted English’ by R. K. Narayan**

|                   |                                                                                                                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vocabulary</b> | : The Concept of Word Formation -The Use of Prefixes and Suffixes - Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives - Synonyms and Antonyms                                                                                                     |
| <b>Grammar</b>    | : Identifying Common Errors in Writing with Reference to Articles and Prepositions                                                                                                                                                                                                 |
| <b>Reading:</b>   | : Reading and Its Importance- Techniques for Effective Reading                                                                                                                                                                                                                     |
| <b>Writing</b>    | : Sentence Structures – Use of Phrases and Clauses in Sentences – Importance of Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents. |

### **UNIT-II**

#### **‘Appro JRD’ by Sudha Murthy**

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vocabulary</b> | : Words Often Mis-spelt – Homophones, Homonyms and Homographs                                                                         |
| <b>Grammar</b>    | : Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement                            |
| <b>Reading</b>    | : Sub-Skills of Reading – Skimming and Scanning – Exercises for Practice                                                              |
| <b>Writing</b>    | : Nature and Style of Writing- Defining /Describing People, Objects, Places and Events – Classifying- Providing Examples or Evidence. |

### **UNIT-III**

#### **‘Lessons from Online Learning’ by F.Haider Alvi, Deborah Hurst et al**

|                   |                                                                                 |
|-------------------|---------------------------------------------------------------------------------|
| <b>Vocabulary</b> | : Words Often Confused – Words from Foreign Languages and their Use in English. |
|-------------------|---------------------------------------------------------------------------------|



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grammar</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses                                                                                           |
| <b>Reading</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Sub-Skills of Reading – Intensive Reading and Extensive Reading – Exercises for Practice                                                                                        |
| <b>Writing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Format of a Formal Letter-Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV/Resume                              |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Art and Literature</b>                                                                                                                                                         |
| <b>Vocabulary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | : Standard Abbreviations in English                                                                                                                                               |
| <b>Grammar</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Redundancies and Clichés in Oral and Written Communication                                                                                                                      |
| <b>Reading</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Survey, Question, Read, Recite and Review (SQ3R Method) - Exercises for Practice                                                                                                |
| <b>Writing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Writing Practices- Essay Writing-Writing Introduction and Conclusion -Précis Writing                                                                                            |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Go, Kiss the World’ by Subroto Bagchi</b>                                                                                                                                      |
| <b>Vocabulary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | : Technical Vocabulary and their Usage                                                                                                                                            |
| <b>Grammar</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Common Errors in English ( <i>Covering all the other aspects of grammar which were not covered in the previous units</i> )                                                      |
| <b>Reading</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Reading Comprehension-Exercises for Practice                                                                                                                                    |
| <b>Writing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | : Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports<br>- Writing a Report |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                   |
| 1. “English: Language, Context and Culture” by Orient Black Swan Pvt. Ltd, Hyderabad. 2022. Print.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                   |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                   |
| 1. Effective Academic Writing by Liss and Davis (OUP)<br>2. Richards, Jack C. (2022) Interchange Series. Introduction, 1,2,3. Cambridge University Press<br>3. Wood, F.T. (2007). Remedial English Grammar. Macmillan.<br>4. Chaudhuri, Santanu Sinha. (2018). Learn English: A Fun Book of Functional Language, Grammar and Vocabulary. (2 <sup>nd</sup> ed.), Sage Publications India Pvt. Ltd.<br>5. (2019). Technical Communication. Wiley India Pvt. Ltd.<br>6. Vishwamohan, Aysha. (2013). English for Technical Communication for Engineering Students. Mc Graw-Hill Education India Pvt. Ltd.<br>7. Swan, Michael. (2016). Practical English Usage. Oxford University Press. Fourth Edition |                                                                                                                                                                                   |

**Course Outcomes:**

1. will be able to acquire language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Use language components to communicate effectively in formal and informal situations

## ENGINEERING MECHANICS

| I Year II Semester: MECHANICAL ENGINEERING & AERONAUTICAL ENGINEERING                                                                                                                                                                                                            |                                                                                |              |   |   |         |               |     |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                      | Category                                                                       | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME05                                                                                                                                                                                                                                                                           | ESC                                                                            | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                  |                                                                                | 3            | 1 | 0 | 4       | 40            | 60  | 100   |
| <b>COURSE OUTCOMES:</b>                                                                                                                                                                                                                                                          |                                                                                |              |   |   |         |               |     |       |
| At the end of the course the student should be able to:                                                                                                                                                                                                                          |                                                                                |              |   |   |         |               |     |       |
| 1. Evaluate different system of forces acting on rigid body under static equilibrium.                                                                                                                                                                                            |                                                                                |              |   |   |         |               |     |       |
| 2. Analyse the frictional forces existing at different contact surfaces of the rigid body under static equilibrium.                                                                                                                                                              |                                                                                |              |   |   |         |               |     |       |
| 3. Determine the centre of gravity and moment of inertia of simple and composite bodies.                                                                                                                                                                                         |                                                                                |              |   |   |         |               |     |       |
| 4. Analyse linear and rotary motion of particle or rigid bodies using Kinematic & Kinetic Equations.                                                                                                                                                                             |                                                                                |              |   |   |         |               |     |       |
| 5. Evaluate solutions for mechanics problems by using virtual work method and also able to distinguish basics of mechanical vibrations.                                                                                                                                          |                                                                                |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                           | INTRODUCTION TO ENGINEERING MECHANICS & EQUILIBRIUM OF PLANAR SYSTEM OF FORCES |              |   |   |         |               |     |       |
| Introduction to engineering mechanics: Basic concepts, Classification of a force system, Parallelogram law, Resultant of coplanar concurrent force system, Moment of a force, its applications, Varignon's theorem, Resultant of coplanar non-concurrent force systems, Couples. |                                                                                |              |   |   |         |               |     |       |
| Equilibrium of Planar System of forces: Equilibrium, Conditions of equilibrium, free body diagrams, Lami's theorem.                                                                                                                                                              |                                                                                |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                          | FRICTION                                                                       |              |   |   |         |               |     |       |
| Friction: Introduction, types of friction, laws of friction, coefficient of friction, angle of friction, angle of repose, analysis of bodies on rough horizontal, inclined planes, wedge friction, ladder friction and screw friction                                            |                                                                                |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                         | CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA                                |              |   |   |         |               |     |       |
| Centroid & Centre of gravity: Introduction, centroids & centre of gravity of simple figures (from first principles), centroid of composite sections, centre of gravity of composite solids, theorems of pappus & Guldinus.                                                       |                                                                                |              |   |   |         |               |     |       |
| Moment of Inertia: Area moment of inertia of plane sections (from first principles), Transfer theorems, Moment of inertia of standard sections and composite sections. Mass moment of inertia of rectangular plate, circular plate, cone, cylinder and sphere.                   |                                                                                |              |   |   |         |               |     |       |
| UNIT-IV                                                                                                                                                                                                                                                                          | KINEMATICS & KINETICS                                                          |              |   |   |         |               |     |       |
| Kinematics of a particle: rectilinear motion, curvilinear motion, Rotary motion.                                                                                                                                                                                                 |                                                                                |              |   |   |         |               |     |       |
| Kinetics of rigid bodies: Analysis of connecting bodies, kinetics of rigid body rotation, D' Alembert's principle, work-energy theorem, impulse-momentum principle, their applications                                                                                           |                                                                                |              |   |   |         |               |     |       |

| UNIT-V                                                                                                                                                                                                                                                                                         | VIRTUAL WORK & MECHANICAL VIBRATIONS |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Virtual work:</b> Concept of virtual work, Principle of virtual work, its applications.                                                                                                                                                                                                     |                                      |
| <b>Mechanical Vibrations:</b> Basic terminology, free and forced vibrations, resonance, simple harmonic motion, simple, compound and torsional pendulums.                                                                                                                                      |                                      |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                             |                                      |
| <ol style="list-style-type: none"><li>1. Reddy Vijay Kumar and K. Suresh Kumar, Singer's Engineering Mechanics</li><li>2. Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company</li><li>3. Bansal R.K, A Text book of Engineering Mechanics, Laxmi Publications</li></ol> |                                      |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                        |                                      |
| <ol style="list-style-type: none"><li>1. Engineering Mechanics/Irving Shames/Prentice Hall</li><li>2. Engineering Mechanics/N.H Dubey/Tata Mc Graw Hill</li><li>3. A Text book of Engineering Mechanics/ R S Khurmi / S Chand</li></ol>                                                        |                                      |

## PROGRAMMING FOR PROBLEM SOLVING

| I/II Semester: Common for all Branches                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                      |              |   |   |         |               |     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Category                                                                             | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6CS02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ESC                                                                                  | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <p><b>Course Outcomes :</b></p> <ol style="list-style-type: none"><li>1) Apply algorithmic thinking to understand, define and solve problems</li><li>2) Develop computer programs using programming constructs and control structures and to use arrays to develop C programs</li><li>3) Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems.</li><li>4) Use Strings and structures to formulate algorithms and programs.</li><li>5) Use FILE to perform read and write operations.</li></ol>                                    |                                                                                      |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING & INTRODUCTION TO C LANGUAGE |              |   |   |         |               |     |       |
| <p><b>Algorithm</b> -Definition, Characteristics of Algorithm. Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum And Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples.</p> <p><b>Introduction To C Language:</b> Structure of C Program, Data Types, data input and output statements, Operators, Precedence and Associativity of operators , Evaluation of Expressions, Type Conversions In Expressions.</p> |                                                                                      |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CONTROL STRUCTURES AND ARRAYS                                                        |              |   |   |         |               |     |       |
| <p><b>Control structures: Decision statements;</b> if and switch statement; <b>Loop control statements:</b> while, for and do while loops, Jump statements: break, continue, goto statements.</p> <p><b>Arrays:</b> Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays</p>                                                                                                                                                                                              |                                                                                      |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FUNCTIONS AND POINTERS                                                               |              |   |   |         |               |     |       |
| <p><b>Functions:</b> Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc.. Storage classes.</p> <p><b>Pointers:</b> Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.</p>            |                                                                                      |              |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>STRINGS AND USER DEFINED DATA TYPES</b>   |
| <p><b>Strings:</b> Arrays of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings</p> <p><b>Structures and Unions:</b> Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.</p>                                                                                                                                                                                                                                                           |                                              |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>FILE HANDLING , SEARCHING AND SORTING</b> |
| <p><b>File Handling:</b> Command Line Arguments, File Modes, Basic File Operations Read, Write And Append, Example Programs. Random Access Using fseek, ftell and rewind Functions.</p> <p><b>Basic Searching And Sorting Algorithms:</b> Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.</p>                                                                                                                                                                                                                                                                                                                                                                 |                                              |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                              |
| <ol style="list-style-type: none"> <li>1) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)</li> <li>2) Byron Gottfried, “Programming with C”, Schaum's Outlines Series, McGraw Hill Education, 3<sup>rd</sup> edition, 2017.</li> <li>3) Programming in C E. Balagurusamy, Tata McGraw-Hill Publishing,</li> </ol>                                                                                                                                                                                                                                                                                                              |                                              |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                              |
| <ol style="list-style-type: none"> <li>1) W. Kernighan Brian, Dennis M. Ritchie, “The C Programming Language”, PHI Learning, 2nd Edition, 1988.</li> <li>2) Yashavant Kanetkar, “Exploring C”, BPB Publishers, 2nd Edition, 2003.</li> <li>3) Schildt Herbert, “C: The Complete Reference”, Tata McGraw Hill Education, 4th Edition, 2014.</li> <li>4) R. S. Bichkar, “Programming with C”, Universities Press, 2nd Edition, 2012.</li> <li>5) Dey Pradeep, Manas Ghosh, “Computer Fundamentals and Programming in C”, Oxford University Press, 2nd Edition, 2006.</li> <li>6) Stephen G. Kochan, “Programming in C”, Addison-Wesley Professional, 4th Edition, 2014.</li> </ol> |                                              |

## ELEMENTS OF MECHANICAL ENGINEERING DESIGN

| I Year II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                           |                                       |              |   |   |         |               |     |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                          | Category                              | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME06                                                                                                                                                                                                                                                                                                                                                                                                                               | ESC                                   | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                       | 1            | 0 | 2 | 2       | 50            | -   | 50    |
| <b>COURSE OVER VIEW:</b><br>The purpose of this course is to introduce the fundamental principles of Mechanical Engineering Design which is very important and relevant in the context of today's engineering professionals.                                                                                                                                                                                                         |                                       |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br>The course should enable the students to:<br>1. Understand the working, usage and applications of different measuring instruments<br>2. Select suitable material fit for the purpose.<br>3. Identify design principles from an engineering perspective.<br>4. Devise visual design using conceptualization techniques<br>5. Choose an appropriate reverse engineering technique for the component design. |                                       |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>MEASUREMENTS</b>                   |              |   |   |         |               |     |       |
| Vernier caliper, micrometer and screw gauge – types, usage and applications; Transducers – Force, Torque, Strain, Temperature and Pressure; Accuracy, precision and resolution; measurement errors; Introduction to fits and Tolerances.<br><b>Practice / Demonstration:</b><br>Measurement of length, height and diameter; Measurement of Force, Temperature and Pressure                                                           |                                       |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>MATERIAL SELECTION</b>             |              |   |   |         |               |     |       |
| Commonly used materials, standard sizes and shapes, Fit for purpose, The role of materials selection in design, Modulus-Density, Strength-Density, Material selection process, Introduction to 3D printer materials;<br><b>Case studies and Brain storming.</b>                                                                                                                                                                      |                                       |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>MECHANICAL DESIGN FUNDAMENTALS</b> |              |   |   |         |               |     |       |
| Fundamental principles – Laws of nature, Simplicity Vs Complexity, Golden rectangle, Functional independence, Reciprocity, Self principles, Stability, Symmetry, Triangulate for stiffness, Load paths, Avoid bending stresses, Managing of friction, Preload, Centers of action, Exact constraints and controlled compliance;<br><b>Case studies and Brain storming.</b>                                                            |                                       |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>DESIGN THINKING</b>                |              |   |   |         |               |     |       |
| Design Conceptualization and Philosophy, Original, Adaptive, Variant and Re-Design, Innovation, Types of innovation, Conceptualization techniques: Idea generation, brain storming, Mind maps, SCAMPER, TRIZ, Biomimicry, Shape mimicry, SWOT analysis; <b>Case studies and Brain storming.</b>                                                                                                                                      |                                       |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>REVERSE ENGINEERING</b>            |              |   |   |         |               |     |       |
| Objectives and common application fields, Existing technologies, Contact systems, Non-contact systems, Introduction to Prototyping – 3D Scanning, 3D Printing, Laser Cutting, Plasma Cutting, Water Cutting, Introduction to CNC Machines.<br><b>Practice / Demonstration on:</b><br>3D Printer, Laser Cutting machine, CNC Lathe and Milling machine; Simple parts - Prototype preparation.                                         |                                       |              |   |   |         |               |     |       |

**Reference Books:**

1. Anand K Bewoor and Vinay A Kulkarni, Metrology & Measurements, McGraw Hill Education, 2017.
2. R. K. Jain, Mechanical and Industrial Measurements, 12th edition, Khanna Publishers, 2011.
3. Michael F. Ashby, Materials Selection in Mechanical Design, 5th Edition, Butterworth-Heinemann, 2016.
4. Ullman. D.G, The Mechanical Design Process, McGraw-Hill, 1997.
5. Otto. K and Wood, K, Product Design, Pearson Education, 2001.
6. Daniel Ling, Complete Design Thinking Guide for Successful Professionals, Create Space Independent Publishing, 2015.
7. Jimmy Jain, Design Thinking for Startups: A Handbook for Readers and Workbook for Practitioners, Notion Press, 2018.



## ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB

| Course Code | Category | Hours / |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|---------|---|---|---------|---------------|-----|-------|
| A6HS02      | HSMC     | L       | T | P | C       | CIE           | SEE | Total |
|             |          | 0       | 0 | 3 | 1.5     | 40            | 60  | 100   |

### COURSE OBJECTIVES:

**The course should enable the students:**

1. To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning
2. To sensitize the students to the nuances of English speech sounds, word accent, intonation and rhythm
3. To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
4. To improve the fluency of students in spoken English and neutralize the impact of dialects.
5. To train students to use language appropriately for public speaking, group discussions and interviews

**English Language and Communication Skills Lab (ELCS) shall have two parts:**

- a. Computer Assisted Language Learning (CALL) Lab
- b. Interactive Communication Skills (ICS) Lab

### Listening Skills

#### Objectives

1. To enable students develop their listening skills to appreciate its role in the LSRW skills approach to language and improve their pronunciation.
2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions.

Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.

- Listening for general content
- Listening to fill up information
- Intensive listening
- Listening for specific information

### Speaking Skills

#### Objectives

1. To involve students in speaking activities in various contexts
2. To enable students express themselves fluently and appropriately in social and professional contexts
  - Oral practice
  - Describing objects/situations/people
  - Role play – Individual/Group activities
  - Just A Minute (JAM) Sessions

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Exercise-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| <b>CALL Lab:</b><br><i>Understand:</i> Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening.<br><i>Practice:</i> Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs- Consonant Clusters- Past Tense Marker and Plural Marker- <i>Testing Exercises</i><br><b>ICS Lab:</b><br><i>Understand:</i> Spoken vs. Written language- Formal and Informal English.<br><i>Practice:</i> Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others.                      |  |
| <b>Exercise-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>CALL Lab:</b><br><i>Understand:</i> Structure of Syllables – Word Stress– Weak Forms and Strong Forms – Stress pattern in sentences – Intonation.<br><i>Practice:</i> Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms- Stress pattern in sentences – Intonation - <i>Testing Exercises</i><br><b>ICS Lab:</b><br><i>Understand:</i> Features of Good Conversation – Strategies for Effective Communication.<br><i>Practice:</i> Situational Dialogues – Role Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette. |  |
| <b>Exercise-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| <b>CALL Lab:</b><br><i>Understand:</i> Errors in Pronunciation-Neutralising Mother Tongue Interference (MTI).<br><i>Practice:</i> Common Indian Variants in Pronunciation – Differences between British and American Pronunciation -Testing Exercises<br><b>ICS Lab:</b><br><i>Understand:</i> Descriptions- Narrations- Giving Directions and Guidelines – Blog Writing<br><i>Practice:</i> Giving Instructions – Seeking Clarifications – Asking for and Giving Directions – Thanking and Responding – Agreeing and Disagreeing – Seeking and Giving Advice – Making Suggestions.             |  |
| <b>Exercise-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>CALL Lab:</b><br><i>Understand:</i> Listening for General Details.<br><i>Practice:</i> Listening Comprehension Tests - Testing Exercises<br><b>ICS Lab:</b><br><i>Understand:</i> Public Speaking – Exposure to Structured Talks - Non-verbal Communication- Presentation Skills.<br><i>Practice:</i> Making a Short Speech – Extempore- Making a Presentation.                                                                                                                                                                                                                              |  |
| <b>Exercise-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| <b>CALL Lab:</b><br><i>Understand:</i> Listening for Specific Details.<br><i>Practice:</i> Listening Comprehension Tests - <i>Testing Exercises</i><br><b>ICS Lab:</b><br><i>Understand:</i> Group Discussion<br><i>Practice:</i> Group Discussion                                                                                                                                                                                                                                                                                                                                              |  |

**Reference Books:**

1. (2022). *English Language Communication Skills – Lab Manual cum Workbook*. Cengage Learning India Pvt. Ltd.
2. Shobha, KN & Rayen, J. Lourdes. (2019). *Communicative English – A workbook*. Cambridge University Press
3. Kumar, Sanjay & Lata, Pushp. (2019). *Communication Skills: A Workbook*. Oxford University Press
4. Board of Editors. (2016). *ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities*. Orient Black Swan Pvt. Ltd.
5. Mishra, Veerendra et al. (2020). *English Language Skills: A Practical Approach*. Cambridge University Press

**COURSE OUTCOMES:**

Students will be able to:

1. Acquire the skill of independent language learning
2. Overcome with the nuances of English speech sounds, word accent, in to nation and rytham.
3. Obtain the consistent accent and intelligibility in pronunciation.
4. Neutralize the impact of dialects.
5. Apply language appropriately public speaking, group discussions and interviews.

## PROGRAMMING FOR PROBLEM SOLVING LAB

**I - II Semester: Common to all branches**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6CS03      | ESC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 3 | 1.5     | 40            | 60  | 100   |

### COURSE OBJECTIVES:

- 1) To be familiarize with flowgorithm to solve simple problems
- 2) To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
- 3) To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings, pointers and structures.

### COURSE OUTCOMES:

**At the end of the course, student will be able to**

- 1) Solve simple mathematical problems using Flowgorithm.
- 2) Correct syntax errors as reported by the compilers and logical errors encountered at run time.
- 3) Develop programs by using decision making and looping constructs.
- 4) Implement real time applications using the concept of array, pointers, functions and structures.
- 5) Solve real world problems using matrices, searching and sorting.

### LIST OF EXPERIMENTS

#### Week-1 INTRODUCTION TO FLOGORITHM

1. Installation and working of Flowgorithm Software.
2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc.
3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle).
4. Draw a flowchart to find the sum of individual digits of a 3 digit number
5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.
6. Draw a flowchart to find roots of a quadratic equation.
7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd
8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| triangle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                               |
| 9. Draw a flowchart to check whether a given number is palindrome or not.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                               |
| <b>Week-2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>BASIC DATA TYPES</b>       |
| 1) Write a C program to find division of 1 integer 1 float numbers.<br>2) Write a C program to find division of 2 integer numbers.<br>3) Write a C program to find average of (1int,1float) numbers.<br>4) Write a C Program to Swap Numbers without Using Temporary Variable<br>5) Write a C Program to Swap two Numbers Using Temporary Variable<br>6) Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line.<br>a) $(x+y+z) / (x-y-z)$<br>b) $(x+y+z) / 3$<br>c) $(x+y) * (x-y) * (y-z)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |
| <b>Week-3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>OPERATORS</b>              |
| 1) Write a C program to convert temperature from Fahrenheit to Celsius and vice versa ( $c=(fh-32)/1.8$ )<br>2) Write a C program to find area and perimeter of a circle. ( $area=\pi r^2$ perimeter= $2\pi r$ )<br>3) Write a C program to calculate area and perimeter of a right angled triangle.<br>a. ( $Area=1/2*b*h$ perimeter= $w+h+sqrt(w*w+h*h)$ )<br>4) Find the sum of natural numbers 1 to n.(read n as input) (Use formula $sum=n(n+1)/2$ )<br>5) Write a C program to calculate Simple interest ( $SI=PTR/100$ )<br>6) Write a C program to calculate area and perimeter of a rectangle. $Area=l*b$<br>Perimeter= $2*(l+b)$<br>7) Write a C program to calculate the value of the third angle of a triangle if two angles are given as input.( $a+b+c=180$ )<br>8) Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. ( $Amt=num\ of\ units*cost$ )<br>9) Write a C Program to calculate area and perimeter of a triangle.<br>a. Perimeter= $(a+b+c)$<br>b. $s=(a+b+c)/2$<br>c. $Area=sqrt(s*(s-a)*(s-b)*(s-c))$<br>10) Write a C program to read five Subject marks and find the average.<br>11) Write a C program to Calculate Compound interest ( $CI=p(1+r/100)^n$ ) |                               |
| <b>Week-4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CONDITIONAL STATEMENTS</b> |

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program which takes two integer operands and one operator from the user(+, -, \*, /, % use switch)
3. Write a program to compute grade of students using if else ladder. The grades are assigned as followed:
 

marks < 50 F  
 50 ≤ marks < 60 C  
 60 ≤ marks < 70 B  
 70 ≤ marks < 80 B+  
 80 ≤ marks < 90 A  
 90 ≤ marks ≤ 100 A+
4. Write a C program to whether given year leap year or not.
5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.

### Week-5      LOOPING STATEMENTS

- 1) Write a C program to find Sum of individual digits of given integer
- 2) Write a C program to generate first n terms of Fibonacci series
- 3) Write a C Program to find the Sum of Series  $SUM = 1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$
- 4) Write a C program to print the Fibonacci sequence up to given value of n.
- 5) Write a C program to print the multiplication table to the given value of n.
- 6) Write a C program to check whether given number is palindrome or not
- 7) Write a C program to check whether given number is perfect or not

### Week-6      NESTED LOOPING STATEMENTS

- 1) Write a C program to generate prime numbers between 1 and n
- 2) Write a C program to generate Pascal's triangle.
- 3) Write a C program to generate the following pyramid of numbers.

```

      1
    1 3 1
  1 3 5 3 1

```

- 4) Write a C program to generate the pattern

```

*****
****
***
**
*

```

### Week-7      ARRAYS

- 1) Write a C Program to implement following searching methods
  - i. Binary Search

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| ii. Linear Search<br>2) Write a C program to find largest and smallest number in a list of integers<br>3) Write a C program<br>i. To add two matrices<br>ii. To multiply two matrices<br>4) Write a C program to find Transpose of a given matrix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |
| <b>Week-8</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>FUNCTIONS</b> |
| 1) Write a C program to find the factorial of a given integer using non recursive functions<br>2) Write a C program to find GCD of given integers using non recursive functions<br>3) Write a C Program to find the power of a given number using non recursive functions<br>4) Write a C program to find sum of natural numbers using non recursive function.<br>5) Write a C program to reverse a given integer number using non recursive functions<br>6) Write a C Program to find binary equivalent of a given decimal number using recursive functions.<br>7) Write a C Program to print Fibonacci sequence using recursive functions.<br>8) Write a C Program to find LCM of 3 given numbers using recursive functions<br>9) Write a C program to find the factorial of a given integer using recursive functions<br>10) Write a C program to print fibonacci series till n terms using recursion. |                  |
| <b>Week-9</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>STRINGS</b>   |
| 1) Write a C program using to Insert a sub string into a given main string from a given position<br>2) Write a C program using to Delete n characters from a given position in a string<br>3) Write a C program to determine if given string is palindrome or not<br>4) Write C Programs to demonstrate the following string handling functions.<br>a. strcat()<br>b. strcmp()<br>c. strrev()<br>d. strcpy()<br>e. strlen()<br>f. strstr()<br>g. strncpy()<br>h. strncat()<br>i. strncmp()                                                                                                                                                                                                                                                                                                                                                                                                                |                  |
| <b>Week-10</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>POINTERS</b>  |
| 1) Write a C program to read the elements of 1-d array using pointers and print them in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>reverse order using pointers.</p> <ol style="list-style-type: none"> <li>2) Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference.</li> <li>3) Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array.</li> <li>4) Write a C Program to print 2-D array using pointers.</li> </ol>                                            |                   |
| <b>Week-11</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>STRUCTURES</b> |
| <ol style="list-style-type: none"> <li>1) Write a C Program using functions to               <ol style="list-style-type: none"> <li>a) Reading a complex number</li> <li>b) Writing a complex number</li> <li>c) Add two complex numbers</li> <li>d) Multiply two complex numbers</li> </ol> <p>Note: represent complex number using structure</p> </li> <li>2) Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.</li> </ol> |                   |
| <b>Week-12</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>FILES</b>      |
| <ol style="list-style-type: none"> <li>1) Write a C program to read and print the content of a file.</li> <li>2) Write a C program copy the content of one file to another file</li> <li>3) Write a C program to merge two file into third file.</li> <li>4) Write a C Program to find the number of lines in a text file</li> </ol>                                                                                                                                                                                                                                                     |                   |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |
| <ol style="list-style-type: none"> <li>1) Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014.</li> <li>2) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13<sup>th</sup> Edition, 2012.</li> </ol>                                                                                                                                                                                                                                |                   |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |
| <ol style="list-style-type: none"> <li>1) Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018</li> <li>2) King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2<sup>nd</sup> Edition, 2015.</li> <li>3) Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3<sup>rd</sup> Edition, 2004.</li> <li>4) Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1<sup>st</sup> Edition, 1994.</li> </ol>                                                    |                   |



# **B.TECH III SEMESTER SYLLABUS**

**PROBABILITY, STATISTICS AND COMPLEX ANALYSIS**

**II B.TECH II Semester: MECH , AERO**

| Course Code:  | Category   | Hours / Week |          |          | Credits  | Maximum Marks |            |              |
|---------------|------------|--------------|----------|----------|----------|---------------|------------|--------------|
| <b>A6BS05</b> | <b>BSC</b> | <b>L</b>     | <b>T</b> | <b>P</b> | <b>C</b> | <b>CIE</b>    | <b>SEE</b> | <b>Total</b> |
|               |            | 3            | 1        | -        | 4        | 40            | 60         | 100          |

**Course Objectives**

1. Evaluation of Probability distribution of Discrete and Continuous random
2. Evaluation of the given data for appropriate test of hypothesis for large samples.
3. Evaluation of the given data for appropriate test of hypothesis for small samples.
4. Evaluation of Limit, continuity, differentiability, analyticity and finding harmonic function
5. Evaluation of the line integrals along piece wise smooth paths& Express the given complex function as a power series using Taylor's series, Maclaurin's series, Residues and improper integrals.

**UNIT-I**

**RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS**

Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation,.

Binomial, Poisson, Normal distributions, and their Properties and hence find mean and variance

**UNIT-II**

**TESTING OF HYPOTHESIS - I**

**Testing of hypothesis:** Null hypothesis, Alternate hypothesis, Type I& Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test.

**Large sample tests:**

- (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances)
- (ii) Tests of significance of difference between sample S.D and population S.D.
- (iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                 |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--|
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>TESTING OF HYPOTHESIS-II</b> |  |
| Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit.                                                                                                                                                                                                                                                                                                                                                         |                                 |  |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>COMPLEX DIFFERENTIATION</b>  |  |
| Limit, Continuity and Differentiation of Complex functions, Analyticity, Cauchy-Riemann equations(without proof), finding harmonic conjugate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                 |  |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>COMPLEX INTEGRATION</b>      |  |
| Line integral, Cauchy's theorem, Cauchy's Integral formula, Zeros of analytic functions, Singularities, Taylor's series, Laurent's series, Residues, Cauchy Residue theorem (All theorems without Proof).                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                 |  |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |  |
| 1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.<br>2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall.<br>3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning                                                                                                                                                                                                                                                                                                                                                      |                                 |  |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                 |  |
| 1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.<br>2. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand<br>3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press                                                                                                                                                                                                                                                                                                                                                                                |                                 |  |
| <b>Course Outcomes</b><br><b>At the end of the course, student will be able to:</b> <ol style="list-style-type: none"> <li>1. Describe the Discontinuous random variables and Find moments of Discrete and continuous random variables.</li> <li>2. Evaluate the given data for appropriate test of hypothesis for Large samples.</li> <li>3. Evaluate the given data for appropriate test of hypothesis for small samples.</li> <li>4. Evaluate Limit, continuity, differentiability, analyticity and harmonic function .</li> <li>5. Express the given complex function as a power series using Taylor's series and Maclaurin's series and Evaluation of Residues</li> </ol> |                                 |  |

## STRENGTH OF MATERIALS

### II Year I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME09      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

Strength of Materials is the branch of continuum mechanics which deals with behaviour of solids under different loading conditions. This course will help students to acquire knowledge on the concepts needed for the design of various machine elements.

#### COURSE OUTCOMES:

By undergoing this course, student can be able to:

1. Analyse various stresses, strains and deformations induced in the body due to application of various loads.
2. Evaluate and sketch the variation of shear force and bending moment along the length of different types beams with various support conditions.
3. Evaluate the resistance of various beam sections under bending and shear loads.
4. Analyse the resultant effect of principal stresses induced in the body and can determine slope and deflection for various beams under different loading and boundary conditions.
5. Analyse the stresses developed and deformation of thin and thick cylinder due to internal Pressure.

#### UNIT-I      SIMPLE STRESSES AND STRAINS

Elasticity and Plasticity, Types of stresses and strains, Hooke's law, stress, strain diagram for ductile and brittle materials, Working stress, Factor of safety, Bars of varying section, composite bars, Temperature stresses, Lateral strain, Poisson's ratio and volumetric strain, Elastic module and the relationship between them, Strain energy, Resilience - Gradual, Sudden, and Impact loadings.

#### UNIT-II      SHEAR FORCE AND BENDING MOMENT

Definition of beam, Types of beams, Support reactions, Concept of shear force and bending moment, Relation between Shear Force, Bending Moment and rate of loading at a section of a beam. Shear Force and Bending Moment diagrams for cantilever, simply supported and overhanging beams subjected to point loads, U.D.L., U.V.L and combination of these loads, Point of contra flexure.

#### UNIT-III      FLEXURAL STRESSES & SHEAR STRESSES

**FLEXURAL STRESSES:** Theory of simple bending, Assumptions, Derivation of bending equation:  $M/I = f/y = E/R$ , Neutral axis, Moment of Resistance, section modulus and determination bending stresses in rectangular and circular sections (Solid and Hollow), I, T, Angle and Channel sections.

**SHEAR STRESSES:** Derivation of formula, Shear stress distribution across various beams sections like rectangular, circular, I, T, angle sections.

#### UNIT-IV      PRINCIPAL STRESSES AND STRAINS & DEFLECTION OF BEAMS

**PRINCIPAL STRESSES AND STRAINS:** Introduction - Stresses on an inclined section of a bar under axial loading - compound stresses - Normal and tangential stresses on an inclined plane for biaxial stresses - Two perpendicular normal stresses accompanied by a state of simple shear - Mohr's circle of stresses - Principle stresses and strains - Analytical and graphical solutions.

**DEFLECTION OF BEAMS:** Bending into a circular arc slope, deflection and radius of curvature, Differential equation for the elastic line of a beam, Double integration and Macaulay's methods, Determination of slope and deflection for cantilever and simply supported beams subjected to point loads and UDL.

**UNIT-V****THIN CYLINDERS & THICK CYLINDERS**

**THIN CYLINDERS:** Thin seamless cylindrical shells, Derivation of formula for longitudinal and circumferential stresses hoop, longitudinal and volumetric strains, changes in dia, and volume of thin cylinders, Riveted boiler shells, Thin spherical shells.

**THICK CYLINDERS:** Introduction, Longitudinal Stress, Circumferential Stresses in Thick Cylinders, Lami's Equations, Variation of stresses across the thickness.

**Text Books:**

1. Dr.S.Sadhu Singh, Strength of materials, Khanna publishers.
2. Ramamrutham S, Strength of materials, Danpath rai publishing company.

**Reference Books:**

1. Strength of materials by R.K.Bansal, Laxmi Publications (P) Ltd.
2. Engineering mechanics of solids by EGOR P. Popov, Pearson publishers
3. Timoshenko and Gere, Mechanics of Materials, CBS Publishers, 2011
4. Strength of Materials by Andrew Pytel and Ferdinand L. Singer Longman

## MATERIAL SCIENCE AND METALLURGY

| II B.TECH I Semester : MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                         |            |   |   |         |               |     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Category                | Hours/Week |   |   | Credits | Maximum Marks |     |       |
| A6ME10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PCC                     | L          | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         | 3          | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br><br>This course is intended to provide basic knowledge on the metallurgical aspects of metals that are being used in our homes to various engineering applications. It covers the fundamental aspects of materials, crystal structures and their representation, and various defects that are present in the materials. Then, the necessity of alloying and the corresponding changes occurring in their phase diagrams are discussed. A special focus is given on the important ferrous and non-ferrous alloys which are extensively used in the industries. Tailoring of material properties through different heat treatment processes with their microstructural changes are included. And, finally the course concludes with discussion on materials other than metals that includes polymers, ceramics and advanced materials such as composites. |                         |            |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br>At the end of course, students will be able to<br>1. Explain basic concepts of crystal structures and their imperfections and also draw crystal points, directions and planes in cubic unit cells.<br>2. Interpret various phases present in the binary phase diagrams of alloys and also calculate mass fraction of phases.<br>3. Recommend heat treatment processes for the desired changes in properties of steels.<br>4. Differentiate ferrous and non-ferrous alloys in terms of their properties and applications.<br>Classify and explain polymers, ceramics and composites in terms of their properties and applications.                                                                                                                                                                                                                       |                         |            |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CRYSTAL STRUCTURE       |            |   |   |         |               |     |       |
| Metallic crystal structures: Unit cells, crystal systems, directions and planes, slip systems.<br>Imperfections in solids: point defects, dislocations, surface and volume defects, determination of grain size, strengthening mechanisms for single phase metals: grain size reduction, Hall-Petch equation, cold working.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |            |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ALLOYS & PHASE DIAGRAMS |            |   |   |         |               |     |       |
| Alloys: Definition, necessity of alloying; Solid solutions: substitutional and interstitial, Hume-Rothery rules. Introduction Phase diagrams: Gibbs phase rule, unary phase diagram, cooling curves of pure metals, construction of phase diagram,<br>Interpretation of binary phase diagrams: Tie line, Lever rule, various invariant reactions, Microstructure development upon cooling, Iron Carbon phase diagram: Phases, Invariant reactions involved.                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |            |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | HEAT TREATMENT OF STEEL |            |   |   |         |               |     |       |
| Transformation Diagrams: Time-Temperature-Transformation (TTT) diagrams Fe-C alloys, bainite and martensite, continuous cooling curves and interpretation of final microstructures and properties.<br>Heat treatment processes: Annealing & their types, normalizing, spheroidizing, hardening and tempering, Surface hardening: flame and induction hardening, Case hardening: carburizing, nitriding and cyaniding.<br>Special heat treatment processes: austempering & its limitations, martempering.                                                                                                                                                                                                                                                                                                                                                                           |                         |            |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>FERROUS AND NON - FERROUS ALLOYS</b>  |
| <p><b>Ferrous alloys:</b> Introduction, Classification, Plain carbon steels: high, medium and low carbon steels, alloying of steels: stainless steels, tool steels; Cast irons: grey, white, malleable and spheroidal cast irons</p> <p><b>Non - ferrous alloys:</b> Copper alloys - brass and bronze; Aluminium and Aluminium alloys.</p>                                                                                                                                                         |                                          |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>POLYMERS, CERAMICS AND COMPOSITES</b> |
| <p><b>Polymers and Ceramics:</b> Introduction, classification, general characteristics, and potential applications, Advanced Ceramics: MEMS, Optical Fibers, Ceramic Ball bearings.</p> <p><b>Composites:</b> Introduction, classification, general characteristics, and potential applications. Advanced Composites: Hybrid Composites, Structural Composites, Laminar and Sandwich panels, Nano - composites.</p>                                                                                |                                          |
| <b>TextBooks:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                          |
| <ol style="list-style-type: none"> <li>1. W.D.Callister, Materials Science and Engineering-An Introduction, Wiley India, 6 Edition.</li> <li>2. V.Raghavan, Material Science and Engineering, Prentice Hall of India Private Limited.</li> <li>3. Sidney H.Averner, Introduction to Physical Metallurgy, Tata McGrawhill education(P)Ltd, 2<sup>nd</sup> Edition</li> </ol>                                                                                                                        |                                          |
| <b>ReferenceBooks:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                          |
| <ol style="list-style-type: none"> <li>1. U. C. Jindal, Engineering Materials and Metallurgy, Pearson, 2011.</li> <li>2. Kenneth G. Budinski and Michael K. Budinski, Engineering Materials, Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.</li> <li>3. V. D. Kodgire, Material Science and Metallurgy for Engineers, Everest, 1st Edition, 2006.</li> <li>4. Ashby, "Engineering Materials Vol I &amp; II" fifth edition authored by M F Ashby and David R H Jones.</li> </ol> |                                          |

## MANUFACTURING PROCESSES

| II B.TECH I Semester : MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                 |              |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Category                        | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PCC                             | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                 | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course is intended to introduce various manufacturing processes such as casting, joining, forming and their relevance in current manufacturing industry. Also focus to explore processing methods of plastics.                                                                                                                                                                                                                                                      |                                 |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br>The course should enable the students to <ol style="list-style-type: none"><li>1. Explain casting, casting methods, preparation of patterns and castings.</li><li>2. Enumerate various welding &amp; cutting techniques.</li><li>3. Explain hot working, cold working, extrusion, rolling and forging processes.</li><li>4. Select equipment for various deformation processes.</li><li>5. Demonstrate various methods used for producing plastic shapes.</li></ol>      |                                 |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CASTINGS & METHODS OF MELTING   |              |   |   |         |               |     |       |
| CASTING: Steps involved in making a casting, Advantage of casting and its applications. Patterns and Pattern making, Types of patterns, Materials used for patterns, pattern allowances and their construction, Principles of Gating, Gating ratio and design of Gating systems Solidification of casting, Concept, Solidification of pure metal and alloys, short and long freezing range alloys. Risers Types function and design, special casting processes 1) Centrifugal 2) Die 3) Investment. |                                 |              |   |   |         |               |     |       |
| METHODS OF MELTING: Crucible melting and cupola operation, steel making processes                                                                                                                                                                                                                                                                                                                                                                                                                   |                                 |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | WELDING & CUTTING OF METALS     |              |   |   |         |               |     |       |
| WELDING: Classification of welding process types of welds and welded joints and their characteristics, design of welded joints, Gas welding, ARC welding, Forge welding, resistance welding, Thermit welding and Plasma (Air and water) welding. TIG and MIG, welding, Friction welding, Induction welding, Explosive welding, Laser welding, Soldering and Brazing. Heat affected zones in welding; welding defects, causes and remedies, destructive, non-destructive testing of welds.           |                                 |              |   |   |         |               |     |       |
| CUTTING OF METALS: Oxy Acetylene Gas cutting, water plasma, cutting of ferrous, non-ferrous metals.                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | HOT AND COLD WORKING, EXTRUSION |              |   |   |         |               |     |       |
| HOT WORKING AND COLD WORKING: Strain hardening, recovery, recrystallisation and grain growth, Comparison of properties of cold and hot worked parts.                                                                                                                                                                                                                                                                                                                                                |                                 |              |   |   |         |               |     |       |
| EXTRUSION OF METALS: Basic extrusion process and its characteristics. Hot extrusion and cold extrusion, Forward extrusion and backward extrusion, Impact extrusion, Hydrostatic extrusion.                                                                                                                                                                                                                                                                                                          |                                 |              |   |   |         |               |     |       |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ROLLING                         |              |   |   |         |               |     |       |



**ROLLING:** Fundamentals, theory of rolling, types of Rolling mills and Forces in rolling and power requirements. Stamping, forming and other cold working processes: Blanking and piercing, Bending and forming,

**DRAWING:** Drawing and its types, wire drawing and Tube drawing, coining, Hot and cold spinning Forces and power requirement in the above operations.

|               |                                             |
|---------------|---------------------------------------------|
| <b>UNIT-V</b> | <b>FORGING &amp; PROCESSING OF PLASTICS</b> |
|---------------|---------------------------------------------|

**FORGING:** Principles of forging, Tools and dies, Types Forging, Smith forging, Drop Forging, Roll forging, Rotary forging, forging defects.

**PROCESSING OF PLASTICS:** Types of Plastics, Properties, applications and their Processing methods and Equipment (blow and injection modelling).

**Text Books:**

1. P. N. Rao (2011), Manufacturing Technology, Vol -1, 3rd edition, Tata McGraw- Hill education (P) Ltd, New Delhi
2. S. Kalpakjain (2005), Manufacturing Engineering and Technology, 4th edition, Pearson Education, New Jersey.

**Reference Books:**

1. R. K. Jain (2010), Production Technology, 16th edition, Khanna publishers, New Delhi, India.
2. B.S. Raghuwanshi (2011), A course in workshop Technology, Vol - II, 3rd Edition, Dhanpat Rai & Co, New Delhi, India
3. Gosh (2004), Manufacturing science, Affiliated East-west press (p) Ltd, New Delhi, India

## THERMODYNAMICS

### II B.TECH I Semester : MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME14      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 1 | 0 | 4       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course intended to introduce thermodynamic laws and their application to analyse boilers, heat pumps, heat engines etc., properties of mixture of gases and pure substances, Psychrometry.

#### COURSE OBJECTIVES:

The course should enable the students to:

- To understand the basic concepts of thermodynamics.
- To comprehend laws of thermodynamics and apply it to the related processes.
- Perform thermal analysis on behaviour and performance of systems.

#### COURSE OUTCOMES:

- CO1 - Apply the first law of thermodynamics for simple open and closed systems under steady and unsteady conditions.
- CO2 – Analyse the second law of thermodynamics to open and closed systems and calculate entropy.
- CO3 - Summarize the thermodynamic properties of gases and steam and apply it to related system analysis.
- CO4 - Calculate the properties of gas mixtures and apply it to related system analysis.
- CO5 - Evaluate the properties of moist air and its use in psychometric processes.

#### UNIT-I Basic Concepts and First Law

Basic concepts – concept of continuum, microscopic and macroscopic approach. Path and point functions. System and their types. Thermodynamic Equilibrium State, Quasi-static, reversible and irreversible processes. Heat and work transfer.

Zeroth law of thermodynamics –First law of thermodynamics –application to closed and open systems – steady and unsteady flow processes.

#### UNIT-II Second Law and Entropy

Second law of thermodynamics – Kelvin-Planck's and Clausius statements. Reversibility and irreversibility. Heat Engine, Refrigerator, Heat pump, Carnot theorem, efficiency, COP. Clausius inequality, concept of entropy, entropy of ideal gas, principle of increase of entropy.

#### UNIT-III Properties of Pure Substances

Properties of pure substances - solid, liquid and vapour phases, phase rule, P-V, P-T, T-V, T-S, H-S diagrams, PVT surfaces.

Thermodynamic properties of steam, Dryness fraction and its determination by separating and throttling calorimeter.

|                |                                                                       |
|----------------|-----------------------------------------------------------------------|
| <b>UNIT-IV</b> | <b>Ideal and Real Gases, Gas Mixtures and Thermodynamic Relations</b> |
|----------------|-----------------------------------------------------------------------|

Properties of ideal and real gases - Equations of state - compressibility factor - compressibility chart. Properties of gas mixture.

Exact differentials – Difference and ratio of heat capacities - Clausius Clapeyron equations - Joule Thomson effect.

|               |                     |
|---------------|---------------------|
| <b>UNIT-V</b> | <b>Psychrometry</b> |
|---------------|---------------------|

Psychrometric properties, Psychrometric charts. Property calculations of air vapour mixtures by using chart and expressions.

Psychrometric process – adiabatic saturation, sensible heating and cooling, humidification, dehumidification, evaporative cooling and adiabatic mixing. Simple Applications.

**Text Books:**

1. Nag.P.K, Engineering Thermodynamics, 6th Edition, Tata McGraw-Hill, 2017.
2. R.K.Rajput, A Textbook of Engineering Thermodynamics, 5 th Edition, Laxmi Publications, 2016.
3. Onkar Singh, Applied Thermodynamics, 4 th Edition, New Age International Private Limited, 2018.
4. Chattopadhyay, P, "Engineering Thermodynamics", Oxford University Press, 2016.

**Reference Books:**

1. Yunus A Cengel and Michael A Boles, Thermodynamics: An Engineering Approach, 8th Edition, McGraw Hill, 2017.
2. Claus Borgnakke, Richard E. Sonntag, Fundamentals of Thermodynamics, 8th Edition, Wiley, 2016.
3. Lynn D. Russell, George A. Adebiyi, Engineering Thermodynamics, 6th Edition, Oxford University Press, New Delhi, 2008.

**MANUFACTURING PROCESSES LAB****II B TECH I Semester : MECHANICAL ENGINEERING**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME13      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This course is designed to impart hands-on practical exposure on manufacturing processes and equipment.

**COURSE OUTCOMES:**

The student will be able to:

1. Design and manufacture different patterns
2. Prepare various castings.
3. Operate arc welding, gas welding and resistance welding equipment
4. Perform operations such as blanking, piercing, deep drawing, extrusion, bending and other operations.
5. Use injection moulding and blow moulding equipment to produce plastic products such as water bottle, water bottle caps etc.

**LIST OF EXPERIMENTS:****METAL CASTING**

1. Pattern Design and making - for one casting drawing.
2. Sand properties testing – 1 Exercise -for strength and permeability
3. Moulding Melting and Casting - 1 Exercise

**II.WELDING**

1. ARC Welding Lap & Butt Joint - 2 Exercises
2. Spot Welding - 1 Exercise
3. TIG Welding - 1 Exercise
4. Plasma welding and Brazing - 2 Exercises (Water Plasma Device)

**III. MECHANICAL PRESS WORKING**

1. Blanking & Piercing operation and study of simple, compound and progressive press tool.
2. Hydraulic Press: Deep drawing and extrusion operation.

**IV. PROCESSING OF PLASTICS**

1. Injection Moulding
2. Blow Moulding

## STRENGTH OF MATERIALS AND MATERIAL SCIENCE AND METALLURGY LAB

### II B TECH I Semester : MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME11      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

### COURSE OUTCOMES:

At the end of the course, students will be able to

1. Analyze mechanical behaviour of test sample under tensile, compressive, torsion and impact load conditions.
2. Find deflection of simply supported and cantilever beams with different cross sections/materials.
3. Calculate stiffness and modulus of rigidity of helical coil spring.
4. Identify the various phases depicted in the microstructure of pure metals, ferrous and non-ferrous alloys.
5. Analyze the significance of cooling rate in the formation of fine grains using Jominy end quench test.

### SYLLABUS:

#### LIST OF EXPERIMENTS: Strength of Materials Lab

1. Direct tension test.
2. Deflection test on simply supported beam and Cantilever beam.
3. Torsion test.
4. Hardness test
  - a. Brinell's hardness test
  - b. Rockwell hardness test
5. Test on springs.
6. Charpy and Izod Impact test.

#### LIST OF EXPERIMENTS: Metallurgy Lab

7. Study of microstructures of pure metals such as Iron, Cu, Al, etc.
8. Study of microstructures of plain carbon steels.
9. Study of microstructure of cast irons.
10. Study of microstructure of alloy steels.
11. Study of microstructures of non-ferrous alloys.
12. Determination of hardenability of steels by Jominy end quench test.

## PYTHON LAB FOR MECHANICAL APPLICATIONS

### II B TECH I Semester : MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME15      | ESC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

Python is a general-purpose, object-oriented programming language with high-level programming capabilities. It has apparent and easily understandable syntax, portability, and easy to learn.

#### COURSE OUTCOMES:

The course should enable the students to

1. Apply conditional statement, loops condition and functions in python program
2. Solve mathematical and mechanical problems using python program
3. Plot various type of chart using python program
4. Analyse the mechanical problem using python program
5. Illustrate programs on various python libraries such as numpy, pandas and matplotlib.

#### List of Python Programs

1. Write a program to find roots of quadratic equation.
2. Write a program to perform equations of uniform motion of kinematics:
  - i.  $v = u + at$
  - ii.  $s = ut + \frac{1}{2}(at^2)$
  - iii.  $v^2 = u^2 - 2as$
3. Write a menu driven program to perform following properties of thermodynamics as given below:
  - i. First Law of thermodynamics (  $U = Q - W$  ), where  $\Delta U$  is the change in the internal energy. Q is the heat added to the system, and W is the work done by the system.
  - ii. Efficiency of Heat Engine =  $(T_H - T_C) / T_H$  where  $T_H$  &  $T_C$  is the temperature of HOT and COLD Reservoirs.
4. Write the menu program to find the relationship between stress and strain curve as given below:
  - i.  $E = \text{Stress} / \text{Strain}$
  - ii.  $G = \text{Shear Stress} / \text{Shear strain}$
  - iii. Relation between E&C, E& K and E,C, and K
5. Write the program to determine the shear force and bending moment in beams.
  - i. Cantilever beam Carrying Point Load at Center
  - ii. Simply Supported beam Carrying UDL
6. Write a Program to find out unknown magnitude of  $T_B$  and  $T_D$  of unknown tensions can be obtained from two scalar equations of equilibrium i.e.  $\sum F_x = 0$  and  $\sum F_y = 0$ .
7. Write a program to perform interpolation of equally and unequally spaced data.
8. Write a program to calculate total pressure exerted in ideal fluid as equation is given below:  
 $P + \frac{1}{2}(\rho v^2) + \rho gh = \text{constant}$   
 Where P is Pressure, V is Velocity of fluid,  $\rho$  is density and h is the height of the container.
9. Write a program to find and delete repeating number in Given List.
10. Write a program to input and print the element sum of user defined matrix.
11. Write a program to input and multiply two different matrices.
12. Write a program to compute eigen value and vector of a given 3\*3 matrix using NumPy.
13. Write a program to draw line using equation  $y=mx+c$
14. Write the program to determine the intersection point of two lines.
15. Draw various types of charts using matplotlib.
16. Write a program for Brasenham's line drawing algorithm.
17. Write a program for geometric transformation of a given object.
18. Write program to perform the following operations on Strings.
  - i. Concatenation
  - ii. String Reverse

## CONSTITUTION OF INDIA

### II B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6HS06      | MC       | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 2            | 0 | 0 | 0       | 50            | -   | 50    |

#### COURSE OBJECTIVES:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

#### COURSE OUTCOMES:

**At the end of the course, the student will be able to:**

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels.
5. Explain the functions and responsibilities of election commission of India and union public service commission.

#### UNIT-I HISTORY OF MAKING OF THE INDIAN CONSTITUTION

Introduction to the constitution of India, the making of the constitution and salient features of the constitution.

#### UNIT-II PHILOSOPHY OF THE INDIAN CONSTITUTION

Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.

#### UNIT-III UNION GOVERNMENT

Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.

#### UNIT-IV STATE GOVERNMENT

State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)

#### UNIT-V ELECTION COMMISSION

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Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.

**Text Books:**

1. M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice – Hall of India Pvt. Ltd. New Delhi
2. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012
3. The constitution of India, P.M.Bakshi, Universal Law Publishing Co.,
4. The Constitution of India, 1950 (Bare Act), Government Publication.
5. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.

**Reference Books:**

1. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis, 2014.
2. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2015.
3. Indian constitution at work, NCERT
4. SubashKashyap, Indian Constitution, National Book Trust
5. J.A. Siwach, Dynamics of Indian Government & Politics
6. D.C. Gupta, Indian Government and Politics
7. H.M. Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal Law Publication)
8. J.C. Johari, Indian Government and Politics Hans
9. J. Raj Indian Government and Politics



# **B.TECH IV SEMESTER SYLLABUS**

## BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

(Common to CSE, IT, AIML, DS, CS and CSIT, Aero, Mech)

### II B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6EE60      | ESC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OBJECTIVES:

1. Develop fundamentals, including Ohm's law, Kirchhoff's laws and be able to solve for currents, voltages and power in electrical circuits.
2. Develop EMF equation and analyze the operation of DC Machines.
3. Analyze the working principle of Transformer.
4. Discuss the operation of AC Machines.
5. Analyze the operation of PN junction diode and rectifiers.
6. Discuss the operation and characteristics of Transistors.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Evaluate current and voltage values in resistive circuits with independent sources.
2. Explain the working of DC machines and solve the numerical problems.
3. Explain the working of AC electrical machines and solve the numerical problems.
4. Analyze the V-I characteristics of PN – junction diode and describe the operation of rectifiers.
5. Analyze the different configurations of Transistors and obtain its characteristics.

#### UNIT-I ELECTRICAL CIRCUITS

Basic definitions-Ohm's Law, types of elements, types of sources, Kirchhoff's Laws – simple problems., series & parallel resistive networks with DC excitation, star to delta and delta to star transformations.

#### UNIT-II DC MACHINES

Principle of Operation of DC Motor, types of DC motor, Torque equation & Losses and problems. DC Generator construction and working Principle, EMF Equation types of generators and problems.

#### UNIT-III AC MACHINES

Working principle and Construction of transformer, Emf Equation & problems. Principle operation of 3-phase induction motor, slip and torque Equation, Torque –slip characteristics & problems.

#### UNIT-IV DIODE AND ITS CHARACTERISTICS

PN JUNCTION DIODE: Operation of PN junction Diode: forward bias and reverse bias, Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers –capacitor filters, inductor filters

#### UNIT-V TRANSISTORS

Bipolar Junction Transistor - NPN & PNP Transistor, CB, CE, CC Configurations and Characteristics.

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**Text Books:**

1. Basic Electrical Engineering by M.S.Naidu and S.Kamakshaiah TMH
2. Electronic Devices and circuits by J.Millman, C.C.Halkias and Satyabrata Jit 2ed.

**Reference Books:**

1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, “Basic Electrical, Electronics and Computer Engineering”, Tata McGraw Hill, Second Edition, (2006).
2. Nagsarkar T K and Sukhija M S, “Basics of Electrical Engineering”, Oxford press (2005).
3. Mehta V K, “Principles of Electronics”, S.Chand & Company Ltd, (1994).
4. Mahmood Nahvi and Joseph A. Edminister, “Electric Circuits”, Schaum’ Outline Series, McGraw Hill, (2002).

## FLUID MECHANICS AND HYDRAULIC MACHINES

| II B.TECH II Semester: MECHANICALENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                       |              |   |   |         |               |     |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Category                              | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | PCC                                   | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                       | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course is introduced with the fundamentals of fluid mechanics such as fluid and fluid flow properties followed by the discussion on the behaviour of fluid at rest and in motion. Then, the basic principles of mass, momentum and energy are emphasized. The differential equations of continuity and momentum are developed and their applications are highlighted in flow measurements. At the end, the basics of turbo machinery are also covered to provide glimps on hydraulic machines which plays an important role in the conversion of hydraulic energy to mechanical energy and vice-versa. |                                       |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br>The course should enable the students to <ol style="list-style-type: none"><li>1. Describe the basic principles of fluid mechanics.</li><li>2. Identify various types of flows and formulate their governing equations.</li><li>3. Analyze the losses in pipe flows with the concepts of flow through pipes.</li><li>4. Evaluate hydrodynamic forces of jet striking different vanes from various angles.</li><li>5. Design the working proportions of hydraulic turbines and pumps.</li></ol>                                                                                                              |                                       |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | FLUID STATICS                         |              |   |   |         |               |     |       |
| <b>Fluid Statics:</b> Units and Dimensions, Properties of fluids: Mass Density, Specific weight, Specific Volume, Specific gravity, Viscosity, Surface tension and Capillarity, Vapour pressure and Cavitation.<br><b>Pressure, Buoyancy and Floatation:</b> Pressure and its Measurements (Manometers). Buoyancy Forces, Buoyancy forces on submerged bodies. Floatation and stability of floating bodies.                                                                                                                                                                                                                            |                                       |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | FLUID KINEMATICS                      |              |   |   |         |               |     |       |
| <b>Fluid Kinematics:</b> Methods to describe fluid motions, Types of fluid flow- steady and unsteady, uniform and non-uniform, laminar and turbulent, rotational and irrotational, One dimensional, Two dimensional and Three Dimensional.<br><b>Discharge and Continuity Equation:</b> Rate of Flow or Discharge. Stream line, path line, streak line and stream tube, basic principles of mass, momentum and energy, continuity equation for one, two and three dimensional flow, fluid acceleration.                                                                                                                                |                                       |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FLUID DYNAMICS AND FLOW THROUGH PIPES |              |   |   |         |               |     |       |
| <b>Fluid Dynamics:</b> Surface and body forces, equations of motion: Euler's and Bernoulli's equations, Dimensional analysis, Bernoulli's equation for incompressible fluids and its practical applications: Venturimeter, Orificemeter, Pitot tube, Momentum equation.<br><b>Flow through Pipes:</b> Elementary turbulent flow (Reynolds's experiment), flow through pipes: head loses in pipes (Major losses & Minor losses). Head loss due to friction: Darcy Weisbach equation, Total Energy Line (TEL), Hydraulic Gradient Line (HGL), Pipes in series and parallel; Basic concepts of boundary layer and its Separation.         |                                       |              |   |   |         |               |     |       |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | BASICS OF TURBO MACHINERY             |              |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Basics of Turbo machinery:</b> Introduction, classification, hydrodynamic force of jets on stationary and moving plates, flat, inclined and curved vanes.<br><b>Workdone and Efficiency:</b> Jet striking centrally and at tip, velocity diagrams, work done and efficiency, flow over radial vanes.                                                                                                                   |                                     |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                             | <b>HYDRAULIC TURBINES AND PUMPS</b> |
| <b>Hydraulic Turbines:</b> Classification of Turbines, Heads and efficiencies, Impulse and Reaction turbines, Pelton wheel, Francis turbine and Kaplan turbine: working proportions, work done, efficiencies, draft tube theory, cavitation, surge tank, water hammer.<br><b>Hydraulic Pumps:</b> Classification, centrifugal and reciprocating pumps: working, work done, heads and efficiencies.                        |                                     |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                        |                                     |
| <ol style="list-style-type: none"> <li>1. Modi and Seth, Hydraulics, Fluid Mechanics and Hydraulic Machinery, Rajsons Publications, 20<sup>th</sup> Edition, 2013.</li> <li>2. YunusCengel, John Cimbala, Fluid Mechanics, McGraw Hill Education.</li> <li>3. S K Som, Gautam Biswas and Suman Chakraborty, Introduction to Fluid Mechanics and Fluid Machines, 3<sup>rd</sup> Edition, McGraw Hill Education.</li> </ol> |                                     |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |
| <ol style="list-style-type: none"> <li>1. Rajput, Fluid Mechanics and Hydraulic Machines, S.Chand&amp; Co, 6<sup>th</sup> Edition, 1998.</li> <li>2. Banga, Sharma, Hydraulic Machines, Khanna Publishers, 6<sup>th</sup> Edition, 2001.</li> <li>3. Dr. R. K. Bansal, A Text Book of Fluid Mechanics and Hydraulic Machines, Laxmi Publications (P) Ltd, 10<sup>th</sup> Edition, 2015.</li> </ol>                       |                                     |

## THERMAL ENGINEERING - I

| II B.TECH II Semester: MECHANICALENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                         |              |   |   |         |               |     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Category                                | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PCC                                     | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                         | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course will make students to understand principles of Internal combustion engines, study testing and performance of compressors.                                                                                                                                                                                                                                                                                                                                                                                |                                         |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br>The course should enable the students to:<br><ol style="list-style-type: none"><li>1. Demonstrate the basic Cycles and operation of IC engines.</li><li>2. Analyse the combustion process and also how it does affect the performance of the IC engines.</li><li>3. Describe procedures of testing and analyses the emission of IC engines.</li><li>4. Apply the modern developments and alternate fuels on IC engines.</li><li>5. Formulate and perform the maintenance and operation of Air compressors.</li></ol> |                                         |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AIR STANDARD CYCLES AND IC ENGINES      |              |   |   |         |               |     |       |
| Air standard Cycles - Otto, Diesel and Dual cycle.<br><br>IC Engine- Classification, Working principle, Valve and Port Timing Diagrams. Comparison of Air Standard and Actual Cycles, Time Loss Factor, Heat Loss Factor, Loss due to Rubbing Friction.                                                                                                                                                                                                                                                                                         |                                         |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | COMBUSTION IN IC ENGINES                |              |   |   |         |               |     |       |
| Combustion in S.I. Engines: Normal Combustion and abnormal combustion, Importance of flame speed and effect of engine variables, Type of Abnormal combustion, pre-ignition and knocking.<br><br>Combustion in C.I. Engines: Four stages of combustion, Delay period and its importance, Effect of engine variables, Diesel Knock, Need for air movement.                                                                                                                                                                                        |                                         |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | IC ENGINE TESTING AND EMISSION          |              |   |   |         |               |     |       |
| Performance of IC Engines: Measurement and analysis of engine performance. Factors effecting efficiency. Variable speed, Air/Fuel ratio. Heat balance test.<br><br>Engine Emissions: SI and CI engine emissions, Harmful effects, Emissions measurement methods, methods for controlling engine emissions.                                                                                                                                                                                                                                      |                                         |              |   |   |         |               |     |       |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MODERN DEVELOPMENTS AND ALTERNATE FUELS |              |   |   |         |               |     |       |
| Super Charging, Stratified charge engine, Dual-fuel engines, CRDI and HCCI concepts<br><br>Alternate Fuels for IC Engines: Need for use of alternate fuels, Use of alcohol fuels, Biodiesel                                                                                                                                                                                                                                                                                                                                                     |                                         |              |   |   |         |               |     |       |
| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AIR COMPRESSORS                         |              |   |   |         |               |     |       |
| Compressors: Classification, positive displacement and roto dynamic machinery, Rotary blower, vane sealed compressor, principle of working, efficiency considerations.                                                                                                                                                                                                                                                                                                                                                                          |                                         |              |   |   |         |               |     |       |

**Reciprocating Air Compressors:** Principle of operation, work required, Isothermal efficiency, volumetric efficiency, clearance, multi stage compression, under cooling, saving of work, minimum work condition for stage compression.

**Text Books:**

1. V.Ganesan (2011), I.C. Engines, 3rd edition, Tata Mc Graw-Hill, New Delhi, India.
2. R.K.Rajput (2011), Thermal Engineering, 18th edition, Lakshmi Publications, New Delhi, India.

**Reference Books:**

1. Mathur, Sharma (2008), IC Engines, 3rd edition, Dhanpat Rai & Sons, New Delhi, India.
2. Pulkrabek (2008), Engineering fundamentals of IC Engines, 2<sup>nd</sup> edition, Pearson Education, New Jersey.
3. Rudramoorthy (2003), Thermal Engineering, 5th edition, Tata McGraw-Hill, New Delhi, India.

## THEORY OF MACHINES - I

| II B.TECH II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                 |              |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Category                                        | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | PCC                                             | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                 | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVER VIEW:</b><br>This course will deal with kinematic analysis of mechanisms and machines. It covers basic concepts and principles of various mechanisms, analysis of velocity and acceleration of different mechanisms, study of steering gear mechanisms, gears, gear trains, cams and followers.                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                 |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br>The course should enable the students to:<br>1. Outline basic concepts and principles of mechanisms for doing useful work.<br>2. Solve velocity and acceleration in different mechanisms.<br>3. Analyze straight line and steering gear mechanisms.<br>4. Select the gears and gear trains for transmission of motion.<br>5. Design cam profile for different follower motions.                                                                                                                                                                                                                                                                                                                                                |                                                 |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | INTRODUCTION                                    |              |   |   |         |               |     |       |
| Introduction: Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria, Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains. Spatial mechanisms, introduction to the kinematics of spatial mechanisms. Limit positions, Mechanical Advantage, Transmission Angle.                                                                                                                                                                                                                                            |                                                 |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | VELOCITY & ACCELERATION ANALYSIS                |              |   |   |         |               |     |       |
| <b>Velocity in Mechanisms:</b> Relative Velocity Method- Relative velocities of two bodies moving in straight line, velocity of a point on a link by relative velocity method, Velocities in four bar and single slider crank mechanisms – Problems.<br>Instantaneous Centre Method: Introduction, Velocity of a point on a link, Types of instantaneous centres, Kennedy's theorem, Method of locating instantaneous centres, determination of angular velocities of points and links – Problems.<br><b>Acceleration in Mechanisms:</b> Acceleration of a point on a link Acceleration in four bar and single slider crank mechanisms– Problems. Introduction to synthesis of linkages (Dimensional synthesis for motion, function and path generation). |                                                 |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | STRAIGHT LINE MOTION & STEERING GEAR MECHANISMS |              |   |   |         |               |     |       |
| <b>Straight line motion mechanisms:</b> Types, Peaucellier, Hart's, Scott Russel, Tchebichef's, Watt's, Grasshopper and Robert Mechanisms - Pantograph.<br><b>Steering gear mechanisms:</b> Introduction, Condition for correct steering, Davis Steering gear mechanism, Ackerman's steering gear mechanism. Hooke's joint: Single and double Hooke's joint – velocity ratio – application – Problems.                                                                                                                                                                                                                                                                                                                                                    |                                                 |              |   |   |         |               |     |       |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>GEARS &amp; GEAR TRAINS</b> |
| Friction wheels and toothed gears – types – spur gear terminology, law of gearing, condition for constant velocity ratio for transmission of motion, phenomena of interference, condition for minimum number of teeth to avoid interference, expressions for arc of contact, path of contact and contact ratio of pinion and gear.<br><b>Gear trains:</b> Introduction – Types – Simple, compound and reverted gear trains, Epicyclic gear train – Methods of finding train value or velocity ratio – Problems. Introduction to differential gear for an automobile. |                                |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CAMS</b>                    |
| Cams: Introduction, classification of followers and cams; Displacement, velocity and acceleration analysis. Construction of motion of the follower for uniform velocity, simple harmonic motion, uniform acceleration and retardation, cycloidal motion. Introduction to pressure angle and under cutting, sizing of cams.                                                                                                                                                                                                                                           |                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| <ol style="list-style-type: none"> <li>1. Theory of Machines / S.S. Rattan /Tata McGraw-Hill Education.</li> <li>2. Theory of Machines/Thomas Bevan/Pearson/ 3rd Edition</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                  |                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                |
| <ol style="list-style-type: none"> <li>1 Kinematics and Dynamics of machinery/Robert L. Norton/Tata Mc Graw Hill</li> <li>2 Mechanisms of machines / Cleghorn W.L. / Oxford University Press</li> <li>3 Theory of Mechanism and Machines/Ghosh and Mallik/Affiliated East-West Pvt.Ltd.</li> <li>4 Theory of Machines by R.S Khurmi &amp; J.K Gupta, S.Chand Publications.</li> </ol>                                                                                                                                                                                |                                |

## DESIGN OF MACHINE ELEMENTS

### II B.TECH II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME20      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW :

This course deals the systematic approach to design, standardization, design and manufacturing considerations, engineering materials properties and their selection, simple and compound stresses in machine elements, theories of failure under static and dynamic loading situations, Design of riveted joints, welded joints, bolted joints, keys and cotter joints, shafts, shaft couplings and springs.

#### COURSE OUTCOMES:

The course should enable the students to

1. Explain various types of design, various considerations for design and theories of failure.
2. Design riveted, welded and bolted joints.
3. Design keys, cotter and knuckle joints.
4. Design shafts and shaft couplings.
5. Design Mechanical Springs.

#### UNIT-I INTRODUCTION & FATIGUE

**Introduction:** General considerations & Manufacturing considerations for design of machine elements, Mechanical properties of materials, and preferred numbers.  
Simple Stresses, Simple bending, Pure Torsion, Combined bending and torsion, Static theories of failure.  
**Fatigue:** Types of Fatigue loading, S-N Curve – Endurance Limit, Factors effecting Endurance Limit, Stress Concentration - Theoretical stress concentration factor - Fatigue stress concentration factor - Notch sensitivity- Design for fluctuating stresses-endurance limit-Goodman line, Soderberg line and Gerber's Parabola.

#### UNIT-II DESIGN OF RIVTED, WELDED & BOLTED JOINTS

**Riveted Joints:** Cold riveting and hot riveting, types of riveted joints, terminology, methods of failure of riveted joints-strength equations-efficiency of riveted joints-eccentrically loaded riveted joints  
**Welded Joints:** Types, Advantages, Dis-advantages, Design of fillet welds-axial loads-circular fillet welds subjected to torsion and bending.  
**Design of bolts:** Design of bolts with pre-stresses, bolts of uniform strength, bolted joints under eccentric loading.

#### UNIT-III DESIGN OF KEYS, COTTER & KNUCKLE JOINTS

**Keys, Cotter & Knuckle Joints:** Design of key-stresses in keys-cotter joints-spigot & socket, sleeve & cotter, Gib & cotter joints & Knuckle Joints.

#### UNIT-IV DESIGN OF SHAFTS & SHAFT COUPLINGS

**Design of shafts:** Design of solid & hollow shafts for strength & rigidity- Design of shafts for complex loads-shaft sizes-Design of shaft for Belt drives.  
**Shaft couplings:** Rigid couplings-Muff, split muff and flange couplings-Flexible couplings-PIN Bush couplings

#### UNIT-V DESIGN OF MECHANICAL SPRINGS

**Mechanical Springs:** Stresses & Deflection of helical springs-extension of compression springs- springs for static & fatigue loading of helical springs- energy storage capacity-helical torsion springs, Laminated Springs

**Text Books:**

1. V. Bandari (2011), *A Text Book of Design of Machine Elements*, 3<sup>rd</sup> edition, Tata McGraw hill education (P) ltd, New Delhi, India.
2. R. L. Norton (2006), *Machine Design (An Integrated approach)*, 2<sup>nd</sup> edition, Pearson Publishers, Chennai, India.

**Reference Books:**

1. Shigley, J.E, (2011), *Mechanical Engineering Design*, 9<sup>th</sup> Edition, Tata McGraw-Hill, New Delhi, India.
2. Dr.P.C Sharma, D.K Agarwal, Machine Design, S.K Kataria& Sons

## BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB

### (Common to II year – AERO, MECH)

#### II B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6EE61      | ESC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

#### COURSE OBJECTIVES:

1. To introduce the concepts of electrical circuits and its components
2. To study and understand the different types of DC/AC machines and Transformers.
3. To introduce the concepts of diodes & transistors, and
4. To impart the knowledge of various transistors configurations, characteristics and applications.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. To analyze and solve electrical circuits using network laws and theorems.
2. To understand and analyze basic Electric and Magnetic circuits
3. To study the working principles of Electrical Machines
4. To identify and characterize diodes and various types of transistors.

#### LIST OF EXPERIMENTS

##### PART A : ELECTRICAL

1. a. Identification, Specifications, Testing of R, L, C Components (Color Codes), Bread Boards, Diodes, BJTs.  
b. Verification of Ohms law
2. Verification of KCL and KVL.
3. Magnetization characteristics of DC shunt generator (Determination of critical field resistance and critical speed)
4. Swinburne's test on dc shunt motor (Predetermination of efficiencies)
5. OC and SC tests on Single-phase transformer.
6. Brake test on 3-phase Induction motor.

##### PART B : ELECTRONICS

7. Study and operation of Function Generator Regulated Power Supply (RPS) CRO.
8. PN Junction Diode Characteristics (Forward bias, Reverse bias)
9. Zener Diode Characteristics
10. Rectifier without Filters (Full wave & half wave).
11. Rectifier with Filters (Full wave & half wave).
12. Transistor CE Characteristics (Input and Output)

#### Text Books:

1. Basic Electrical Engineering by M.S.Naidu and S.Kamakshaiah TMH
2. Electronic Devices and circuits by J.Millman, C.C.Halkias and Satyabrata Jit 2ed.

**Reference Books:**

1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, “Basic Electrical, Electronics and Computer Engineering”, Tata McGraw Hill, Second Edition, (2006).
2. Nagsarkar T K and Sukhija M S, “Basics of Electrical Engineering”, Oxford press (2005).
3. Mehta V K, “Principles of Electronics”, S.Chand & Company Ltd, (1994).
4. Mahmood Nahvi and Joseph A. Edminister, “Electric Circuits”, Schaum’ Outline Series, McGraw Hill, (2002).

## FLUID MECHANICS AND HYDRAULIC MACHINES LAB

### II B.TECH II Semester: MECHANICALENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME17      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

#### COURSE OVERVIEW :

This course will provide a basic understanding of calibration of flow measuring devices such as venturi meter, orifice meter, notches, water meter, rotameter etc used in channels and pipes and Losses in pipes by major and minor loss apparatus. Measurements of critical Performance parameters like efficiency, flow rate etc of various hydraulic machines such as turbines, pumps are also measured in this lab. Students can perform tests and find the flow rate of equipment like venturimeter, orifice meter, and notches and can calibrate them. Darcy's as well as Chezy's coefficient of friction for different pipes can be found out in pipe friction apparatus. Also students can find out the performance of centrifugal and reciprocating pumps.

#### COURSE OUTCOMES:

At the end of course students are able to

1. Develop procedure for standardization of experiments.
2. Calibrate flow discharge measuring devices used in pipes.
3. Determine the major and minor losses in a given pipe.
4. Prove that the total head at any point along the fluid flow is same.
5. Test the performance of pumps and turbines.

#### LIST OF EXPERIMENTS:

1. Impact of jets on Vanes.
2. Performance Test on Pelton Wheel.
3. Performance Test on Francis Turbine.
4. Performance Test on Kaplan Turbine.
5. Performance Test on Single Stage Centrifugal Pump.
6. Performance Test on Multi Stage Centrifugal Pump.
7. Performance Test on Reciprocating Pump.
8. Calibration of Venturimeter.
9. Calibration of Orifice meter.
10. Determination of friction factor for a given pipe line.
11. Determination of loss of head due to sudden contraction in a pipeline.

## COMPUTER AIDED MACHINE DRAWING LAB

| II B.TECH II Semester: MECHANICALENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                     |              |   |   |         |               |     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Category                                            | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PCC                                                 | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                     | 0            | 0 | 2 | 1       | 40            | 60  | 100   |
| <p><b>COURSE OVERVIEW:</b></p> <p>The purpose of this course is to introduce conventional representation of materials and machine components, drawing of various machine elements, joints as well as assembly drawing of engine parts and machine parts through Computer Aided Drafting. Machine drawing can be used as a communicating medium in industries to furnish all details needed for the manufacturing and assembling of a machine.</p> <p><b>COURSE OUTCOMES:</b></p> <p>The course should enable the students to:</p> <ol style="list-style-type: none"><li>1. Understand and apply Indian and international standards while drawing machine components.</li><li>2. Illustrate various machine elements and simple parts in drawings.</li><li>3. Construct the drawing of riveted joints, couplings and bearings.</li><li>4. Discover the engine part drawings for apt assembled views.</li><li>5. Select the elements of machine parts for appropriate assembled views.</li></ol> |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>MACHINE DRAWING CONVENTIONS</b>                  |              |   |   |         |               |     |       |
| <p>Need for drawing conventions – Introduction to standard conventions:</p> <ol style="list-style-type: none"><li>1. Types of Drawings – Introduction, Drawing area, Title block size, location and details.</li><li>2. Conventional representation of materials and machine elements.</li><li>3. Methods of dimensioning, representation of geometric dimensioning &amp; tolerance on drawings.</li><li>4. Types of sections – drawing of full and half section.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>DRAWING OF MACHINE ELEMENTS AND SIMPLE PARTS</b> |              |   |   |         |               |     |       |
| <p>Selection of Views, additional views for the following machine elements and parts with every drawing proportions. Popular forms of screw threads, bolts, nuts, set screws, keys, cotter joint and knuckle joint.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>RIVTED JOINTS, COUPLINGS AND BEARINGS</b>        |              |   |   |         |               |     |       |
| <p>Riveted joints for plates, flanged coupling, universal coupling.</p> <p>Journal, pivot, collar and foot step bearings.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ASSEMBLED VIEWS OF ENGINE PARTS</b>              |              |   |   |         |               |     |       |
| <p>Drawing of assembled views for the part drawings of the following using conventions.</p> <p>Engine parts – Stuffing box, eccentric, connecting rod, piston.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ASSEMBLED VIEWS OF MACHINE PARTS</b>             |              |   |   |         |               |     |       |
| <p>Drawing of assembled views for the part drawings of the following using conventions.</p> <p>Machine parts - Screws jack, drill jig, Plummer block, clapper block.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                     |              |   |   |         |               |     |       |

**Text Books:**

1. Machine Drawing – K.L. Narayana, P. Kannaiah & K. Venkata Reddy / New Age Publishers.
2. Machine Drawing with AutoCAD – Goutham Pohit, Goutham Ghosh / Pearson.
3. Machine Drawing – N. Sidheswar, P. Kanniah and V.V.S. Sastry / Tata McGraw Hill.

**Reference Books:**

1. Machine Drawing – N.D. Bhatt / Charotar Publishing House.
2. Machine Drawing – P.S. Gill / S.K. Kataria & Sons.
3. Machine Drawing – R.K. Dhawan / S.Chand Publications
4. IS code SP 46(2003): Engineering Drawing Practice for schools and colleges by Bureau of Indian Standards.
5. Bureau of Indian Standards



## REAL TIME PROJECT / FIELD BASED PROJECT

### II B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME22      | PWC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 4 | 2       | 50            | -   | 50    |

#### Evaluation Procedures

Real-Time (or) Field-based Research Project course: The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination.

The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e., 20 marks out of 50 marks from average of the two examinations.

There shall be NO external evaluation.

The student is deemed to have failed, if he

- (i) does not submit a report on the Project, or
- (ii) does not make a presentation of the same before the internal committee as per schedule, or
- (iii) secures less than 40% marks in this course.

#### COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Design engineering solutions to complex problems utilizing a systems approach
3. Conduct an experiments in the engineering project and analysis the data results
4. Communicate with engineers and the community at large in written an oral form.
5. Demonstrate the knowledge, skills and attitudes of a professional engineer

## GENDER SENSITIZATION

| II B. Tech. I Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                             |   |   |         |               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Category | Hours / Week                |   |   | Credits | Maximum Marks |     |       |
| A6HS05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MC       | L                           | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          | 0                           | 0 | 2 | 0       | 50            | -   | 50    |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Students will have developed a better understanding of important issues related to gender in contemporary India.<br>2. Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender. This will be achieved through discussion of materials derived from research, facts, everyday life, literature and film.<br>3. Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.<br>4. Students will acquire insight into the gendered division of labour and its relation to politics and economics.<br>5. Men and women students and professionals will be better equipped to work and live together as equals.<br>6. Students will develop a sense of appreciation of women in all walks of life. Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence. |          |                             |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          | <b>UNDERSTANDING GENDER</b> |   |   |         |               |     |       |
| Gender: Why Should We Study It? (Towards a World of Equals: Unit -1) Socialization: Making Women, Making Men (Towards a World of Equals: Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                             |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          | <b>GENDER AND BIOLOGY</b>   |   |   |         |               |     |       |
| Missing Women: Sex Selection and Its Consequences (Towards a World of Equals: Unit -4) Declining Sex Ratio. Demographic Consequences. Gender Spectrum: Beyond the Binary (Towards a World of Equals: Unit -10) Two or Many? Struggles with Discrimination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |                             |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          | <b>GENDER AND LABOUR</b>    |   |   |         |               |     |       |
| Housework: the Invisible Labour (Towards a World of Equals: Unit -3)<br>“My Mother doesn’t Work.” “Share the Load.”<br>Women’s Work: Its Politics and Economics (Towards a World of Equals: Unit -7)<br>Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                             |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          | <b>ISSUES OF VIOLENCE</b>   |   |   |         |               |     |       |
| Sexual Harassment: Say No! (Towards a World of Equals: Unit -6)<br>Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading: “Chupulu”.<br>Domestic Violence: Speaking Out (Towards a World of Equals: Unit -8)<br>Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                             |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                         |                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <p>New Forums for Justice.</p> <p>Thinking about Sexual Violence (Towards a World of Equals: Unit -11)</p> <p>Blaming the Victim - “I Fought for my Life....” - Additional Reading: The Caste Violence.</p> <p>Sustainable Industrialization and Urbanization – Sustainable urbanization, industrialization, material selection, pollution prevention, industrial ecology, industrial symbiosis, poverty reduction.</p> |                               |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                           | <b>GENDER: CO – EXISTENCE</b> |
| <p>Just Relationships: Being Together as Equals (Towards a World of Equals: Unit -12)</p> <p>Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers Additional Reading: Rosa Parks - The Brave Heart</p>                                                                                                                                                                                  |                               |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                      |                               |
| <p>1. All the five Units in the Textbook, “Towards a World of Equals: A Bilingual Textbook on Gender” written by A. Suneetha, Uma Bhargubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad, Telangana State in the year 2015.</p>                                                                         |                               |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                 |                               |
| <p>1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012</p> <p>2. Abdulali Sohaila. “I Fought For My Life...and Won.” Available online at:<br/> <a href="http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/">http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/</a></p>                                                  |                               |

## **III B.TECH. I SEMESTER SYLLABUS**

## BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6HS08      | HSMC     | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

  

**COURSE OVERVIEW:**  
 This Course enables the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

**COURSE OUTCOMES:**  
**At the end of the course, the student should be able to:**

1. Gain the conceptual knowledge on Business Economics and practical applications of Demand and Supply viz. Laws, Types, Elasticity, and Forecasting and Equilibrium.
2. Familiarize the concepts and applications related to Production and Cost of a firm.
3. Assess the features of different Market Structures and forms of Business Organizations existing in the modern business.
4. Analyze how Capital Budgeting decisions are carried out.
5. Elaborate the concepts and principles of Financial Accounting & interpret financial statements through ratio analysis.

|                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                    | <b>INTRODUCTION TO BUSINESS ECONOMICS &amp; DEMAND AND SUPPLY ANALYSIS</b> |
| <b>Business Economics:</b> Definition, Nature and Scope of Business Economics. <b>Demand Analysis:</b> Demand Determinants, Law of Demand and its exceptions, Elasticity of Demand, Types, Measurement and its Significance, Demand Forecasting and its Methods. <b>Supply Analysis:</b> Law of Supply, Supply Function, Equilibrium of Demand and Supply.                                       |                                                                            |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>PRODUCTION &amp; COST ANALYSIS</b>                                      |
| <b>Production Function</b> – Law of Variable Proportions and Laws of Returns to Scale, Isoquants and Isocosts, MRTS, Cobb-Douglas Production function, Economies of Scale. <b>Cost Analysis:</b> Cost concepts, Break-even Analysis (BEA).                                                                                                                                                       |                                                                            |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                  | <b>MARKET STRUCTURE, PRICING &amp; BUSINESS ORGANISATIONS</b>              |
| <b>Market Structure:</b> Classification of market structures, Features of Perfect competition, Monopoly, Oligopoly and Monopolistic Competition. Pricing: Objectives and Policies, and Methods. <b>Business Organisations:</b> Features and evaluation of different forms of Business Organisations- Sole Proprietorship, Partnership, Joint Stock Company, Government, Private and Cooperative. |                                                                            |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>CAPITAL BUDGETING</b>                                                   |
| Capital and its significance, Types of Capital, Methods and sources of raising capital - <b>Capital Budgeting:</b> features of capital budgeting proposals, Methods of Capital Budgeting: Payback Period- Accounting Rate of Return- Net Present Value- Internal Rate of Return and Profitability Index (simple problems).                                                                       |                                                                            |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                           | INTRODUCTION TO FINANCIAL ACCOUNTING & ANALYSIS |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Financial Accounting:</b> Accounting concepts and Conventions - Double - Entry Book Keeping, Journal, Ledger, and Trial Balance - Final Accounts with simple adjustments. <b>Financial Analysis:</b> Analysis and Interpretation of Financial Statements, Financial Analysis through Liquidity, Leverage, Solvency and Profitability ratios.                                                                                                  |                                                 |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                 |
| 1. D. D. Chaturvedi, S. L. Gupta, Business Economics - Theory and Applications, International Book House Pvt. Ltd. 2013.<br>2. Prasanna Chandra, Financial Management, 10 <sup>th</sup> Edition, Mc Graw Hill, 2019.<br>3. Dhanesh K Khatri, Financial Accounting, Tata Mc –Graw Hill, 2011.                                                                                                                                                     |                                                 |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                 |
| 1. Aryasri, Managerial Economics and Financial Analysis, TMH, 2012.<br>2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age International Publishers, Hyderabad, 2013.<br>3. S. N. Maheshwari, Sunil K Maheshwari, Sharad K Maheshwari, Financial Accounting, 5 <sup>th</sup> Edition, Vikas Publications, 2013.<br>4. I. M. Pandey, Financial Management, 11 <sup>th</sup> Edition, Vikas Publications, 2015. |                                                 |

## MACHINE DESIGN

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Category | Hours / Week               |   |   | Credits | Maximum Marks |     |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------|---|---|---------|---------------|-----|-------|
| A6ME23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PCC      | L                          | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          | 3                          | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>Machine Design focus mainly on design of power transmitting elements like gears, Connecting rod, crankpin, crankshafts, pistons, cylinders, bearings, belts, ropes, chain's, pulleys, Power Screws and nuts.                                                                                                                                                                                                                                                                                                                                                          |          |                            |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Identify design variables and performance factors in the study of journal bearings<br>2. Apply the basic principles for design of piston, connecting rod and crankshaft based on maximum bending and twisting moments<br>3. Design screws that are applied in hosting of heavy weights like screw jack etc.<br>4. Explain power transmission elements like belts, chains and ropes and able to design on stress criteria.<br>5. Apply the design concepts to evaluate the strength of the gear. |          |                            |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          | <b>BEARINGS</b>            |   |   |         |               |     |       |
| Types of Journal bearings – lubrication – Bearing modulus – Full and partial bearings, Partial bearings – Clearance ratio – Heat dissipation of bearings, bearing materials – journal bearing design – Ball and roller bearings – Static loading of ball and roller bearings, bearing life.                                                                                                                                                                                                                                                                                                      |          |                            |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          | <b>ENGINE PARTS</b>        |   |   |         |               |     |       |
| Connecting rod: Thrust in connecting rod – stress due to whipping action on connecting rod ends – Cranks and Crank shafts, strength and proportions of over hung and center cranks – Crank pins, Crank shafts. Pistons, Forces acting on piston – Construction Design and proportions of piston, Cylinder, Cylinder liners.                                                                                                                                                                                                                                                                      |          |                            |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          | <b>POWER SCREWS</b>        |   |   |         |               |     |       |
| Types of screw threads for power screws, Torque required to raise and lower the loads by square threaded screws, Efficiency of square threaded screws, Overhauling and self-locking, ACME threads, Stresses in power screws, Design of screw jack.                                                                                                                                                                                                                                                                                                                                               |          |                            |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          | <b>BELTS &amp; PULLEYS</b> |   |   |         |               |     |       |
| Transmission of power by Belt and Rope drives, Transmission efficiencies, Belts – Flat and V types – Ropes – Pulleys for belt and rope drives, Materials, Chain drives.                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                            |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          | <b>GEARS</b>               |   |   |         |               |     |       |
| Spur gears – Helical gears – Load concentration factor – Dynamic load factor, Surface compressive strength, Bending strength – Design analysis of spur gears – Estimation of centre distance, module and face width, check for plastic deformation, check for dynamic and wear considerations.                                                                                                                                                                                                                                                                                                   |          |                            |   |   |         |               |     |       |

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**Text Books:**

1. V. Bandari, A Text Book of Design of Machine Elements, 3<sup>rd</sup> Edition, Tata McGraw Hill Education (P) Ltd., New Delhi, India, 2011.
2. R. L. Norton, Machine Design (An Integrated approach), 2<sup>nd</sup> Edition, Pearson Publishers, Chennai, India, 2006.

**Reference Books:**

1. R. S. Khurmi, A Text book of Machine Design, 14<sup>th</sup> Edition, S. Chand & Company, New Delhi, India, 2018.
2. S. MD. Jalaludin,(2011), Machine Design, 3<sup>rd</sup> Edition, Anuradha Publishers, Chennai, India, 2011.



## THEORY OF MACHINES - II

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME24      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### **COURSE OVERVIEW:**

This course is intended to deal with the forces and their effects, while acting upon the machine parts in motion. The knowledge of this subject is very essential for an engineer in designing the various parts of a machine. Study of gyroscopes is very much helpful to learn the precession and stability of moving vehicle such as aircrafts, Naval ships and automobiles. This course helps to analyse and evaluate the force and torques that causes of motions in mechanisms. It provides knowledge on clutches, brakes and dynamometers, balancing of machines and application of vibrations.

#### **COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Analyse the effect of gyroscopic couples on the stability of aeroplane, naval ship and automobiles moving in a curved path and Static and dynamic analysis of 4-link mechanism.
2. Analyse the frictional forces and torques transmitted by various components such as clutches, brakes and dynamometers.
3. Demonstrate governing systems that control the speed of engines using the inertia phenomena and centrifugal forces and analyse different types of governors.
4. Find the position and magnitude of unbalanced masses for rotating and reciprocating parts of the engine
5. Determine frequency of various machine members subjected to free and forced vibrations.

#### **UNIT-I      GYROSCOPIC COUPLE, STATIC & DYNAMIC FORCE ANALYSIS**

Precessional angular motion: Gyroscopic couple – effect of precessional motion on the stability of moving vehicles such as motorcycle – motorcar – aeroplanes and ships.

Static and Dynamic Force Analysis: Static force analysis of planar mechanisms – Analytical Method – D'Alembert's principle, Dynamic Analysis of 4-link mechanism, Single Slider Crank Mechanism. Engine Force Analysis – Piston Effort, Crank Effort of reciprocating Engine.

#### **UNIT-II      FRICTION, BRAKES & DYNAMOMETERS**

Friction: Pivots and collars – uniform pressure, uniform wear – friction circle and friction axis: lubricated surfaces – boundary friction – film lubrication. Clutches – Types – Single plate, multi-plate, cone and centrifugal clutches.

Brakes and Dynamometers: Types of brakes: Simple block brake, band and block brake – internal expanding shoe brake – effect of braking of a vehicle. Dynamometers – absorption and transmission types.

#### **UNIT-III      TURNING MOMENT DIAGRAM AND FLYWHEELS, GOVERNORS**

Turning Moment Diagram and Flywheels: Turning moment diagram – fluctuation of energy – flywheels – energy stored and their design – flywheel in punching press.

Governors: Types of governors – Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. Sensitiveness, isochronisms and hunting – stability of governors.

|                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                | <b>BALANCING</b>  |
| Balancing of rotating masses – Analytical and graphical methods.<br>Balancing of reciprocating masses – Primary, Secondary, and higher balancing. Examination of multi cylinder-in-line engines for primary and secondary balancing- locomotive balancing – hammer blow – Swaying couple – variation of tractive effort.                                                                                                      |                   |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>VIBRATIONS</b> |
| Vibrations: Types of vibratory motion, types of free vibrations, natural frequency of free longitudinal and transverse vibrations – natural frequency of free transverse vibrations over a simply supported shaft and fixed shaft due to concentrated and distributed loads. Critical or whirling speed of a shaft – Natural frequency of torsional vibrations of single rotor, two rotor and three rotor systems.            |                   |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                            |                   |
| <ol style="list-style-type: none"> <li>1. S.S. Rattan, Theory of Machines, 4<sup>th</sup> Edition, Tata McGraw-Hill Education India Pvt. Ltd., 2017.</li> <li>2. Sadhu Singh, Theory of Machines: Kinematics and Dynamics, 3<sup>rd</sup> Edition, Pearson Education India, 2012.</li> </ol>                                                                                                                                  |                   |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                       |                   |
| <ol style="list-style-type: none"> <li>1. J.S. Rao and R.V. Duggipati, Mechanism and Machine Theory, New Age International (P) Ltd., 1992.</li> <li>2. John J. Uicker, Jr., Gordon R. Pennock, Joseph E. Shigley, Theory of Machines, 5<sup>th</sup> Edition, Oxford University Press, 2017.</li> <li>3. R.S. Khurmi, Theory of Machines, 14<sup>th</sup> Edition, S. Chand &amp; Company, New Delhi, India, 2018.</li> </ol> |                   |

## CAD/CAM

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME25      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 2            | 0 | 0 | 2       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This course in CAD/CAM provides a comprehensive exploration of the fundamentals of CAD, CAM, and Automation, covering topics such as geometric modelling, NC control, computer-aided process planning, computer-aided manufacturing, computer-aided quality control, and Computer Integrated Manufacturing (CIM), with a focus on the benefits of utilizing these technologies in the product design and development cycle.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Describe basic concepts of computer hard ware and software connected to CAD & CAM.
2. Explain wire frame modelling, surface modelling and solid modelling.
3. Describe about NC, CNC, DNC, adaptive control systems, Group Technology and can develop NC part programs.
4. Describe computer aided process planning and manufacturing resource planning.
5. Explain concepts of computer aided quality control and computer integrated manufacturing.

| UNIT-I                                                                                                                                                                                                                                                                 | INTRODUCTION |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Fundamentals of CAD, CAM, Automation, design process, Application of computers for design, Benefits of CAD, Computer peripherals for CAD, CAD software- definition of system software and application software, CAD database and structure. Product development cycle. |              |

| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | GEOMETRIC MODELING |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Introduction to Geometric Modelling, <b>Wire frame modelling:</b> Definition, advantages, disadvantages, wire frame entities. <b>Surface modelling:</b> Definition, advantages, disadvantages, surface entities. <b>Solid Modelling:</b> Definition, constructive solid geometry, advantages. <b>Curve representation:</b> Implicit and explicit forms of straight line, circle. <b>Parametric representation of synthetic curves:</b> Cubic Spline, Bezier Curve, B Spline Curve. |                    |

| UNIT-III                                                                                                                                                                                                                                                                                                                                                                         | NC CONTROL & GROUP TECHNOLOGY |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>NC Control:</b> Introduction, Elements of NC system, NC part programming: Methods of NC part programming, manual part programming, Computer assisted part programming, CNC, DNC, and Adaptive Control Systems.<br><b>Group Technology:</b> Part families, Parts classification & coding. Production flow analysis, Machine cell design using rank order clustering technique. |                               |

| UNIT-IV                                                                                                                                                                                                                                                                                                                                         | COMPUTER AIDED PROCESS PLANNING & COMPUTER AIDED MANUFACTURING |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Computer aided process planning:</b> Difficulties in traditional process planning, Computer aided process planning: Retrieval type and Generative type.<br><b>Computer aided manufacturing:</b> Material Resource Planning, inputs to MRP, MRP output records, Benefits of MRP, Enterprise Resource Planning, Capacity Requirements Planning |                                                                |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | COMPUTER AIDED QUALITY CONTROL AND CIM |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Computer aided quality control:</b> Automated inspection – Off-line, On-line, contact, Non-contact; Coordinate measuring machines, Machine vision, integration of CAQC with CAD/CAM.<br><b>Computer Integrated Manufacturing:</b> Types of systems in CIM, CIM advantages and disadvantages.                                                                                                                                                                                                                                                                                                                                                                                              |                                        |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |
| <ol style="list-style-type: none"><li>1. P.N. Rao, CAD/CAM Principles and Applications, 3<sup>rd</sup> Edition, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2017.</li><li>2. Mikell P. Groover, Emory W. Zimmers, Jr., CAD/CAM Computer-Aided Design and Manufacturing, 1<sup>st</sup> Edition, Pearson Education, 2003.</li></ol>                                                                                                                                                                                                                                                                                                                                                      |                                        |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |
| <ol style="list-style-type: none"><li>1. Ibrahim Zeid, R. Sivasubramanian, CAD/CAM Theory and Practice, 2<sup>nd</sup> Edition, McGraw Hill Education, 2009.</li><li>2. P. Radhakrishnan, S. Subramanian and V. Raju, CAD/CAM/CIM, 4<sup>th</sup> Edition, New Age International Publishers, 2018.</li><li>3. Chennakesava R. Alavala, CAD/CAM Concepts and Applications, Prentice Hall India Learning Pvt. Ltd., 2008.</li><li>4. Farid Amirouche, Principles of Computer Aided Design and Manufacturing, 2<sup>nd</sup> Edition, Pearson, 2004.</li><li>5. Warren S Seames, Computer Numerical Control Concepts and programming, 4<sup>th</sup> Edition, Thomson Learning, 2022.</li></ol> |                                        |

## THERMAL ENGINEERING - II

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Category | Hours / Week                                |   |   | Credits | Maximum Marks |     |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------|---|---|---------|---------------|-----|-------|
| A6ME27                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PCC      | L                                           | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          | 3                                           | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>The primary focus of this course is on application of laws of Thermodynamics to analyse steam and gas turbine cycles. It also emphasizes on analysis of the major components of steam and gas turbine plants and their applications.                                                                                                                                                                                                                                                                                                                                                |          |                                             |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate the Rankine cycle and apply thermodynamic concepts on boilers.<br>2. Figure out the types and applications of steam nozzles and condensers.<br>3. Classify the steam turbines and describe it's working.<br>4. Explain the working of gas turbines and analyse the performance.<br>5. Demonstrate the principles of operation of refrigeration and Air- conditioning systems.                                                                                                                     |          |                                             |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          | <b>BASIC CONCEPTS AND BOILERS</b>           |   |   |         |               |     |       |
| <b>Basic Concepts:</b> Rankine cycle - Schematic layout, Thermodynamic Analysis, Concept of Mean Temperature of Heat addition, Methods to improve cycle performance – Regeneration & reheating- Cycles.<br><b>Boilers:</b> Classification – Working principles with sketches including H.P. Boilers – Mountings and Accessories – Working principles – Boiler horse power, Equivalent Evaporation, Efficiency and Heat balance.                                                                                                                                                                                |          |                                             |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          | <b>STEAM NOZZLES &amp; STEAM CONDENSORS</b> |   |   |         |               |     |       |
| <b>Steam Nozzles:</b> Stagnation Properties – Function of nozzle – Applications and Types – Flow through nozzles, Thermodynamic analysis – Assumptions -Velocity of nozzle at exit – Ideal and actual expansion in nozzle – Velocity coefficient – Condition for maximum discharge – Critical pressure ratio – Criteria to decide nozzle shape.<br><b>Steam Condensers:</b> Requirements of steam condensing plant, Classification of condensers, working principle of different types, vacuum efficiency and condenser efficiency, air leakage, sources and its affects, air pump, cooling water requirement. |          |                                             |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          | <b>STEAM TURBINES</b>                       |   |   |         |               |     |       |
| <b>Impulse turbine:</b> Mechanical details, Velocity diagram, effect of friction, power developed, axial thrust, blade or diagram efficiency, condition for maximum efficiency. De-Laval Turbine – its features. Methods to reduce rotor speed – Velocity compounding and pressure compounding, Velocity and Pressure variation along the flow.<br><b>Reaction Turbine:</b> Mechanical details, principle of operation, thermodynamic analysis of a stage, degree of reaction, velocity diagram, Parson's reaction turbine, condition for maximum efficiency.                                                  |          |                                             |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          | <b>GAS TURBINES</b>                         |   |   |         |               |     |       |
| Simple gas turbine plant, Ideal cycle, essential components, parameters of performance, actual cycle, regeneration, inter cooling and reheating, Closed and Semi-closed cycles, merits and demerits, Combustion chambers and turbines of Gas Turbine Plant- Brief Concepts.                                                                                                                                                                                                                                                                                                                                    |          |                                             |   |   |         |               |     |       |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                | REFRIGERATION AND AIR CONDITONING |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Refrigeration:</b> Reversed Carnot cycle – Performance Evaluation – Vapour Compression Cycle, Vapour absorption cycle – mechanical details, working principles, use of P-H charts for calculations.<br><b>Air- Conditioning:</b> Introduction, classification and working principles of air-conditioning systems, requirements, schematic layout of a typical plant.                                                                               |                                   |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |
| <ol style="list-style-type: none"><li>1. R. K. Rajput, Thermal Engineering, 11<sup>th</sup> Edition, Lakshmi Publications, New Delhi, India, 2020.</li><li>2. V. Ganesan, Gas Turbines, 3<sup>rd</sup> Edition, Tata McGraw Hill Education (P) Ltd, New Delhi, India, 2017.</li><li>3. C.P. Arora, Refrigeration and Air Conditioning, 3<sup>rd</sup> Edition, McGraw Hill Education (P) Ltd, New Delhi, India, 2017.</li></ol>                       |                                   |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |
| <ol style="list-style-type: none"><li>1. P. L. Ballaney, Thermal Engineering, 25<sup>th</sup> Edition, Khanna Publishers, New Delhi, India, 2015.</li><li>2. H.I.H. Saravanamuttoo, C.F.C. Rogers, H. Cohen, P.V. Strznicky, A.C. Nix, Gas Turbine Theory, 7<sup>th</sup> Edition, Pearson Education, 2019.</li><li>3. R. Yadav, Thermodynamics and Heat Engines, 6<sup>th</sup> Edition, Central Publishing House, Allahabad, India, 2012.</li></ol> |                                   |

## CAD LAB

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code                                                                                                                                                                                                                                                                                                           | Category | Hours / Week                  |   |   | Credits | Maximum Marks |     |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|---|---|---------|---------------|-----|-------|
| A6ME26                                                                                                                                                                                                                                                                                                                | PCC      | L                             | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                       |          | 0                             | 0 | 2 | 1       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course is framed to have hands on experience on design software so that students are able to model various mechanical engineering components.                                                                                                                                         |          |                               |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Draw 2D drawings by sketcher module using design software.<br>2. Model 3D objects by Part Design module using design software.<br>3. Sketch, model and assemble various machine part drawings using design software. |          |                               |   |   |         |               |     |       |
| <b>Part A</b>                                                                                                                                                                                                                                                                                                         |          | <b>Sketcher Module</b>        |   |   |         |               |     |       |
| 1. Draw given 2D diagrams by using sketcher module. (Any 3 diagrams)                                                                                                                                                                                                                                                  |          |                               |   |   |         |               |     |       |
| <b>Part B</b>                                                                                                                                                                                                                                                                                                         |          | <b>Part Design Module</b>     |   |   |         |               |     |       |
| 1. Model given 3D objects by using part module. (Any 4 models)                                                                                                                                                                                                                                                        |          |                               |   |   |         |               |     |       |
| <b>Part C</b>                                                                                                                                                                                                                                                                                                         |          | <b>Assembly Design Module</b> |   |   |         |               |     |       |
| 1. Model and Assemble parts of knuckle joint.<br>2. Model and Assemble parts of stuffing box.<br>3. Model and Assemble parts of universal coupling.                                                                                                                                                                   |          |                               |   |   |         |               |     |       |

## THERMAL ENGINEERING LAB

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A6ME28      | PCC      | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

#### **COURSE OVERVIEW:**

The primary purpose of the ‘Thermal Engineering Lab’ is to show students the experimental methods on thermal energies on various engines and demonstrate their operational procedures. These values can further be used to determine other fuel properties. In order that students have good understanding of the theory underlying the experiments, the entire course is designed such that classroom lectures precede lab work. Students are advised to pay close attention in class so that they can perform well in the lab.

#### **COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Illustrate the Valve timing and port timing diagrams of CI & SI Engines.
2. Examine the performance parameters of internal combustion engines.
3. Examine the heat balance sheet in an internal combustion engine
4. Plot the performance characteristics of reciprocating air compressor.
5. Understand the working of steam boilers.

#### **LIST OF EXPERIMENTS:**

1. I.C. Engines Valve Timing Diagram
2. I.C. Engines Port Timing Diagram
3. I.C. Engines Performance Test 4-Stroke Diesel Engine
4. I.C. Engines Performance Test 4-Stroke Petrol Engine
5. I.C. Engines Heat Balance
6. I.C. Engines Air/Fuel Ratio and Volumetric Efficiency
7. Performance Test on Variable Compression Ratio Engine
8. Performance Test on Reciprocating Air-Compressor Unit
9. Dis-assembly/ Assembly of Engines
10. Study of Boilers



## INDEPENDENT STUDY / MOOCS

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME30      | PWC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 2 | 1       | -             | 100 | 100   |

#### Evaluation Procedures

There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization: Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be NO internal marks for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.

#### COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO

1. Demonstrate a sound technical knowledge of their selected certified course.
2. Discuss the various engineering solutions to solve the complex problems in selected certified course.
3. Illustrate the core engineering skill from reputed institutions
4. Develop skills in core engineering such critical thinking in the field of research.
5. Demonstrate the knowledge, skills and attitudes of a professional engineer.

**PROFESSIONAL ELECTIVE – I**

## PROFESSIONAL ELECTIVE - I

### ARTIFICIAL INTELLIGENCE IN MECHANICAL ENGINEERING

#### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME51      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course covers mechanical engineering AI and machine learning concepts like design optimisation, computer vision, robotics, and data analysis. Python programming, statistics, linear algebra, supervised and unsupervised learning, neural networks, deep learning, and reinforcement learning are covered in the course. Students learn TensorFlow, PyTorch, OpenCV, and ROS, among other AI and ML tools. The course trains students to use AI and ML to solve complex engineering problems.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Explain the basic concepts and principles of artificial intelligence and machine learning and their relevance to mechanical engineering.
2. Apply Python programming and mathematical tools to implement and evaluate AI and ML algorithms and models.
3. Use various AI and ML frameworks and libraries to develop and deploy applications for engineering design, computer vision, robotics, and data analysis.
4. Compare and contrast different AI and ML methods and techniques, and select the appropriate ones for different engineering scenarios and challenges.
5. Demonstrate ethical and professional awareness and responsibility when using AI and ML for engineering purposes.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING</b> |
| History and evolution of AI and ML, applications and challenges of AI and ML in mechanical engineering, AI and ML terminology and concepts, Python programming basics, data structures, and libraries.                                                                                                                                                                                                                                         |                                                                     |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>MATHEMATICAL FOUNDATIONS OF AI AND ML</b>                        |
| Statistics, probability, linear algebra, calculus, optimization, and numerical methods for AI and ML, data preprocessing, visualization, and exploration, descriptive and inferential statistics, hypothesis testing, and confidence intervals.                                                                                                                                                                                                |                                                                     |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>SUPERVISED AND UNSUPERVISED LEARNING</b>                         |
| Supervised learning concepts and algorithms, such as regression, classification, decision trees, k-nearest neighbors, support vector machines, and ensemble methods, unsupervised learning concepts and algorithms, such as clustering, dimensionality reduction, principal component analysis, and k-means, evaluation metrics and methods, such as accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and cross-validation. |                                                                     |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>NEURAL NETWORKS AND DEEP LEARNING</b>   |
| Neural network concepts and architectures, such as perceptron, multilayer perceptron, feedforward, and feedback networks, activation functions, loss functions, and gradient descent, deep learning concepts and architectures, such as convolutional neural networks, recurrent neural networks, long short-term memory, and generative adversarial networks, TensorFlow and PyTorch frameworks for building and training neural networks and deep learning models, applications of neural networks and deep learning in engineering design, computer vision, robotics, and data analysis. |                                            |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>REINFORCEMENT LEARNING AND ROBOTICS</b> |
| Reinforcement learning concepts and algorithms, such as Markov decision processes, value functions, policy iteration, value iteration, Q-learning, and deep Q-networks, robotics concepts and algorithms, such as kinematics, dynamics, motion planning, control, localization, mapping, and navigation, OpenCV and ROS libraries for computer vision and robotics applications, integration of reinforcement learning and robotics for solving complex engineering tasks and problems.                                                                                                     |                                            |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                            |
| <ol style="list-style-type: none"> <li>1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, Pearson India Education Services Pvt Ltd., 3<sup>rd</sup> Edition, 2015.</li> <li>2. Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.</li> <li>3. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, The MIT Press, 2016.</li> </ol>                                                                                                                                                                                    |                                            |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            |
| <ol style="list-style-type: none"> <li>1. Christopher Tong and Duvvuru Sriram, Artificial Intelligence in Engineering Design: Volume III, Academic Press, 1992.</li> <li>2. Machine Learning in Engineering by Yunus A. Cengiz</li> <li>3. Tshilidzi Marwala, Artificial Intelligence Techniques for Rational Decision Making, Springer, 2014.</li> <li>4. Machine Learning for Mechanical Engineering by Rui Xue, Zhenyu Liu, and Xiangyu Wang</li> </ol>                                                                                                                                  |                                            |
| <b>MOOC Course:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                            |
| NPTEL: Mechanical Engineering - NOC: Fundamentals Of Artificial Intelligence<br><a href="https://archive.nptel.ac.in/courses/112/103/112103280/#">https://archive.nptel.ac.in/courses/112/103/112103280/#</a>                                                                                                                                                                                                                                                                                                                                                                               |                                            |

## MECHATRONICS

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME52      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This program, developed in direct response to industrial demand for engineers with multi-disciplinary skills, is a combination of mechanical, electronics, control, computer and systems design engineering streams. The program allows engineers to design construct and run factory production lines and automated processes, where they use their skills in computers, micro- controllers, programmable logic controllers, programming, industrial sensors, hydraulic, pneumatic and electric drives, design of mechanical structures and mechanisms and knowledge of manufacturing processes.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate basic concepts of mechatronics.
2. Explain actuators and elements of motion control.
3. Classify and discuss about sensors.
4. Explain elements and working of various transducers.
5. Explain concepts of PLC’s and artificial intelligence.

|               |                     |
|---------------|---------------------|
| <b>UNIT-I</b> | <b>INTRODUCTION</b> |
|---------------|---------------------|

Introduction: Evolution of Mechatronics, Modules in Mechatronic Systems, Design process, Control Systems - Open and Closed loop systems, Analog and digital control systems, Mechatronics in Manufacturing – Adaptive and distributed control systems–Modelling and simulation of mechatronics.

|                |                                       |
|----------------|---------------------------------------|
| <b>UNIT-II</b> | <b>ACTUATORS &amp; MOTION CONTROL</b> |
|----------------|---------------------------------------|

Actuators and Motion Control: Characteristics of Mechanical, electrical, Hydraulic and pneumatic actuators and their limitations. Control parameters and system objectives. Mechanical configurations, Popular control system configurations, S-curve, Motor/load inertia machining, design with linear studies.

|                 |                |
|-----------------|----------------|
| <b>UNIT-III</b> | <b>SENSORS</b> |
|-----------------|----------------|

Sensors: Definition of sensor, Classification of sensors by function , Classification of performance, Performance parameters, Classification of sensors by output, Pressure sensors, flow sensors , Hall-effect sensors-Linear and threshold type, Light sensors-photo diodes, Proximity Sensors-Inductive and Ultrasonic type; Optical sensor.

|                |                    |
|----------------|--------------------|
| <b>UNIT-IV</b> | <b>TRANSDUCERS</b> |
|----------------|--------------------|

Transducers: Difference between transducer and sensor, Transducer principle, Photo electric transducers-photo emissive transducer, photo conductive transducer, photovoltaic transducer; Inductive transducers, Capacitive Transducers, pyro electric transducers, piezo electric transducer, Ionization transducers, strain gauge, load cells and flow transducers.

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | PLCs & ARTIFICIAL INTELLIGENCE |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>PLCs:</b> Programmable Logic Controllers Architecture of Programmable Logic Controllers – Input/output modules – programming methods – Timers and counters – Master control – Branching – Data handling – Analog input/output – Selection of PLC and troubleshooting.<br><b>AI:</b> Intelligent Mechatronics and Case Studies Fuzzy logic control and Artificial Neural Networks in mechatronics – Algorithms – Computer based instrumentation – Real-time Data Acquisition and Control – Software integration – Man Machine interface –Vision system. |                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |
| <ol style="list-style-type: none"><li>1. Introduction to Mechatronics and Measurement Systems, David G. Alciatore, Michael B. Histan, Tata McGraw Hill, 4th Edition, 2012.</li><li>2. Mechatronics, Ganesh S.Hegde, Laxmi Publications; Second edition, 2016.</li></ol>                                                                                                                                                                                                                                                                                   |                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| <ol style="list-style-type: none"><li>1. Designing Intelligent Machines, Michel B.Histan and David G.Alciatore, Open University London, 2nd edition, 2003.</li><li>2. Control Sensors and Actuators, D.A. Hall and C.E. Millar CRC Press, 1st Edition, 1999.</li></ol>                                                                                                                                                                                                                                                                                    |                                |

## ADVANCED WELDING TECHNIQUES AND NDT METHODS

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME53      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

The aim of this course is to introduce students about the advancements in Welding Technology and the overview of the non destructive testing methods of materials. The course covers NDE, Ultrasonic, MPI testing of metal parts. It gives an idea about selection of the testing criteria.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Describe the various types of NDT tests like LPT, MPT, UT.
2. Demonstrate principles of special NDT methods such as RT, Acoustic Emission Testing, Infrared Thermograph.
3. Description of advance welding methods and welding technology.
4. Determine the various welding defects associated with welding processes
5. Understanding the Importance of weld joint design

#### UNIT-I INTRODUCTION TO NDT

NDT Versus Mechanical testing, Overview of the Non-Destructive Testing Methods for the detection of manufacturing and processing defects., Area of application of common NDT methods, Advantages and limitations of common NDT methods.

#### UNIT-II SURFACE NDT METHODS

**Liquid Penetrant Testing:** Principles, Pre Cleaning methods, types and properties of liquid penetrants, Different type penetrant systems and developers, advantages and limitations of various methods, Visible and Fluorescent method, Testing Procedure, True and false indication, Lighting requirement, Interpretation and evaluation indication and results.

**Magnetic Particle Testing:** Principles and Theory of magnetism, Hysteresis curve, Inspection materials Magnetization methods, circular magnetization, longitudinal magnetization, Flemings right hand rule, AC/DC current magnetization, Dry and wet methods, Visible and fluorescent methods, Relevant and non relevant, Interpretation and evaluation of test indications.

#### UNIT-III ULTRASONIC TESTING AND RADIOGRAPHY

**Ultrasonic testing and acoustic emission:** sound and sound waves, modes of sound waves, Velocity, frequency, wavelength, Snells law, Reflection, refraction, mode conversion Transducers, transmission and pulse-echo method, straight beam and angle beam, instrumentation, data representation, Difference between A-Scan, B-scan and C-scan. Various applications of ultrasonic testing.

**Radiography:** History of Radiography, Industrial radiography: Principles, imaging techniques (Film radiography), X-Rays and Gamma Rays, types and use of filters and screens, geometric factors, Time distance and shielding concepts characteristics of films –graininess, density, speed, contrast, Image quality, characteristic curves, Penetrameters, Exposure charts.

#### Other NDT Methods:

Overview of Eddy current, visual testing, acoustic emission, thermography, Fluoroscopy-neutron

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| radiography, Computed Radiography.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>WELDING TECHNOLOGY INTRODUCTION</b>         |
| <b>Welding Technology Introduction:</b><br>Introduction to welding, various welding process, SMAW, GTAW, FCAW, GMAW, weld joints and design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>WELDING METALLURGY AND ADVANCED WELDING</b> |
| <b>Welding metallurgy and advanced welding</b><br>welding parameters, Welding metallurgy, welding defects, Submerged arc welding, Electron beam and laser welding, Friction welding, Resistance welding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                |
| 1. Baldev Raj, T.Jayakumar, M.Thavasimuthu, Practical Non-Destructive Testing, Narosa Publishing House, 2009.<br>2. Ravi Prakash, Non-Destructive Testing Techniques, 1st revised edition, New Age International Publishers, 2010.<br>3. O.P Khanna, Text book of welding technology, Dhanpath Rai publication, 1999.                                                                                                                                                                                                                                                                                                                                                                                     |                                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                |
| 1. ASM Metals Handbook, Non-Destructive Evaluation and Quality Control, American Society of Metals Volume-17, Metals Park, Ohio, USA, 2000.<br>2. Paul E Mix, Introduction to Non-destructive testing: a training guide, Wiley, 2 <sup>nd</sup> Edition New Jersey, 2005.<br>3. Charles, J.Hellier, Hand book of Non-destructive evaluation, McGraw Hill, New York, 2001.<br>4. Baldev raj, V. Shankar, A.K Bhaduri, Welding technology for engineers, Welding technology: Process and application, Alpha Science International Ltd., 2006.<br>5. Dr.KS Yadav, Advance welding technology, Rajson publications pvt limited, 2017.<br>6. Roger timings, Fabrication and welding engineering, Newnes, 2008. |                                                |



## NON-CONVENTIONAL ENERGY SOURCES

### III B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                      | Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| A6ME54                                                                                                                                                                                                                                                                                                                                                                                                                                           | PEC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course looks at the working principle of non-conventional energy resources, and key performance characteristics. The technologies and applications of Solar, Wind, Wave, Tidal, OTEC and Geothermal conversion will be focused. The advantages and limitations of these technologies in comparison to conventional sources of energy will also be examined.                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate various energy sources and their availability.<br>2. Explain concepts and applications of solar energy.<br>3. Discuss elaborately the principles of wind energy conversion.<br>4. Explain Biomass conversion techniques and Geothermal Sources and resources<br>5. Explain the principles of tidal energy and Ocean Thermal Energy. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>INTRODUCTION</b><br>Statistics on conventional energy sources and supply in developing countries, Definition-Concepts of NCES, Limitations of RES, Criteria for assessing the potential of NCES. Classification of NCES- Solar, Wind, Geothermal, Bio-mass, Ocean Energy Sources, comparison of these energy sources.                                                                                                                                                                                                   |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>SOLAR ENERGY</b><br>Solar Energy-Energy available from Sun, Solar radiation data, Solar energy conversion into heat, Flat plate and Concentrating collectors, Mathematical analysis of Flat plate collectors and collector efficiency, Principle of Natural and Forced convection, Solar engines-Stirling, Brayton engines, Photovoltaic, p-n junction, solar cells, PV systems, Stand-alone, Grid connected solar power satellite.                                                                                     |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>WIND ENERGY</b><br>Wind energy conversion, General formula -Lift and Drag- Basis of wind energy conversion -Effect of density, frequency variances, angle of attack, and wind speed. Windmill rotors- Horizontal axis and vertical axis rotors. Determination of torque coefficient, Induction type generators- working principle.                                                                                                                                                                                      |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>GEO THERMAL AND BIOMASS ENERGY</b><br>Nature of Geothermal sources, Definition and classification of resources, Utilization for electric generation and direct heating, Well Head power generating units, Basic features- Atmospheric exhaust and condensing, exhaust types of conventional steam turbines.<br>Pyrolysis of Biomass to produce solid, liquid and gaseous fuels, Biomass gasification, Constructional details of gasifier, usage of biogas for chulhas, various types of chulhas for rural energy needs. |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>WAVE, TIDAL AND OTEC ENERGY</b><br>Wave, Tidal and OTEC energy- Difference between tidal and wave power generation, Principles of tidal and wave power generation, OTEC power plants, Operational of small cycle experimental facility, Design of 5 Mw OTEC pro-commercial plant, Economics of OTEC, Environmental impacts                                                                                                                                                                                              |              |   |   |         |               |     |       |

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of OTEC. Status of multiple product OTEC systems.

**Text Books:**

1. Ashok V Desai, Non-Conventional Energy, Wiley Eastern Ltd, New Delhi, 2003.
2. Mittal K M, Non-Conventional Energy Systems, Wheeler Publishing Co. Ltd, New Delhi, 2003.

**Reference Books:**

1. Ramesh R & Kumar K U, Renewable Energy Technologies, Narosa Publishing House, New Delhi, 2004.
2. Wakil MM, Power Plant Technology, Mc Graw Hill Book Co, New Delhi, 2004.
3. G.D. Rai. Non - Conventional Energy Sources, Khanna Publishers, New Delhi, 2017.

## **III B.TECH. II SEMESTER SYLLABUS**

## HEAT TRANSFER

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME31      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This course will provide knowledge about application of conduction, convection and radiation heat transfer concepts to different practical applications.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate the basic modes of heat transfer and compute temperature distribution on steady and unsteady state heat conduction.
2. Analyse the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces.
3. Interpret and analyse the free and forced convection heat transfer.
4. Comprehend the concepts of Boiling, Condensation and Heat exchangers.
5. Explain the principles of radiation heat transfer and apply the radiation laws on black and gray bodies.

|                                                                                                                                                                                                                                                                                                                                            |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                              | <b>INTRODUCTION</b> |
| Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier Law of conduction, general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions. |                     |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                             | <b>ONE DIMENSIONAL HEAT CONDUCTION</b> |
| Heat transfer through homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy and Critical radius of insulation. Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips. Significance of Biot and Fourier Numbers, Chart solutions of transient conduction systems |                                        |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CONVECTION</b> |
| CONVECTIVE HEAT TRANSFER: Concepts of Continuity, Momentum and Energy Equations. Dimensional analysis – Buckingham’s Pi Theorem – Application for developing non-dimensional correlation for convective heat transfer.<br>FORCED CONVECTION: External Flows, Concepts of hydrodynamic and thermal boundary layer and use of empirical correlations for Flat plates and Cylinders. Internal Flows, Concepts about Hydrodynamic and Thermal Entry Lengths, use of empirical correlations for Horizontal Pipe Flow and annulus flow.<br>FREE CONVECTION: Development of Hydrodynamic and thermal boundary layer along a vertical plate, Use of empirical relations for Vertical plates and pipes. |                   |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                    | <b>BOILING &amp; CONDENSATION, HEAT EXCHANGERS</b> |
| BOILING & CONDENSATION: Regimes of Pool boiling and Flow boiling, Critical heat flux, Calculations on Nucleate Boiling. Film wise and drop wise condensation, Nusselt’s theory of condensation on a vertical plate – Film condensation on vertical and horizontal cylinders using |                                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                       |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| empirical correlations.<br>HEAT EXCHANGERS: Classification of heat exchangers, overall heat transfer Coefficient and fouling factor, Concepts of LMTD and NTU methods, Problems using LMTD and NTU methods.                                                                                                                                                                                                           |                                |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                         | <b>RADIATION HEAT TRANSFER</b> |
| Emission characteristics, Laws of black-body radiation, Irradiation, Total and monochromatic quantities, Laws of Planck, Wien, Kirchhoff, Lambert, Stefan and Boltzmann, Heat exchange between two black bodies, concepts of shape factor, Emissivity, heat exchange between grey bodies, radiation shields.                                                                                                          |                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                    |                                |
| <ol style="list-style-type: none"> <li>1. Yunus A. Cengel, Heat Transfer a Practical Approach, 4<sup>th</sup> Edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2012.</li> <li>2. R. C. Sachdeva, Fundamentals of Engineering, Heat and Man Transfer, 3<sup>rd</sup> Edition, New Age, New Delhi, India, 2012.</li> </ol>                                                                                |                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                               |                                |
| <ol style="list-style-type: none"> <li>1. J.P. Holman, Heat Transfer (SI Units), 10<sup>th</sup> Edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2012.</li> <li>2. P. S. Ghoshdastidar, Heat Transfer, 2<sup>nd</sup> Edition, Oxford University Press, New Delhi, India, 2012.</li> <li>3. Incropera, Dewitt, Fundamentals of Heat Transfer, 6<sup>th</sup> Edition, John Wiley, UK, 2012.</li> </ol> |                                |

## MACHINE TOOLS

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME33      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This course will give an insight over different machining processes and tools. In the metal-working industry work pieces of most different shapes and dimensions and of different materials are worked. The shaping of materials is done by either non-cutting process or cutting process. Such relative motion is produced by combination of rotary and translating movements either of work piece or the cutting tool or of both. Machining process also include other processes like grinding, slotting, shaping, honing, planning, lapping, and broaching operations.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Knowledge on cutting tool geometry, mechanism of chip formation and mechanics of orthogonal cutting.
2. Apply the fundamentals and principles of metal cutting to practical applications using lathes.
3. Demonstrate the fundamentals of machining processes and machine tools of Shaping, Slotting, Planing and Drilling machines.
4. Demonstrate the fundamentals of machining processes and gear cutting using Milling Machine.
5. Select appropriate super finishing process for an application and understand the impact of such process in environmental context.

|                                                                                                                                                                                                                                                                                                                                                                                          |                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                            | <b>METAL CUTTING THEORY</b>         |
| Elementary treatment of metal cutting theory – Elements of cutting process – Geometry of single point cutting tool and angles. Chip formation and types of chips – built up edge and its effects. Mechanics of orthogonal cutting, – Merchant’s Force diagram – cutting speeds, feed, depth of cut, tool life, coolants, machinability – Tool materials.                                 |                                     |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                           | <b>LATHE</b>                        |
| Engine lathe – Principle of working, specification of lathe – types of lathes – work & tool holding devices-Taper turning, thread- turning – for Lathes and attachments. Turret and capstan lathes – Principal features of automatic lathes – classification – Single spindle and multi-spindle automatic lathes – tool layout                                                           |                                     |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>SHAPING SLOTTING AND PLANING</b> |
| <b>Shaping slotting and planing machines</b> – Principles of working – Principal parts – specification classification, operations performed.<br><b>Drilling and Boring Machines</b> – Principles of working, specifications, types, operations performed – tool holding devices – twist drill – Boring machines – Fine boring machines – Jig Boring machine. Deep hole drilling machine. |                                     |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                           | <b>MILLING MACHINE</b>              |
| <b>Milling machine</b> – Principles of working – specifications – classifications of milling machines – Principal features of horizontal, vertical and universal milling machines – machining operations , Geometry of milling cutters – Types of milling cutters – Accessories to milling machines.                                                                                     |                                     |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                          | SUPER FINISHING PROCESS |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Grinding machine</b> – Fundamentals – Theory of grinding – classification of grinding machine – cylindrical and surface grinding machine – Tool and cutter grinding machine – special types of grinding machines – Different types of abrasives – bonds specification of a grinding wheel and selection of a grinding wheel.<br><b>Lapping, honing and broaching machines</b> – comparison to grinding – lapping and honing. |                         |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                              |                         |
| 1. R.K. Jain and S.C. Gupta, Production Technology, S Chand Publication, 2009.<br>2. O P Khanna, Production Technology, Danpat Rai Publications, 2015.                                                                                                                                                                                                                                                                          |                         |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                         |                         |
| 1. G. Boothroyd G. Boothroyd, Fundamentals of Metal Machining and Machine Tools, Taylor and Francis, 2005.<br>2. C. Elanchezhian and M. Vijayan, Machine Tools, Anuradha Agencies, 2009.<br>3. Bhattacharya A and Sen.G.C., Principles of machine Tools, New central Book Agency, 2007.                                                                                                                                         |                         |

## FINITE ELEMENT ANALYSIS

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME35      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### **COURSE OVERVIEW:**

The aim of this course to enable students to understand fundamental theory of the FEA method, generate the governing FE equations for systems governed by partial differential equations, understand the use of the basic finite elements for structural applications using truss, beam and plane elements under static and dynamic loading; understand the application and use of the FE method for heat transfer problems.

#### **COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate basic concepts of FEM, Solve the basic spring problems
2. Formulate and solve finite element equations for 1-D elements and Trusses
3. Derive stiffness matrices, analyze beams and 2D CST Elements subjected to different load conditions
4. Formulate and solve axi-symmetric problems using constant strain triangular elements and heat transfer problems,
5. Analyze axial bar, beam and truss subjected to dynamic loads

#### **UNIT-I**

#### **INTRODUCTION TO FEM**

Basic concept, historical background, applications of FEM, Basic equations of elasticity-strain-displacement relations and stress-strain relations, Convergence requirements, Principle of Minimum Potential Energy-Spring Problems.

#### **UNIT-II**

#### **ONE DIMENSIONAL PROBLEMS & ANALYSIS OF TRUSSES**

ONE DIMENSIONAL PROBLEMS: Formulation of Stiffness Matrix for a Bar Element by the Principle of Minimum Potential Energy, Properties of Stiffness Matrix, Characteristics of Shape Functions, Quadratic shape functions. Problems on uniform stepped and tapered bars subjected to point loads, body force and temperature loads.

ANALYSIS OF TRUSSES: Local and Global co-ordinates, Transformation matrix, Stiffness Matrix for Plane Truss, Stress-Strain Calculations and Calculation of reaction forces.

#### **UNIT-III**

#### **ANALYSIS OF BEAMS & 2-D PROBLEMS**

ANALYSIS OF BEAMS: Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, slope and deflection. Simple problems-Point loads & UDL.

2-D PROBLEMS: CST Element-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems. Shape functions of Two dimensional four noded isoparametric element.

#### **UNIT-IV**

#### **AXI-SYMMETRIC SOLIDS & HEAT TRANSFER**

AXI-SYMMETRIC SOLIDS: Axi-symmetric solids subjected to Axi-symmetric loading with triangular elements-simple problems.



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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| STEADY STATE HEAT TRANSFER ANALYSIS: One Dimensional Finite Element analysis of fin and composite slabs.                                                                                                                                                                                                                                                                                                                                                  |                         |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>DYNAMIC ANALYSIS</b> |
| DYNAMIC ANALYSIS: Dynamic Equations of motion, Lumped and consistent mass matrices, Evaluation of Eigen values and Eigen vectors for a stepped bar and beam.                                                                                                                                                                                                                                                                                              |                         |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |
| 1. Introduction to Finite Elements in Engineering by Chandrupatla, Ashok and Belegundu, Prentice, Hall, Pearson.<br>2. The Finite Element Methods in Engineering by SS Rao, Pergamon.                                                                                                                                                                                                                                                                     |                         |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |
| 1. Finite Element Methods: Basic Concepts and applications by Alavala, PHI<br>2. Finite Element Analysis Revised and Enlarged Edition by S.Md.Jalaluddin, Anuradha Publications<br>3. Finite Element Analysis by P.Seshu, PHI<br>4. Finite Element Analysis by Bathe, PHI<br>5. Finite Element Method by Krishna Murthy, TMH<br>6. An Introduction to the Finite Element Method (McGraw-Hill Mechanical Engineering) 3rd Edition. by J Reddy, McGraw-Hill |                         |

## HEAT TRANSFER LAB

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A6ME32      | PCC      | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

The Heat Transfer Lab enables the student to understand the concepts of conduction, convection and radiation practically in various thermal systems through a series of experiments. Students have a fairly good understanding of the theory underlying the experiments and the entire lab course is designed such that classroom lectures precede the lab work.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Apply the principles of various modes of heat transfer to evaluate thermal conductivity of different metals and non-metal objects under steady state conditions.
2. Apply the principles of various modes of heat transfer to evaluate thermal conductivity of different objects under unsteady state conditions.
3. Estimate heat transfer coefficients in forced convection, free convection, condensation and correlate with theoretical values.
4. Obtain variation of temperature along the length of the pin fin under forced and free convection.
5. Apply the principles of radiation to evaluate the surface emissivity of a test plate and Stefan-Boltzmann's constant and compare with theoretical values.

#### LIST OF EXPERIMENTS:

1. Composite Slab Apparatus – Overall heat transfer co-efficient.
2. Heat transfer through lagged pipe.
3. Thermal Conductivity of given metal rod.
4. Heat transfer through pin-fin
5. Heat transfer in forced convection apparatus.
6. Heat transfer in natural convection
7. Parallel and counter flow heat exchanger.
8. Emissivity apparatus.
9. Stefan Boltzmann Apparatus.
10. Heat transfer in drop and film wise condensation.
11. Critical Heat flux apparatus.
12. Study of heat pipe and its demonstration.

## MACHINE TOOLS LAB

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A6ME34      | PCC      | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

#### **COURSE OVERVIEW:**

This course is introduced to create awareness on various machining operations, equipment and their working to create prototype models.

#### **COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. To identify various cutting tools and its uses.
2. To understand the working of various machines such as Lathe, Shaper, Slotter and Planing.
3. To understand the working of various operations on Milling and Drilling Machines
4. To make students to create prototype models by CNC machining.

#### **LIST OF EXPERIMENTS:**

1. Step turning and taper turning using lathe machine.
2. Thread cutting and knurling using lathe machine.
3. Drilling and step boring using lathe machine.
4. Drilling and Tapping using drilling machine.
5. Shaping of V groove using shaper.
6. Slotting of a keyway using slotter machine.
7. Milling of gear.
8. Surface Grinding.
9. Planing Machine operations
10. Cylindrical grinding Operations.
11. Tool and cutter grinding operations
12. To perform different operations by using CNC lathe machine
13. To perform different operations by using CNC Milling machine

## PRODUCTION DRAWING PRACTICE AND COMPUTER AIDED ENGINEERING LAB

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Category                                     | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| A6ME36                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ESC                                          | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                              | 0            | 0 | 2 | 1       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course is intended to impart knowledge on fundamentals of GD&T and analysis software and its applications in solving problems related to Structural and Thermal analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                              |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Recognize the representation of various mechanical elements in the production drawing.<br>2. Acquire knowledge on limits & fits, tolerances, and surface roughness.<br>3. Draw part drawing and assembly drawing showing all limits fits and tolerances, machining operations on the production drawing using AutoCAD.<br>4. Apply analysis software to find the deflections, stresses and strains of the structures subjected various types of boundary and loading conditions.<br>5. Analyse steady state heat transfer problems using analysis software. |                                              |              |   |   |         |               |     |       |
| <b>PART A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Production Drawing Practice (PDP) Lab</b> |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                              |              |   |   |         |               |     |       |
| Conventional representation of materials, conventional representation of machine components, method of indicating notes on drawings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                              |              |   |   |         |               |     |       |
| Limits and Fits: Limit system, Types of fits, exercises involving selection / interpretation of fits and estimation of limits from tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                              |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                              |              |   |   |         |               |     |       |
| Geometric Dimensioning and Tolerancing (GD&T): Need for GD&T, Features and rules of GD&T, datum's control, Form tolerances, Orientation tolerances and Position tolerances.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                              |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                              |              |   |   |         |               |     |       |
| Surface roughness and its indication: Definitions – finishes obtainable from various manufacturing processes, recommended surface roughness on mechanical components. Heat treatment and surface treatment symbols used on drawings.                                                                                                                                                                                                                                                                                                                                                                                                                         |                                              |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                              |              |   |   |         |               |     |       |
| Detailed and Part drawings: Drawing of parts from assembly drawings with indications of size, tolerances, roughness, form and position features etc. Part drawing using computer aided drafting by CAD software.                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                              |              |   |   |         |               |     |       |

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| <b>PART B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Computer Aided Engineering (CAE) Lab</b> |
| <b>Perform the following experiments using analysis software.</b> <ol style="list-style-type: none"><li>1. 2D beam analysis with different sections for different materials for different loading and boundary conditions.</li><li>2. Analysis of Plane Truss with various cross sections and materials to determine member forces, member strains &amp; stresses, joint deflections under static loading.</li><li>3. Static Analysis of flat shell due to internal pressure to estimate the strains, stresses and reactions forces with different boundary conditions.</li><li>4. Static analysis of plate with a hole to determine the deformations and the stresses to study the failure behaviour.</li><li>5. Steady state heat transfer Analysis.</li></ol> |                                             |
| <b>Text Books:</b> <ol style="list-style-type: none"><li>1. K L Narayana, K. Venkata Reddy, P Kannaiah, Production Drawing, 2<sup>nd</sup> Edition, New Age International Pvt. Ltd., 2012.</li><li>2. Goutam Pohit and Goutam Ghosh, Machine Drawing with AutoCAD, Pearson Education, India, 2004.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                             |
| <b>Reference Books:</b> <ol style="list-style-type: none"><li>1. James D. Meadows, Geometric dimensioning and tolerancing, B.S Publications, 2010.</li><li>2. R.K. Jain, Engineering Metrology, 22<sup>nd</sup> Edition, Khanna Publishers, 2022.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                             |

## INDUSTRY ORIENTED MINI PROJECT / INTERNSHIP

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME40      | PWC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 4 | 2       | -             | 100 | 100   |

#### Evaluation Procedures

There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization: Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation /semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal(or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc., Internal Supervisor and a Senior Faculty Member of the Department. There shall be NO internal marks for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.

#### COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Design engineering solutions to complex problems utilizing a systems approach
3. Conduct an experiments in the engineering project and analysis the data results
4. Communicate with engineers and the community at large in written an oral form.
5. Demonstrate the knowledge, skills and attitudes of a professional engineer

**PROFESSIONAL ELECTIVE – II**

## PROFESSIONAL ELECTIVE - II

### SUSTAINABLE ENGINEERING

#### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME56      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course is to inculcate in students an awareness of environmental issues and the global initiatives towards attaining sustainability. The student should realize the potential of technology in bringing in sustainable practices.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Explain sustainable development and different environmental agreements and protocols
2. Discuss real time activities causing environmental issues and different methods to use renewable energy resources
3. Explain local and global environmental issues
4. Differentiate between carbon emissions for regular and sustainable cities and explain different practices to move industries towards sustainability
5. Discuss different renewable energy resources and explain methods to implement green technology

#### UNIT-I

#### INTRODUCTION TO SUSTAINABLE ENGINEERING

Sustainable development, concepts of sustainable development: three pillar model, egg of sustainability model, Atkisson's pyramid model, prism model, principles of sustainable development, sustainable engineering, threats for sustainability. Environmental Ethics and Legislations – Environmental ethics and education, multilateral environmental agreements and protocols, enforcement of environmental laws in India – The Water Act, The Air Act, The Environment Act.

#### UNIT-II

#### LOCAL AND GLOBAL ENVIRONMENTAL ISSUES

Local Environmental Issues- Solid waste, impact of solid waste on natural resources, zero waste concept and three R concept, waste to energy technology: thermo-chemical conversion, biochemical conversion. Global Environmental Issues- Resource degradation: deterioration of water resources, land degradation, air pollution, climate change and global warming, ozone layer depletion, carbon footprint, carbon trading.

#### UNIT-III

#### TOOLS FOR SUSTAINABILITY

Tools for Sustainability - Environmental Management System (EMS), concept of ISO14000, life cycle assessment (LCA): basic components, advantages, disadvantages, case study. Environmental impact assessment (EIA), environmental auditing, bio mimicking, case studies.

#### UNIT-IV

#### SUSTAINABLE HABITAT, SUSTAINABLE INDUSTRIALIZATION AND URBANIZATION

Sustainable Habitat - Concept of green building, green building materials, green building certification and rating: green rating for integrated habitat assessment (GRIHA), leadership in energy and environmental design (LEED) rating, energy efficient buildings, sustainable cities, sustainable



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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <p>transport, sustainable pavements, case studies in sustainability engineering: Green building, sustainable city, sustainable transport system.</p> <p>Sustainable Industrialization and Urbanization – Sustainable urbanization, industrialization, material selection, pollution prevention, industrial ecology, industrial symbiosis, poverty reduction.</p>                                                        |                                                                        |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                           | <b>RENEWABLE ENERGY RESOURCES, GREEN TECHNOLOGY AND GREEN BUSINESS</b> |
| <p>Renewable energy resources- Conventional and non- conventional forms of energy, solar energy, fuel cells, wind energy, small hydro plants, biogas systems, biofuels, energy from ocean, geothermal energy, conservation of energy.</p> <p>Green technology and Green Business: Sustainable business, green technology, green energy, green construction, green transportation, green chemistry, green computing.</p> |                                                                        |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                        |
| <ol style="list-style-type: none"> <li>1. R. L. Rag and Lekshmi Dinachandran Remesh. Introduction to Sustainable Engineering. 2<sup>nd</sup> Edition, PHI Learning Pvt. Ltd., 2016.</li> <li>2. A.S. Bradley, A. O. Adebayo, P. Maria. Engineering applications in sustainable design and development, 1st Edition, Cengage learning, 2016.</li> </ol>                                                                  |                                                                        |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                        |
| <ol style="list-style-type: none"> <li>1. D. T. Allen and D. R. Shonnard. Sustainability Engineering: Concepts, Design and Case Studies, 1st Edition, Prentice Hall, 2011.</li> <li>2. Mackenthun, K.M., Basic Concepts in Environmental Management, Lewis Publication, London, 1998.</li> <li>3. Purohit, S. S., Green Technology - An approach for sustainable environment, Agrobios Publication.</li> </ol>          |                                                                        |

## MACHINE DYNAMICS & VIBRATIONS

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |     |
|-------------|----------|--------------|---|---|---------|---------------|-----|-----|
|             |          | L            | T | P |         | C             | CIE | SEE |
| A6ME57      | PEC      | 3            | 0 | 0 | 3       | 40            | 60  | 100 |

**COURSE OVERVIEW:**

The primary focus of this course is to make the students understand the concepts of vibrations, limitations of vibrations in industry and to the concepts how to minimize the vibrations for the better performance of the machines.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate the various systems of vibrations in a machine and to compute them, can apply the concepts to differentiate various classifications.
2. Can formulate the relations for various types of vibrations conditions using FFT
3. Solve the problems on rotor dynamic issues.
4. Explain the various factors for vibrations and can demonstrate them.
5. Demonstrate various instruments used to measure vibrations.

|                                                                                                                                                                                                                                                       |                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                         | <b>MULTI DEGREE OF FREEDOM SYSTEMS (MDOF)</b> |
| Derivation of equations of motion for mechanical lumped parameter systems forced responses<br>Forced response of Multi DOF systems; vibration absorber Free and forced response of continuous systems (beams, shafts in torsion and bending)          |                                               |
| <b>UNIT-II</b>                                                                                                                                                                                                                                        | <b>VIBRATIONS IN THE FREQUENCY DOMAIN</b>     |
| Frequency response function (FRF) and their main features Fourier Series, Analysis and the Fourier Transform; FFT time domain factors Leakage, Windowing and Aliasing, Time domain response prediction using Fourier transform (Du Hamel’s principle) |                                               |
| <b>UNIT-III</b>                                                                                                                                                                                                                                       | <b>ROTOR DYNAMICS</b>                         |
| Response to out of balance Whirling of shafts Critical speeds Balancing, static balance and dynamic imbalance.                                                                                                                                        |                                               |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                        | <b>VIBRATION INDUCED FATIGUE</b>              |
| Calculation of stress from vibration mode shape and amplitude Input to fatigue calculations; Goodman diagrams.                                                                                                                                        |                                               |
| <b>UNIT-V</b>                                                                                                                                                                                                                                         | <b>VIBRATION MEASUREMENTS</b>                 |
| Transducers: linear displacement, angular displacement, angular velocity, pressure Data acquisition: resolution, aliasing.                                                                                                                            |                                               |

**Text Books:**

1. Mechanical Vibrations by William J. Palm, 2007.
2. Dynamics and Vibrations, An introduction, by Magd Abdel Wahab, Wiley, 2008.

|                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reference Books:</b>                                                                                                                                                                                                                       |
| <ol style="list-style-type: none"><li>1. Cheli F., Diana G., Advanced Dynamics of Mechanical Systems, Editor: Springer, Annoedizione: 2015.</li><li>2. Singiresu S.Rao, Mechanical Vibrations, Editor: Pearson, Annoedizione, 2016.</li></ol> |

## DESIGN FOR MANUFACTURING AND ASSEMBLY

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME58      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

DFM techniques are focused on individual parts and components with a goal of reducing or eliminating expensive, complex or unnecessary features which would make them difficult to manufacture. This course will give insight on design for machining, casting, welding, brazing, forging, extrusion, sheet metal and assembly.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate steps in design process, design rules for manufacturability, material selection considerations and developments in material technology.
2. Demonstrate various machining processes and explain design considerations for machining and casting.
3. Describe various welding processes and explain design considerations for welding, brazing and forging.
4. Explain design considerations for extrusion and sheet metal work.
5. Explain objectives and design considerations for assembly.

|                                                                                                                                                                                                                                                                                                                                                                                                           |                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                             | <b>INTRODUCTION</b>                                |
| Design philosophy steps in Design process – General Design rules for manufacturability – basic principles of designing for economical production, Materials-parameters to be considered for the selection of materials for design – criteria for material selection, material selection process for new product design, developments in material technology.                                              |                                                    |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                            | <b>DESIGN FOR MACHINING &amp; CASTING</b>          |
| Overview of various machining processes – general design rules for drilling, milling and turning – General design recommendations for machined parts. Metal Casting: Appraisal of various casting processes, selection of casting process, – general design considerations for casting – casting tolerances – use of solidification simulation in casting design – product design rules for sand casting. |                                                    |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                           | <b>DESIGN FOR WELDING, BRAZING &amp; FORGING</b>   |
| Appraisal of various welding processes, Factors in design of weld ments – general design guidelines – pre and post treatment of welds – effects of thermal stresses in weld joints – design of brazed joints. Design considerations for Forging.                                                                                                                                                          |                                                    |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                            | <b>DESIGN FOR EXTRUSION &amp; SHEET METAL WORK</b> |
| Extrusion-Design recommendations, design guidelines, Sheet Metal Work-cost effective design principles for Punching, Blanking and bending, Design principles for successful deep drawing, Keeler Goodman Forming Line Diagram.                                                                                                                                                                            |                                                    |

| UNIT-V                                                                                                                                                                                                                                                                                                                             | DESIGN FOR ASSEMBLY |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Design for assembly: objectives, link to design, General design rules for manual assembly, Development of systematic design for assembly methodology, Assembly efficiency, Effect of part symmetry, part thickness, size, weight on handling time, Effect of chamfer design on insertion operations, Estimation of insertion time. |                     |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                 |                     |
| 1. A K Chitale and R C Gupta, Product Design and Manufacturing, PHI, New Delhi, 2003<br>2. Geoffrey Boothroyd, Assembly Automation and Product Design, Marcel Dekker Inc., NY, 1992.                                                                                                                                               |                     |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                            |                     |
| 1. Engineering Design – Material & Processing Approach – George E. Deiter, McGraw Hill Intl. 2 <sup>nd</sup> Edition, 2000.<br>2. Geoffrey Boothroyd, Hand Book of Product Design, Marcel and Dekken, N.Y. 1990.                                                                                                                   |                     |

## COMPUTATIONAL FLUID DYNAMICS

### III B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME59      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

The primary focus of this course is to differentiate between different types of partial differential equations, to apply appropriate numerical techniques and solve heat transfer and fluid flow problems using different numerical techniques.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate modes of heat transfer, methods to solve a physical problem, differentiate analytical and numerical methods.
2. Differentiate between different types of Partial Differential Equations and to apply appropriate numerical techniques
3. Solve 1-Dimensional and 2-Dimensional steady state heat conduction problems
4. Explain explicit, implicit, semi-implicit methods and application of FDM to solve 1D transient heat conduction equations
5. Formulate and solve governing equations for incompressible flow.

| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | INTRODUCTION |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Review of Governing Equations in Heat Transfer and Fluid Flow: Conservation Laws – Differential Form of Equations – Characteristics of Governing Equations - Solution Methods: Analytical, Experimental and Numerical Methods – Review of Boundary Conditions. Relative advantages and disadvantages of experimental, analytical and numerical methods–Scope of CFD–Its applications and limitations–Brief Comparison between different numerical methods, viz., FDM, FEM & FVM Methods |              |

| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                | CLASSIFICATION OF PDE & FDM |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Classification of PDE – Elliptic, parabolic and hyperbolic PDE as governing equations – Examples and their physical significance<br>FDM – Discretization of Partial Derivative Terms using Taylor’s series of approximation –Finite Difference Formulae – Application and implementation aspects of finite-difference equations – Consistency Application of FDM to elliptic equations |                             |

| UNIT-III                                                                                                                                                                                                                                                                   | 1-D & 2-D STEADY STATE HEAT CONDUCTION |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Application of FDM to solve 1D steady state heat conduction in Curvilinear geometry–Singularities–Treatment of singularities. Application of FDM to solve 2D steady state heat conduction– with and without heat generation and subjected to different boundary conditions |                                        |

| UNIT-IV                                                                                                                                                                                                                   | PARABOLIC EQUATIONS & 1-D TRANSIENT HEAT CONDUCTION |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Parabolic Equations – Use of Explicit, implicit and semi implicit methods – Errors and Stability analysis -application of FDM to solve 1D transient heat conduction equations – ADI Scheme – Treatment and Implementation |                                                     |

| UNIT-V | NUMERICAL METHODS FOR INCOMPRESSIBLE FLOW |
|--------|-------------------------------------------|
|--------|-------------------------------------------|

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Numerical methods for incompressible flow – Governing equations –Difficulties in solving N-S equations–Stream function and Vorticity method –Advantages and disadvantages –treatment of boundary conditions– Determination of Pressure for viscous flows – Disadvantages – Staggered Grid–SIMPLE algorithm for pressure liked equations.

**Text Books:**

1. K. Muralidhar, T. Sundarajan, Computational Fluid Flow and Heat Transfer, Narosa Publications, 2009.
2. H.K.Versteeg, W. Malalasekhara, An Introduction to Computational Fluid Dynamics – The Finite Volume Method, PHI, 2<sup>nd</sup> Edition, 2007.
3. S.V. Patankar, Numerical heat transfer and fluid flow, CRC Press Inc, 2017.

**Reference Books:**

1. John.D,Anderson, Computational Fluid Dynamics basics with applications, McGraw Hill Education, 2017.
2. Joel H. Ferziger, Milovan Peric, Computational Methods for Fluid Dynamics, Springer, 2002.
3. T.J. Chung, Computational Fluid Dynamics, Cambridge University Press, 2<sup>nd</sup> Edition, 2010.

## **IV B.TECH. I SEMESTER SYLLABUS**



## COMPOSITE MATERIALS

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |     |
|-------------|----------|--------------|---|---|---------|---------------|-----|-----|
|             |          | L            | T | P |         | C             | CIE | SEE |
| A6ME41      | PCC      | 3            | 0 | 0 | 3       | 40            | 60  | 100 |

**COURSE OVERVIEW:**

This course is designed to provide a basic and important knowledge about the advanced class of materials, i.e. composites where its definitions, classification, fabrication, design and testing of composite materials are discussed. In the end, the current state-of-the-art research work happening in the field of multi-functional polymer nanocomposites, its preparation and applications are discussed.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Explain and differentiate composite materials based on specific applications.
2. Describe the various fabrication methods of polymer composites.
3. Choose suitable manufacturing route for generating metal and ceramic matrix composites.
4. Demonstrate various methods to test the Composites through NDT methods.
5. Elucidate the purpose and significance of Multi-Functional Composites.

|               |                                                 |
|---------------|-------------------------------------------------|
| <b>UNIT-I</b> | <b>INTRODUCTION AND REINFORCEMENT MATERIALS</b> |
|---------------|-------------------------------------------------|

**Introduction:** Definition, functions of matrix and reinforcement, composites merits and drawbacks, classification, polymer matrix composites, metal matrix composites, ceramic matrix composites, carbon-carbon composites, fibre reinforced composites and natural composites.

**Reinforcement Materials:** Glass, Carbon, Kevlar and Boron fibres.

|                |                                                  |
|----------------|--------------------------------------------------|
| <b>UNIT-II</b> | <b>POLYMER COMPOSITES: MANUFACTURING METHODS</b> |
|----------------|--------------------------------------------------|

**Thermoset Composites Manufacturing:** Hand layup, Spray up processes, Resin Transfer Moulding (RTM), Filament winding, Pultrusion, Prepreg Technique, Automated Fibre Placement (AFP) and Automated Tape Laying (ATL).

**Thermoplastic Composites Manufacturing:** Thermoplastic tape winding, Hot press technique, Diaphragm forming process, Injection moulding.

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| <b>UNIT-III</b> | <b>METAL AND CERAMIC COMPOSITES: MANUFACTURING METHODS</b> |
|-----------------|------------------------------------------------------------|

**Metal Matrix Composites:** Solid state methods - hot isostatic pressing (HIP), Foil diffusion bonding. Liquid state methods - Stir casting, Squeeze casting, Centrifugal casting, Melt infiltration.

**Ceramic Matrix Composites:** Sintering and CVD.

**Manufacturing Process Selection Criteria:** Production Rate/Speed, Cost, Performance, Size, Shape.

|                |                                  |
|----------------|----------------------------------|
| <b>UNIT-IV</b> | <b>COMPOSITES DESIGN AND NDT</b> |
|----------------|----------------------------------|

**Composites Design:** Laminate theory, Rule of mixtures, symmetry and balance.

**Non-Destructive Testing (NDT) of Composites:** Visual inspection, Tap testing, Ultrasonic inspection, X-ray inspection, Thermography.

|               |                                                |
|---------------|------------------------------------------------|
| <b>UNIT-V</b> | <b>MULTI-FUNCTIONAL POLYMER NANOCOMPOSITES</b> |
|---------------|------------------------------------------------|

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**Carbon Nanotubes (CNTs) based Multi-Functional Polymer Nanocomposites:** Introduction and necessity of multi-functional composites, CNTs based multi-functional composites fabrication methods: Ultrasonication, In-Situ method, Grafting CNTs onto fibre surfaces - spraying method, and chemical grafting. Properties and Applications of CNTs based multi-functional composites.

**Graphene based Multi-Functional Polymer Nanocomposites:** Introduction, properties and applications.

**Text Books:**

1. Sanjay K. Mazumdar, Composites Manufacturing: Materials, Product, and Process Engineering, CRC press, 2001, ISBN: 0-8493-0585-3.
2. M. Balasubramanian, Composite materials and processing, CRC press, 2013, ISBN: 9781439880548.
3. Madhujit Mukhopadhyay, Mechanics of Composite Materials, Universities Press, 2005, ISBN: 9788173714771.
4. Ashutosh Tiwari and S. K. Shukla, Advanced Carbon Materials and Technology, Wiley, 2014, ISBN: 9781118686232.

**Reference Books:**

1. Flake Campbell Jr, Manufacturing processes for advanced composites, Elsevier, 2003, ISBN: 9781856174152.
2. Ehsan Bafekrpour, Advanced Composite Materials: Properties and Applications, De Gruyter Open Poland, 2017, ISBN: 9783110574401.

## ENGINEERING METROLOGY & INSTRUMENTATION

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME43      | PCC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This course is introduced to impart knowledge of Engineering Metrology and its principle, concepts of limits, fits and tolerances. This course will focus on the basic characteristics of a typical instrument, identification of errors and their types that would occur in an instrument and the concept of transducer, various types and their character.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Identify techniques to minimize the errors in measurement.
2. Demonstrate methods and devices for measurement of length, angle, gear & amp; thread parameters, surface roughness and geometric features of parts.
3. Explain constructional details and working principles of various instruments and their purpose.
4. Demonstrate the instruments used for temperature, pressure and level.
5. Explain measurement of flow, speed, acceleration and vibration measuring instruments

| UNIT-I                                                                                                                                                                                                                                                                                                        | LINEAR AND ANGULAR MEASUREMENTS |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system. Interchangeability and selective assembly. Limit Gauges: Taylor’s principle, Design of GO and NO-GO gauges Measurement of angles, Bevel protractor, and Sine bar. Measurements by using auto Collimator. |                                 |

| UNIT-II                                                                                                                                                                                                                                                                       | SURFACE ROUGHNESS MEASUREMENT |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Roughness, Waviness. CLA, RMS, Rz Values. Methods of measurement of surface finish, Talysurf. Screw thread measurement, Gear measurement; Machine Tool Alignment Tests on lathe, Milling and Drilling machines. Coordinate Measuring Machines: Types and Applications of CMM. |                               |

| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | INTRODUCTION & MEASUREMENT OF DISPLACEMENT |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Introduction:</b> Definition – Basic principles of measurement – Measurement systems, generalized configuration and functional description of measuring instruments – examples. Static and Dynamic performance characteristics – sources of errors, Classification and elimination of errors.<br><b>Measurement of Displacement:</b> Theory and construction of various transducers to measure displacement – Piezo electric, Inductive, capacitance, resistance, ionization and Photo electric Transducers, Calibration procedures. |                                            |

| UNIT-IV                                                                                                                                                                                                                                                                                                                                                                                                               | MEASUREMENT OF TEMPERATURE , PRESSURE & LEVEL |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Measurement of Temperature:</b> Various Principles of Measurement-Classification: Expansion Type: Bimetallic Strip- Liquid in glass Thermometer; Electrical Resistance Type: Thermistor, Thermocouple, RTD.<br><b>Measurement of Pressure:</b> Different principles used – Classification: Manometers, Dead weight pressure gauge, Tester (Piston gauge), Bourdon pressure gauges; Bellows – Diaphragm gauges. Low |                                               |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| pressure measurement – Thermal conductivity gauges, ionization pressure gauges, McLeod pressure gauge.<br><b>Measurement of Level:</b> Direct methods – Indirect methods – Capacitive, Radioactive, Ultrasonic, Magnetic level indicators – Bubbler level indicators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                 |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>MEASUREMENT OF FLOW, SPEED, ACCELERATION &amp; VIBRATION</b> |
| <b>Flow measurement:</b> Rotameter, magnetic, Ultrasonic, Turbine flow meter,<br><b>Measurement of Speed:</b> Mechanical Tachometers, Electrical tachometers, Non-contact type of tachometers<br><b>Measurement of Acceleration and Vibration:</b> Different simple instruments – Principles of Seismic instruments – Vibrometer and accelerometer using this principle – Piezo electric accelerometer.                                                                                                                                                                                                                                                                                                                                 |                                                                 |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                 |
| <ol style="list-style-type: none"> <li>1. M. Mahajan, A Text Book of Metrology, 1<sup>st</sup> Edition, Dhanpat Rai &amp; Co. [P] Ltd., 2012.</li> <li>2. D.S. Kumar, Mechanical Measurements and Controls, 4<sup>th</sup> Edition, Metropolitan Book Co. Pvt. Ltd. New Delhi, India, 2011.</li> <li>3. K. Tayal, Instrumentation and mechanical Measurements, 2<sup>nd</sup> Edition, Galgoe Publications, New Delhi, India, 2004.</li> </ol>                                                                                                                                                                                                                                                                                          |                                                                 |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                 |
| <ol style="list-style-type: none"> <li>1. Connie Dotson, Fundamentals of Dimensional Metrology, 6<sup>th</sup> Edition, Thomson Delmar, 2015.</li> <li>2. Anand K Bewoor, Vinay kulakarni, Metrology &amp; Measurement, Tata McGraw Hill, New Delhi, 2009.</li> <li>3. R. K. Jain, Mechanical and Industrial Measurements, 12<sup>th</sup> Edition, Khanna Publishers, New Delhi, India, 2011.</li> <li>4. Chennakesava R. Alavala, Principles of Industrial Instrumentation and Control Systems, 1<sup>st</sup> Edition, Cengage Learning, New Delhi, India, 2010.</li> <li>5. B. C. Nakra, K. K. Choudhary, Instrumentation, Measurement and Analysis, 4<sup>th</sup> Edition, Tata McGraw-Hill Education Pvt. Ltd., 2016.</li> </ol> |                                                                 |

## COMPOSITE MATERIALS LAB

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|---|---|---------|---------------|-----|-------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          | L            | T | P |         | CIE           | SEE | Total |
| A6ME42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PCC      | 0            | 0 | 2 | 1       | 40            | 60  | 100   |
| <p><b>COURSE OVERVIEW:</b></p> <p>Composite Material Lab is intended to teach the basics of composites manufacturing including polymer matrix composites and metal matrix composites. Experiments are framed to guide the learning process and impart technical knowledge in selecting the matrix and reinforcement material for fabricating the composites.</p> <p><b>COURSE OUTCOMES:</b></p> <p><b>At the end of the course, the student should be able to:</b></p> <ol style="list-style-type: none"> <li>1. Fabricate the metal and polymer matrix composites.</li> <li>2. Determine the mechanical properties of laminated composites.</li> <li>3. Determine the voids and flaws that may present in the composite materials.</li> <li>4. Prepare the metal matrix composites using stir casting machine.</li> <li>5. Find the mechanical properties of MMCs.</li> </ol>                                                |          |              |   |   |         |               |     |       |
| <p><b>LIST OF EXPERIMENTS:</b></p> <ol style="list-style-type: none"> <li>1. Fabrication of a laminated composite using hand lay-up process.</li> <li>2. Study on the flexural properties of laminated composites using three point bend fixture.</li> <li>3. Determining the Barcol Hardness of Laminated Composites.</li> <li>4. Detecting the Voids and Flaws in Laminated Composites using Ultrasonic Flaw Detector.</li> <li>5. Preparation of Aluminium Metal Matrix composites using Stir Casting Machine.</li> <li>6. Determining the tensile properties of MMCs using UTM.</li> <li>7. Determining the hardness of MMCs using Rockwell Hardness Tester.</li> <li>8. Study on the flexural properties of MMCs using three point bend fixture.</li> <li>9. Study the Impact properties of MMCs using impact test machine.</li> <li>10. Detecting the Voids and Flaws in MMCs using Ultrasonic Flaw Detector</li> </ol> |          |              |   |   |         |               |     |       |

## ENGINEERING METROLOGY & INSTRUMENTATION LAB

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A6ME44      | PCC      | 0            | 0 | 2 | 1       | 40            | 60  | 100   |

**COURSE OVERVIEW:**  
This course is introduced to create awareness on various mechanical measuring instruments. This course intended to learn different types of transducers.

**COURSE OUTCOMES:**  
**At the end of the course, the student should be able to:**

1. Perform the measurements Length, Depth, Diameter measuring using Vernier calipers & micrometer.
2. Perform Bore measurement using bore gauge.
3. Use the gear teeth caliper for checking the chordal addendum and chordal height of spur gear.
4. Ability to calibrate and analyze the measuring system for the measurement of displacement and pressure.
5. Ability to calibrate and analyze the measuring system for the measurement of speed.

**LIST OF EXPERIMENTS:**

1. Measurement of outside & inside diameter using vernier callipers.
2. Measurement of outside & inside diameter using micrometer.
3. Height measurement by Vernier height gauge.
4. Angle measurement by using Bevel protractor.
5. Angle measurement by using Sine bar & Slip Gauges.
6. Calibration of Pressure Gauges.
7. Study and calibration of LVDT transducer for displacement measurement.
8. Calibration of capacitive transducer for angular displacement.
9. Study and calibration of photo and magnetic speed pickups for the measurement of speed.
10. Study and calibration of a rotometer for flow measurement.

## RESEARCH PROJECT PHASE - I

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME45      | PWC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 6 | 3       | 100           | -   | 100   |

#### Evaluation Procedures

The UG research project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Phase – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his research project work.

UG research project work shall be carried out in two phases: Research Project Phase – I for approval of project before Mid-II examinations in IV Year I Semester and Research Project Phase – II during IV Year II Semester. Student has to submit research project work report at the end of IV Year II Semester. The research project shall be evaluated for 100 marks before commencement of SEE Theory examinations.

For Research Project Phase – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Research Project Phase – I or does not make a presentation of the same before the evaluation committee as per schedule. A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such “one reappearance” evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

For Research Project Phase – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/ Internship/SDC etc. and the research Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

#### COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Design engineering solutions to complex problems utilising a systems approach
3. Conduct an experiments in the engineering project and analysis the data results
4. Communicate with engineers and the community at large in written an oral forms.
5. Demonstrate the knowledge, skills and attitudes of a professional engineer.

**PROFESSIONAL ELECTIVE – III**



## PROFESSIONAL ELECTIVE - III

### AUTOMATION IN MANUFACTURING

#### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME61      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course is useful to impart knowledge of different types of automation and production system used in industries. Understand the basic concept of the automated material handling system AGV, AS/RS and also to learn concepts of Group technology, Cellular manufacturing, Automated assembly and inspection.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Recognize the steps involved in the process of automation and different types of automations systems used by industry.
2. Get exposure on workstation, Worker-and-machine combinations, and work transport system used in an automated industry.
3. Understand the various types of automated material handling equipment used for transport and storage in a factory.
4. Know the design, layout and benefits of Cellular & Flexible manufacturing systems.
5. Apply quantitative analysis to assembly and also can know the facts about Lean manufacturing and JIT.

#### UNIT I

#### INTRODUCTION TO AUTOMATION

**Introduction to Automation:** Automation in Production Systems-Automated Manufacturing Systems, Computerized Manufacturing Support Systems, Reasons for Automation, Automation Principles and Strategies.

**Automation in Manufacturing:** Manufacturing operations, Production Concepts and Mathematical Models. Costs of Manufacturing Operations, Basic Elements of an Automated Systems, Advanced Automation Functions, Levels of automation.

#### UNIT II

#### AUTOMATED FLOW LINES & LINE BALANCING

**Automated Flow Lines:** General terminology, Configuration of automated flow lines, Methods of work-part transfer, Analysis of Transfer Lines with no Internal Storage, Buffer Storage, Reasons for using storage buffers.

**Line Balancing:** Line balancing methods, ways of improving line balance.

#### UNIT III

#### AUTOMATED MATERIAL HANDLING & STORAGE SYSTEMS

**Automated Material Handling Systems:** Overview of Material Handling Equipment, the 10 Principles of Material Handling, Material Transport Systems, Automated Guided Vehicle Systems, Conveyor Systems, Analysis of Material Transport Systems.

**Material Storage Systems:** Introduction to Storage Systems, Storage System Performance, Storage Location Strategies, Automated Storage Systems, Engineering Analysis of Storage Systems.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>UNIT IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CELLULAR MANUFACTURING &amp; FLEXIBLE MANUFACTURING SYSTEMS</b> |
| <p><b>Cellular manufacturing:</b> Objectives, Composite part concept, Machine cell design, Quantitative analysis-grouping parts and machines by rank order clustering.</p> <p><b>Flexible Manufacturing Systems: introduction to FMS,</b> FMC/FMS Components, FMS Application Considerations, Analysis of Flexible Manufacturing Systems.</p>                                                                                                                         |                                                                    |
| <b>UNIT V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>AUTOMATED ASSEMBLY SYSTEMS</b>                                  |
| <p><b>Automated Assembly Systems:</b> Fundamentals of Automated Assembly Systems, Design for Automated Assembly, and Quantitative Analysis of Assembly Systems.</p> <p><b>Just-In-Time and Lean Production:</b> Lean production and waste management, Just in time (JIT) production systems, worker involvement.</p>                                                                                                                                                  |                                                                    |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                    |
| <ol style="list-style-type: none"> <li>1. Automation, Production Systems and Computer Integrated Manufacturing, M.P. Groover, Pearson Education.</li> <li>2. Industrial Automation: W.P. David, John Wiley and Sons.</li> </ol>                                                                                                                                                                                                                                       |                                                                    |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                    |
| <ol style="list-style-type: none"> <li>1. Handbook of Design, Manufacturing &amp; Automation: R.C. Dorf, John Wiley and Sons.</li> <li>2. An Introduction to Automated Process Planning Systems, Tiess Chiu Chang &amp; Richard A. Wysk</li> <li>3. Performance Modeling of Automated Manufacturing Systems, Viswanandham, PHI</li> <li>4. CAD CAM: Principles, Practice and Manufacturing Management / Chris Mc Mohan, Jimmie Browne / Pearson edu. (LPE)</li> </ol> |                                                                    |

## UNCONVENTIONAL MACHINING PROCESSES

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME62      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

The objective of this course is to introduce the student on advanced topics in the machining processes. The concept of material removal by an edged tool, involving plastic deformation and formation of chips, has been known to man for several hundred years.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Demonstrate knowledge about the modern manufacturing processes and their applications in various industries.
2. Explain the need for unconventional machining processes in comparison with conventional manufacturing processes.
3. Classify of unconventional machining processes
4. Describe the principle, mechanism of metal removal rate of various unconventional machining processes.
5. Identify the selection of processes

#### UNIT I

#### INTRODUCTION

Need for non-traditional machining methods – Classification of modern machining processes – considerations in process selection. Materials. Applications.

Ultrasonic machining – Elements of the process, mechanics of metal removal process parameters, economic considerations, applications and limitations, recent development.

#### UNIT II

#### ABRASIVE JET MACHINING

**ABRASIVE JET MACHINING, WATER JET MACHINING AND ABRASIVE WATER JET MACHINE:** Basic principles, equipments, process variables, mechanics of metal removal, MRR, application and limitations.

**ELECTRO-CHEMICAL PROCESSES:** Fundamentals of electro chemical machining, electrochemical grinding, electro chemical honing and de-burring process, metal removal rate in ECM, Tool design, Surface finish and accuracy economic aspects of ECM –Fundamentals of chemical, machining, advantages and applications.

#### UNIT III

#### THERMAL METAL REMOVAL PROCESSES

**THERMAL METAL REMOVAL PROCESSES:** General Principle and applications of Electric Discharge Machining, Electric Discharge Grinding and electric discharge wire cutting processes – Power circuits for EDM, Mechanics of metal removal in EDM, Process parameters, selection of tool electrode and dielectric fluids, methods surface finish and machining accuracy. Wire EDM, principle, applications.

#### UNIT IV

#### ELECTRON BEAM MACHINING

Generation and control of electron beam for machining, theory of electron beam machining, comparison of thermal and non-thermal processes –General Principle and application of laser beam

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| machining – thermal features, cutting speed and accuracy of cut.                                                                                                                                                                                                                                                                                           |                                                |
| <b>UNIT V</b>                                                                                                                                                                                                                                                                                                                                              | <b>PLASMA MACHINING AND CHEMICAL MACHINING</b> |
| Application of plasma for machining, metal removing mechanism, process parameters, accuracy and surface finish and other applications of plasma in manufacturing industries.<br>Chemical machining - principle-maskants-applications.<br>Magnetic abrasive finishing, Abrasive flow finishing, Electro stream drilling, shaped tube electrolyte machining. |                                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                         |                                                |
| 1. VK Jain, Advanced Machining Processes, Allied publishers, 2009.<br>2. P. C. Pandey, H. S. Shan, Modern Machining Processes, McGraw Hill Education, 2017.                                                                                                                                                                                                |                                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                    |                                                |
| 1. Serope Kalpak jain, Manufacturing Engineering and Technology, Pearson Publications, 2001.<br>2. Kalpak jain, Steven R. Schmid, Manufacturing Engineering & Technology, Pearson Publications, 2018.<br>3. Singh M.K., Unconventional Manufacturing Processes, New Age Publisher, 2007.                                                                   |                                                |

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## ADDITIVE MANUFACTURING

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME63      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course covers the principles and applications of additive manufacturing (AM) in engineering domains. It explores fundamental aspects of AM, including reverse engineering, data formats, slicing algorithms, and process-path generation. The curriculum delves into various AM technologies. Students learn about material selection, modelling and simulation of AM processes, and quality and reliability considerations like defects, accuracy, and surface finish.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Explain the basic concepts and principles of additive manufacturing and its advantages and limitations over conventional manufacturing methods.
2. Compare and contrast different additive manufacturing technologies and select the appropriate one for a given engineering problem or application.
3. Choose suitable materials for additive manufacturing processes and understand their properties and behaviour during and after fabrication.
4. Model and simulate additive manufacturing processes and analyse their thermal, mechanical, and metallurgical aspects.
5. Evaluate the quality and reliability of additive manufacturing products and identify the sources and effects of defects, errors, and variations.

#### UNIT I

#### INTRODUCTION TO ADDITIVE MANUFACTURING

History and evolution of additive manufacturing, applications and challenges of additive manufacturing in various engineering domains, additive manufacturing terminology and concepts, additive manufacturing process chain, classification of additive manufacturing processes, common additive manufacturing technologies and their capabilities, materials, costs, and limitations.

#### UNIT II

#### ADDITIVE MANUFACTURING TECHNOLOGIES

Detailed description and working principle of various additive manufacturing technologies, such as fused deposition modelling, selective laser sintering, stereolithography, direct metal laser sintering, jetting, 3D printing, laminated object manufacturing, and electron beam melting, process parameters and variables, advantages and disadvantages, applications, and examples of each technology.

#### UNIT III

#### MATERIALS FOR ADDITIVE MANUFACTURING

Types and properties of materials used for additive manufacturing, such as polymers, metals, ceramics, and composites; mechanisms of material consolidation and bonding, such as coalescence, sintering, photopolymerization, and solidification; effects of process parameters and variables on material properties and behaviour, such as temperature, composition, phase transformation, and cooling rate; post-processing and finishing techniques, such as heat treatment, machining, and polishing, material testing and characterization methods, such as tensile, hardness, and microstructure tests.

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| <b>UNIT IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>SOFTWARE TOOLS AND PLATFORMS FOR ADDITIVE MANUFACTURING</b>             |
| 3D scanning and digitization methods and devices, data handling and reduction techniques, additive manufacturing software tools and platforms, such as SolidWorks, Cura, and Repetier, data formats and standardisation, such as STL, AMF, and 3MF, slicing algorithms and methods, such as uniform flat layer slicing and adaptive slicing, process-path generation algorithms and methods, such as rasterization and contouring, part orientation and support generation techniques and criteria, such as overhang angle and surface quality.                                                                                                                                                      |                                                                            |
| <b>UNIT V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>MODELING, SIMULATION, AND QUALITY ASPECTS OF ADDITIVE MANUFACTURING</b> |
| Modelling and simulation of additive manufacturing processes, such as heat transfer, phase change, and residual stress, using various methods and tools, such as finite element method, finite difference method, and ANSYS; analysis and optimisation of process parameters and variables, such as temperature, speed, and layer thickness; quality and reliability aspects of additive manufacturing products, such as defects, accuracy, and surface finish; sources and effects of defects, errors, and variations, such as thermal distortion, porosity, and warping; evaluation and measurement methods and tools, such as dimensional analysis, surface profilometry, and optical microscopy. |                                                                            |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                            |
| <ol style="list-style-type: none"> <li>1. Gibson, I., Rosen, D.W. and Stucker, B., Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing, Springer, New York, 2015.</li> <li>2. Chua, C.K., Leong, K.F. and Lim, C.S., Rapid Prototyping: Principles and Applications, World Scientific, Singapore, 2010.</li> <li>3. Gebhardt, A., Understanding Additive Manufacturing: Rapid Prototyping, Rapid Tooling, Rapid Manufacturing, Hanser, Munich, 2012.</li> <li>4. Hopkinson, N., Hague, R. and Dickens, P., Rapid Manufacturing: An Industrial Revolution for the Digital Age, Wiley, Chichester, 2006.</li> </ol>                                  |                                                                            |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                            |
| <ol style="list-style-type: none"> <li>1. Killi, S.W., Additive Manufacturing: Design, Methods, and Processes, Jenny Stanford Publishing, New York, 2017.</li> <li>2. Yang, L., Hsu, K., Baughman, B., Godfrey, D., Medina, F., Menon, M., and Wiener, S., Additive Manufacturing of Metals: The Technology, Materials, Design and Production, Springer, Cham, 2017.</li> </ol>                                                                                                                                                                                                                                                                                                                      |                                                                            |

## PRODUCTION PLANNING AND CONTROL

### IV B. Tech. I Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |     |
|-------------|----------|--------------|---|---|---------|---------------|-----|-----|
|             |          | L            | T | P |         | C             | CIE | SEE |
| A6ME64      | PEC      | 3            | 0 | 0 | 3       | 40            | 60  | 100 |

**COURSE OVERVIEW:**

Production planning is concerned with (i) deciding which products to make, how many of each and when should be completed; (ii) Scheduling the delivery and/or production of the parts and (iii) Planning the man power and equipment resources needed to accomplish the production plan. Production Control is concerned with determining whether the necessary resources to implement the production plan have been provided and if not, it attempts to take corrective action to address the deficiencies.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Demonstrate basic concepts of production planning and control and forecasting methods.
2. Explain inventory management.
3. Explain routing procedure and bill of materials.
4. Discuss scheduling and scheduling policies.
5. Describe dispatching procedure and activities of a dispatcher

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| <b>UNIT I</b> | <b>INTRODUCTION</b> |
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**INTRODUCTION:** Definition – Objectives and functions of production planning and control – Elements of production control – Types of production – Organization of production planning and control department – Internal organization of department.

Forecasting – Importance of forecasting – Types of forecasting, their uses – General principles of forecasting – Forecasting techniques – qualitative methods and quantitative methods.

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| <b>UNIT II</b> | <b>INVENTORY MANAGEMENT</b> |
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Inventory management – Functions of inventories – relevant inventory costs – ABC analysis – VED analysis– EOQ model –Inventory control systems– P–Systems and Q-Systems–Introduction to MRP I, MRP II, ERP, LOB (Line of Balance), JIT and KANBAN system.

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| <b>UNIT III</b> | <b>ROUTING</b> |
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Routing – definition – routing procedure –route sheets – bill of material – factors affecting routing procedure, schedule –definition – difference with loading

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| <b>UNIT IV</b> | <b>SCHEDULING POLICIES</b> |
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Scheduling policies – techniques, standard scheduling methods.

Line Balancing, aggregate planning, chase planning, expediting, controlling aspects.

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| <b>UNIT V</b> | <b>DISPATCHING</b> |
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Dispatching–Activities of dispatcher–dispatching procedure–follow up–definition–Reason for existence of functions–types of follow up, applications of computer in production planning and control.

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**Text Books:**

1. Samuel Eilon, Elements of Production Planning and Control, MacMillan, 2005.
2. Partik Jonsson Stig-Arne Mattsson, Manufacturing, Planning and Control, Tata Mc Graw Hill, 2009.

**Reference Books:**

1. Fundamentals of Production Planning and Control, Chapman, Pearson Education, 2008.
2. Martin K. Starr and David W. Miller, Inventory Control Theory and Practice, Englewood Cliffs, N.J., Prentice-Hall, 1962..
3. Mukhopadhyay, Production Planning and Control, PHI Third Edition, 2015.



**PROFESSIONAL ELECTIVE – IV**

## PROFESSIONAL ELECTIVE - IV

### AUTOMOBILE ENGINEERING

| IV B. Tech. I Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                    |              |   |   |         |               |     |       |
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| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Category                                           | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME66                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | PEC                                                | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                    | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>The curriculum of the automobile engineering course focuses on imparting knowledge about types of automobiles, layout, fuel supply system in CI and SI engines, cooling and ignition systems, transmission systems along with braking systems, steering and suspension systems.                                                                                                                                                                                                                                                               |                                                    |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Explain the basic lay-out of automobile, types of automobile engines, basic conceptslike engine construction, oil filters, pumps etc.<br>2. Discuss various fuels supply systems of S.I and C.I engines<br>3. Explain the operation of engine cooling, lubrication and ignition systems.<br>4. Demonstrate the principles, elements and working of transmission and braking systems<br>5. Explain the elements and working of steering and suspension systems.          |                                                    |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>INTRODUCTION</b>                                |              |   |   |         |               |     |       |
| Types of Automobiles, Layout of four wheeler automobile, chassis and body, power unit , power transmission -types, engine service, engine lubrication,splash and pressure lubrication systems, oil filters, oil pumps , crank case ventilation, reboring, decarburization, Nitriding of crank shaft.                                                                                                                                                                                                                                                                     |                                                    |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>FUEL SUPPLY SYSTEMS IN S.I AND C.I. ENGINES</b> |              |   |   |         |               |     |       |
| <b>FUEL SUPPLY SYSTEM:</b> S.I. Engine: Fuel supply systems, Mechanical and electrical fuel pump, filters,carburetor, types, air filters, GDI.<br><b>C.I. ENGINES:</b> Requirements of diesel injection systems, types of injection systems, fuel pump, nozzle, spray formation, injection timing, testing of fuel pumps.                                                                                                                                                                                                                                                |                                                    |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>COOLING SYSTEM AND IGNITION SYSTEM</b>          |              |   |   |         |               |     |       |
| <b>COOLING SYSTEM:</b> Cooling Requirements, Air Cooling, Liquid Cooling, Thermosyphon and Forced Circulation System, Radiators, Types, Cooling Fan - water pump, thermostat, antifreeze solutions.<br><b>IGNITION SYSTEM:</b> Function of an ignition system, battery ignition system, constructional features of storage, battery, auto transformer, contact breaker points, condenser and spark plug, Magneto coil ignition system, electronic ignition system using contact breaker, electronic ignition using contact triggers, spark advance and retard mechanism. |                                                    |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>TRANSMISSION SYSTEM AND BRAKING SYSTEM</b>      |              |   |   |         |               |     |       |

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| <p><b>TRANSMISSION SYSTEM:</b> Clutches, principle, types, cone clutch, single plate clutch, multi plate clutch, magnetic and centrifugal clutches, fluid fly wheel. Gear boxes, types, sliding mesh, constant mesh, synchro mesh gear boxes, epicyclic gear box, over drive torque converter. Universal joint, Propeller shaft with sliding joint, Hotch Kiss drive, Torque tube drive, universal joint, differential rear axles, wheels and tires.</p> <p><b>BRAKING SYSTEM:</b> Mechanical brake system, Hydraulic brake system, Master cylinder, wheel cylinder, tandem master cylinder, Requirement of brake fluid, Pneumatic and vacuum brakes</p> |                                                |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>STEERING SYSTEM &amp; SUSPENSION SYSTEM</b> |
| <p><b>STEERING SYSTEM:</b> steering linkages, Steering geometry, camber, caster, king pin rake, combined angle toe in, center point steering. Types of steering mechanism, Ackerman steering mechanism, Davis steering mechanism, steering gears, types.</p> <p><b>SUSPENSION SYSTEM:</b> Objectives of suspension systems, rigid axle suspension system, torsion bar, shock absorber, independent suspension system.</p>                                                                                                                                                                                                                                |                                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                |
| <ol style="list-style-type: none"> <li>1. Kirpal Singh, Automobile Engineering - Vol. 1 &amp;2, 12<sup>th</sup> Edition, Standard Publishers, New Delhi, India, 2012.</li> <li>2. William Crouse, Automobile Engineering (SIE), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2012.</li> </ol>                                                                                                                                                                                                                                                                                                                                     |                                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                |
| <ol style="list-style-type: none"> <li>1. B.S. Narang, Automobile Engineering, 5<sup>th</sup> Edition, Karman publishers, New Delhi, India, 2011.</li> <li>2. R. B. Gupta, Automobile Engineering, Satya Prakashan, New Delhi, India, 2012.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                   |                                                |

## POWER PLANT ENGINEERING

| IV B. Tech. I Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |   |   |         |               |     |       |
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| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME67                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | PEC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course attempts to provide the students basic knowledge of the technologies available at plant level and would also acquaint them with the latest technological advances taking place in the energy sector.                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate concepts of coal based thermal power plants.<br>2. Explain principle of working and components of diesel, gas turbine and combined cycle power plants.<br>3. Demonstrate about Hydel Power and also generation of power from renewable energy.<br>4. Explain principle of working and components of nuclear power plant<br>5. Illustrate economic and environmental aspects of power plant. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>INTRODUCTION</b><br><b>Introduction to the Sources of Energy:</b> Resources and Development of Power in India.<br><b>Steam Power Plant:</b> Plant Layout, Design and construction, Feed water treatment, Types of coals, Properties of coal, Coal storage, Coal handling, Fuel and Oil Systems, FBC Boilers, Turbines, Condensers, Working of different Circuits, Draught system, Dust collectors, Cooling towers, Ash handling systems, Safety Measures.                                    |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>DIESEL, GAS TURBINE AND COMBINED CYCLE POWER PLANTS</b><br><b>Internal Combustion Engine Plant:</b> Diesel Power Plant: Introduction, IC Engines Types, Construction, Plant layout with auxiliaries, fuel supply system, air starting equipment, lubrication and cooling system, super charging.<br><b>Gas Turbine Plant:</b> Introduction, Classification, Construction, Layout With Auxiliaries, Principles of Working of Closed and Open Cycle Gas Turbines, Combined Cycle Power Plants. |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>HYDEL POWER &amp; POWER FROM RENEWABLE ENERGY SOURCES</b><br><b>Hydro Electric Power Plant:</b> Water power, Hydrological Cycle / Flow Measurement, Drainage Area Characteristics, Hydrographs, Storage and Pondage, Classification of dams and spill ways. Typical layouts, Plant Auxiliaries, Plant Operation Pumped Storage Plants.<br><b>Renewable Energy Sources:</b> Basic Introduction About Working of Wind, Tidal, Solar Photo Voltaic (SPV), Geo Thermal and Biogas Power Systems. |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>NUCLEAR POWER PLANTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |   |   |         |               |     |       |

**Nuclear Power Station: Nuclear fuel:** Breeding and Fertile Materials, Nuclear Reactor, Reactor Operation.

**Types of Reactors:** Pressurized Water Reactor, Boiling Water Reactor, Sodium, Graphite Reactor, Fast Breeder Reactor, Homogeneous Reactor, Gas cooled Reactor, Radiation Hazards and Shielding, Radioactive Waste Disposal.

#### UNIT-V

#### ECONOMIC AND ENVIRONMENTAL ISSUES OF POWER PLANT

**Power Plant Economics and Environmental Considerations:** Capital cost, Investment of Fixed Charges, Operating Costs, General Arrangement of Power Distribution, Load curves, Load Duration Curve. Definitions of Connected Load, Maximum Demand, Demand Factor, Average Load, Load Factor, Diversity Factor.

**Pollution Control:** Effluents From Power Plants and Impact On Environment, Pollutants and Pollution Standards, Methods of Pollution Control.

#### Text Books:

1. Rajput, A Text Book of Power Plant Engineering, 4<sup>th</sup> Edition, Laxmi Publications, New Delhi, India, 2011.
2. Elanchezhian, L. Saravana Kumar, B. Vijaya Ramkanth, Power plant Engineering, 1<sup>st</sup> Edition, I.K International Publishing House, New Delhi, India, 2007.

#### Reference Books:

1. Arora and S. Domkundwar, A Course in Power Plant Engineering, 5<sup>th</sup> Edition, Dhanpat Rai & Co. Delhi, 2008.
2. P. K. Nag, Power Plant Engineering, 3<sup>rd</sup> Edition, Tata McGraw- Hill Publishing Company Ltd., New Delhi, 2008.

## REFRIGERATION AND AIR CONDITIONING

| IV B. Tech. I Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                |                                                          |              |   |   |         |               |     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Category                                                 | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME68                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PEC                                                      | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>The course is designed to give fundamental knowledge of types of refrigeration, refrigeration cycles, refrigerants and their behavior under various conditions, air conditioning load calculation and designing of components of air distribution system.                                                                                                                                                                          |                                                          |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate the basic concepts of refrigeration.<br>2. Explain the elements and working of vapour compression refrigeration system<br>3. Analyze the vapour absorption refrigeration system.<br>4. Discuss the basic concepts of air conditioning, psychrometric properties and processes.<br>5. Study the air conditioning systems and analyze their loads. |                                                          |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>INTRODUCTION TO REFRIGERATION</b>                     |              |   |   |         |               |     |       |
| Introduction: Necessity and applications, unit of refrigeration, C.O.P and Mechanical refrigeration.<br>Air Refrigeration: Air Refrigeration Cycles-reversed Carnot cycle, Bell-Coleman cycle analysis, Air Refrigeration systems-merits and demerits, analysis.                                                                                                                                                                                              |                                                          |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>VAPOUR COMPRESSION REFRIGERATION</b>                  |              |   |   |         |               |     |       |
| Working principle and essential components of the plant – simple vapour compression refrigeration cycle –COP – representation of cycle on T-S and p-h charts – effect of sub cooling and super heating– cycle analysis – actual cycle influence of various parameters on system performance – use of p-h charts – numerical problems.                                                                                                                         |                                                          |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>VAPOUR ABSORPTION SYSTEMS</b>                         |              |   |   |         |               |     |       |
| Vapour Absorption systems: Simple Vapour Absorption Refrigeration System, Absorbent Refrigerant combinations, Water-Ammonia Systems, Water Lithium Bromide System, Contrast between the two systems, Modified Version of Aqua-Ammonia System with Rectifier and Analyser Assembly. Brief description of Refrigerants, Classification, Desirable properties, Ozone Depletion and Global warming.                                                               |                                                          |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>INTRODUCTION TO AIR CONDITIONING AND PSYCHROMETRY</b> |              |   |   |         |               |     |       |
| Psychrometry – Air-water vapour mixtures, Psychrometric Properties, Psychrometric Processes, Psychrometric Charts, Problems using Psychrometric charts Human Comfort, Factors affecting Human Comfort, Effective Temperature.<br>Introduction to Air-Conditioning, Basic Definition, ASHRAE Nomenclature, Applications of Air-Conditioning.                                                                                                                   |                                                          |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>LOAD CALCULATIONS</b>                                 |              |   |   |         |               |     |       |
| Different Air-Conditioning Systems Classifications –Central –Zoned –Unitary –Winter –Summer –                                                                                                                                                                                                                                                                                                                                                                 |                                                          |              |   |   |         |               |     |       |

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| Year Round, Comfort AC, Industrial AC, Factors affecting Selection of System, Need For Ventilation, Infiltration, Load concepts of RSHF, GSHP- Problems, Cooling Load Estimate, Heating Load Estimate- Problems. |
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| <b>Text Books:</b> |
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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>1. Arora, C.P., Refrigeration and Air Conditioning, 3<sup>rd</sup> Edition, McGraw Hill, New Delhi, 2010.</li><li>2. R.S. Khurmi &amp; J.K. Gupta., A textbook of Refrigeration and Air Conditioning, S Chand Publications, 2010.</li><li>3. Domkundwar Arora., A Course in Refrigeration &amp; Air Conditioning”, Dhanpat Rai &amp; Co.(P) Ltd- Delhi, 2012.</li></ol> |
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| <b>Reference Books:</b> |
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| <ol style="list-style-type: none"><li>1. Dossat R.J., Principles of refrigeration, John Wiley, S.I. Version, Texas, 2001.</li><li>2. Klaus D. Timmerhaus and Thomas M. Flynn, Cryogenic Process Engineering, Plenum Press, New York, 1989.</li></ol> |
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## ELECTRIC AND HYBRID VEHICLES

| IV B. Tech. I Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                            |              |   |   |         |               |     |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Category                                                                   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME69                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PEC                                                                        | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                            | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course introduces the fundamental concepts, principles, analysis, design, and control systems of electric and hybrid vehicles.                                                                                                                                                                                                                                                                                                                                                                             |                                                                            |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Analyze the impact of conventional vehicles on the society and different types of drive train topologies<br>2. Evaluate the performance and energy consumption in electric vehicles<br>3. Assess the configuration and architecture in hybrid vehicles<br>4. Choose the type of motor used in electric and hybrid electric vehicles<br>5. Design the drive train suitable for electric and hybrid vehicles                                |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>INTRODUCTION TO ELECTRIC AND HYBRID VEHICLES</b>                        |              |   |   |         |               |     |       |
| History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies                                                                                                                                                                                                                                                                                                                                                                             |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ELECTRIC VEHICLES</b>                                                   |              |   |   |         |               |     |       |
| Configurations of EVs, Performance of EVs, Traction Motor Characteristics, Tractive Effort and Transmission Requirement, Vehicle Performance, Tractive Effort in Normal Driving, Energy Consumption.                                                                                                                                                                                                                                                                                                                                       |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>HYBRID ELECTRIC VEHICLES</b>                                            |              |   |   |         |               |     |       |
| Concept of Hybrid Electric Drive Trains, Architectures of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains (Electrical Coupling), Parallel Hybrid Electric Drive Trains (Mechanical Coupling), Parallel Hybrid Drive Train with Torque Coupling, Parallel Hybrid Drive Train with Speed Coupling, Hybrid Drive Trains with Both Torque and Speed Coupling.                                                                                                                                                                |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ELECTRIC PROPULSION SYSTEMS</b>                                         |              |   |   |         |               |     |       |
| Introduction, DC Motor Drives, Principle of Operation and Performance, Combined Armature Voltage and Field Control, Chopper Control of DC Motors, Multi-Quadrant Control of Chopper-Fed DC Motor Drives, Two-Quadrant Control of Forward Motoring and Regenerative Braking, Four-Quadrant Operation, Induction Motor Drives, Basic Operation Principles of Induction Motors, Steady-State Performance, Constant Volt/Hertz Control, Power Electronic Control, Field Orientation Control, Permanent Magnetic BLDC Motor Drives, SRM Drives. |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>DESIGN PRINCIPLE OF SERIES AND PARALLEL HYBRID ELECTRIC DRIVE TRAIN</b> |              |   |   |         |               |     |       |
| Design Principles of Parallel (Mechanically Coupled) Hybrid Electric Drive Train: Drive Train Configuration and Design Objectives, Control Strategies, Parametric Design of a Drive Train. Design                                                                                                                                                                                                                                                                                                                                          |                                                                            |              |   |   |         |               |     |       |



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| Principles of Parallel (Mechanically Coupled) Hybrid Electric Drive Train: Drive Train Configuration and Design Objectives, Control Strategies, Parametric Design of a Drive Train. |
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| <b>Text Books:</b> |
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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004</li><li>2. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.</li></ol> |
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| <b>Reference Books:</b> |
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- |                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>1. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.</li><li>2. Allen Fuhs, Hybrid Vehicles and the future of personal transportation, CRC Press, 2011</li></ol> |
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## **IV B.TECH. II SEMESTER SYLLABUS**

## RESEARCH PROJECT PHASE - II

### IV B. Tech. II Semester: MECHANICAL ENGINEERING

| Course Code | Category | Hours / Week |   |    | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|----|---------|---------------|-----|-------|
| A6ME50      | PWC      | L            | T | P  | C       | CIE           | SEE | Total |
|             |          | 0            | 0 | 22 | 11      | 40            | 60  | 100   |

#### Evaluation Procedures

The UG research project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Phase – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his research project work.

UG research project work shall be carried out in two phases: Research Project Phase – I for approval of project before Mid-II examinations in IV Year I Semester and Research Project Phase – II during IV Year II Semester. Student has to submit research project work report at the end of IV Year II Semester. The research project shall be evaluated for 100 marks before commencement of SEE Theory examinations.

For Research Project Phase – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Research Project Phase – I or does not make a presentation of the same before the evaluation committee as per schedule. A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such “one reappearance” evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

For Research Project Phase – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/ Internship/SDC etc. and the research Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

#### COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Design engineering solutions to complex problems utilising a systems approach
3. Conduct an experiments in the engineering project and analysis the data results
4. Communicate with engineers and the community at large in written an oral forms.
5. Demonstrate the knowledge, skills and attitudes of a professional engineer.

**PROFESSIONAL ELECTIVE – V**

## PROFESSIONAL ELECTIVE - V

### INDUSTRY 4.0 AND IOT

| IV B. Tech. II Semester: MECHANICAL ENGINEERING |          |              |   |   |         |               |     |       |
|-------------------------------------------------|----------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                     | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME71                                          | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|                                                 |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

**COURSE OVERVIEW:**

This course introduces the concepts and applications of Industry 4.0 and Industrial Internet of Things (IOT) in various industrial sectors, such as manufacturing, healthcare, energy, and transportation. The course covers the fundamentals of Industry 4.0 and IOT, such as sensing, actuation, communication, networking, cloud computing, big data analytics, machine learning, cybersecurity, and smart business models. The course also exposes the students to various IOT platforms, tools, and frameworks, such as Arduino, Raspberry Pi, MQTT, CoAP, AWS IOT, and IBM Watson. The course aims to equip the students with the skills and knowledge to design, implement, and evaluate IOT solutions for industrial problems and challenges.

**COURSE OUTCOMES:**

**At the end of the course, the student should be able to:**

1. Explain the basic concepts and principles of Industry 4.0 and IOT and their relevance to industrial processes and systems.
2. Apply sensing, actuation, communication, and networking technologies to collect, transmit, and process data from industrial devices and machines.
3. Use various cloud computing, big data analytics, and machine learning techniques to store, analyze, and visualize industrial data and derive insights and decisions.
4. Implement and deploy IOT applications for industrial scenarios using various IOT platforms, tools, and frameworks.
5. Evaluate the performance, security, and scalability of IOT solutions and identify the opportunities and challenges of Industry 4.0 and IOT.

| UNIT-I                                                                                                                                                                                                                                                                                                                                                   | INTRODUCTION TO INDUSTRY 4.0 AND IOT                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| History and evolution of Industry 4.0 and IOT, applications and challenges of Industry 4.0 and IOT in various industrial sectors, IOT terminology and concepts, IOT architecture and components, IOT standards and protocols.                                                                                                                            |                                                                  |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                  | SENSING, ACTUATION, AND COMMUNICATION IN IOT                     |
| Sensing and actuation technologies and devices, such as sensors, actuators, RFID, NFC, and QR codes, communication technologies and protocols, such as Wi-Fi, Bluetooth, ZigBee, LoRa, MQTT, CoAP, and HTTP, networking technologies and protocols, such as TCP/IP, IPv6, 6LoWPAN, and RPL, data formats and serialization, such as JSON, XML, and CBOR. |                                                                  |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                 | CLOUD COMPUTING, BIG DATA ANALYTICS, AND MACHINE LEARNING IN IOT |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Cloud computing concepts and services, such as IaaS, PaaS, SaaS, and FaaS, cloud computing platforms and providers, such as AWS, Azure, and Google Cloud, big data analytics concepts and techniques, such as MapReduce, Hadoop, Spark, and Kafka, machine learning concepts and algorithms, such as regression, classification, clustering, and neural networks, machine learning frameworks and libraries, such as TensorFlow, PyTorch, and scikit-learn.                                                                                      |                                                   |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>IOT PLATFORMS, TOOLS, AND FRAMEWORKS</b>       |
| IOT platforms and providers, such as AWS IOT, Azure IOT, IBM Watson, and Google Cloud IOT, IOT tools and devices, such as Arduino, Raspberry Pi, ESP32, and NodeMCU, IOT frameworks and libraries, such as Node-RED, Johnny-Five, and Paho, IOT development and deployment, such as programming languages, IDEs, APIs, and SDKs, IOT testing and debugging, such as simulators, emulators, and loggers.                                                                                                                                          |                                                   |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IOT APPLICATIONS, SECURITY, AND CHALLENGES</b> |
| IOT applications in various industrial domains, such as smart manufacturing, smart healthcare, smart energy, and smart transportation, IOT security concepts and techniques, such as encryption, authentication, authorization, and blockchain, IOT challenges and opportunities, such as scalability, interoperability, reliability, and sustainability, IOT case studies and best practices, such as Industry 4.0, LEAN production, and smart business models.                                                                                 |                                                   |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                   |
| <ol style="list-style-type: none"> <li>1. Alasdair Gilchrist, Industry 4.0: The Industrial Internet of Things, Apress, 1<sup>st</sup> Edition, 2017.</li> <li>2. Sabina Jeschke, Christian Brecher, Houbing Song, and Danda B. Rawat, Industrial Internet of Things: Cybermanufacturing Systems, Springer, 1<sup>st</sup> Edition, 2017.</li> <li>3. G.R. Kanagachindamaresan, R. Anand, E. Balasubramanian, V. Mahima, Internet of Things for Industry 4.0: Design Challenges and Solutions, Springer, 1<sup>st</sup> Edition, 2020.</li> </ol> |                                                   |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                   |
| <ol style="list-style-type: none"> <li>1. Alp Ustundag and Emre Cevikcan, Industry 4.0: Managing The Digital Transformation, Springer, 1<sup>st</sup> Edition, 2018.</li> <li>2. Industrial Internet of Things for Engineers and Scientists by Sabrie Soloman</li> <li>3. Industry 4.0: Concepts, Technologies, and Applications by R. K. Mittal and Inderdeep Singh</li> <li>4. Industrial Internet of Things and Cyber-Physical Systems: Transforming the Conventional to Digital by Parag Kulkarni and Prachi Joshi</li> </ol>                |                                                   |
| <b>MOOC Course:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                   |
| Introduction to Industry 4.0 and Industrial Internet of Things - Course (nptel.ac.in)<br><a href="https://onlinecourses.nptel.ac.in/noc20_cs69/preview">https://onlinecourses.nptel.ac.in/noc20_cs69/preview</a>                                                                                                                                                                                                                                                                                                                                 |                                                   |

## NANO TECHNOLOGY

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            |              |   |   |         |               |     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Category                                   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME72                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PEC                                        | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>Nanotechnology is one of the core subjects of multidisciplinary nature. It finds their applications in the various domains such as medicine, health, energy, electronics, etc. This course will give insights into basic conceptual differences from macro to the nano world, size effects, unique properties of nanomaterials, their characterization and applications in the engineering sectors.                                                                                      |                                            |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate various concepts of nanotechnology and explain effect of size on different properties of nanomaterials,<br>2. Describe the properties, microstructure and defects of nanomaterials,<br>3. Explain how nanomaterials can be synthesized,<br>4. Enumerate on the characterization techniques of nanomaterials, and<br>5. Discuss the potential applications of nanomaterials in the engineering sectors. |                                            |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTRODUCTION</b>                        |              |   |   |         |               |     |       |
| <b>Introduction:</b> Definitions, Importance of Nanotechnology, nano-scale comparison, Challenges involved in nanotechnology. Classification of nanomaterials according to their origin, dimensions & structural configuration.<br><b>Size Effect:</b> Structural differences on the nanoscale, Factors influencing the properties of nanomaterials, Surface Area and Energy.                                                                                                                                       |                                            |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>UNIQUE PROPERTIES OF NANO MATERIALS</b> |              |   |   |         |               |     |       |
| <b>Microstructure and Defects in Nanocrystalline Materials:</b> Dislocations, Twins, stacking faults and voids, Grain Boundaries, triple, and disclinations,<br><b>Effect of Nano-dimensions on Materials Behavior:</b> Elastic properties, Melting Point, Diffusivity, Grain growth characteristics, enhanced solid solubility,<br><b>Other properties:</b> Magnetic, Electrical, Optical, Thermal, and Mechanical properties.                                                                                     |                                            |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>SYNTHESIS OF NANO MATERIALS</b>         |              |   |   |         |               |     |       |
| <b>Physical Methods:</b> Mechanical Methods, Methods based on Evaporation, Sputter Deposition, Chemical Vapour Deposition, Electric Arc Deposition, Ion Beam Technique, Molecular Beam Epitaxy.<br><b>Chemical Methods:</b> Synthesis of Metal nanoparticles by colloidal route, Sol-Gel Method, Hydrothermal synthesis, Sonochemical Synthesis, Microwave Synthesis                                                                                                                                                |                                            |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>NANOMATERIALS CHARACTERIZATION</b>      |              |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Nanomaterials characterization:</b> X-Ray Diffraction (XRD), Zeta potential, Microscopic techniques: Scanning electron microscopy (SEM), Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM), and Scanning Tunneling Microscopy (STM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                      |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>APPLICATIONS OF NANOMATERIALS</b> |
| Nanomaterials in medical and healthcare applications, Textiles, Paints, Energy, Defence and Space Applications, Future transportation benefits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                      |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                      |
| <ol style="list-style-type: none"> <li>1. B.S. Murthy, P. Shankar, Baldev Raj, B.B. Rath and James Munday, Text Book of Nanoscience and Nanotechnology, Springer, Universities Press, 1<sup>st</sup> Edition, 2013.</li> <li>2. Introduction to Nanotechnology – Charles P. Poole, Jr., and Frank J. Owens, Wiley India Edition, 2012.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                      |
| <ol style="list-style-type: none"> <li>1. T. Pradeep, Nano: The Essentials: Understanding Nanoscience and Nanotechnology, Mc Graw-Hill Education, 1<sup>st</sup> Edition, 2007.</li> <li>2. Michael F. Ashby, Paulo J. Ferreira and Daniel L. Schodek, Nanomaterials, Nanotechnologies and Design, Butterworth-Heinemann Ltd., 2009.</li> <li>3. David Ferry, Transport in Nano structures, Cambridge University press, 2000.</li> <li>4. Challa S., S. R. Kumar, Josef Hormes, Carola Leuschner, Nanofabrication towards biomedical application: Techniques, tools, Application and impact, Wiley-VCH Verlag GmbH, 1<sup>st</sup> Edition, 2005.</li> <li>5. Michael J. O'Connell, Carbon Nanotubes: Properties and Applications, CRC Press, 1<sup>st</sup> Edition, 2006.</li> <li>6. S. Dutta, Electron Transport in Mesoscopic systems, Cambridge University press, 1995.</li> </ol> |                                      |



## ROBOTICS ENGINEERING

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                  |              |   |   |         |               |     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Category                                         | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME73                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | PEC                                              | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                  | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>Today robot finds applications in industries, medical and other fields. For example, in eye surgery (replacement of retina), where a cylindrical portion needs to be replaced, the operation is best done by robots. Mobile robots like walking machines, hopping machines are examples of robots, Nuclear and power plants uses fish like robots which move inside pipes for purpose of inspection. This course will give an insight on various types of industrial robots, their kinematic and kinetic aspects, different types of grippers, mechanics of grippers, trajectory planning etc. |                                                  |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate different types of robots, specifications of robots and different end effectors used in robots.<br>2. Analyze gripper forces and evaluate composite rotation and transformation matrices.<br>3. Model forward, inverse and differential kinematics of robot manipulators.<br>4. Explain trajectory planning techniques; formulate Lagrangian - Euler and newton - Euler equations.<br>5. Explain about actuators, feedback components used in robots and industrial applications.                            |                                                  |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>INTRODUCTION</b>                              |              |   |   |         |               |     |       |
| Introduction: Automation and Robotics, Asimov’s laws, Robot Architecture, Components, Anatomy of robot, Factors to be considered in the selection of robot, present and future applications. Specifications - Degree of freedom, Pay load, Parts per hour, Accuracy, Repeatability, Speed, Work space, Work volume, Work envelope, classification of robots based on configuration and control systems<br>End effectors: Mechanical and Non-mechanical grippers, Types of actuation mechanisms, requirements for the design of grippers, considerations for the selection of grippers.                                    |                                                  |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>MOTION ANALYSIS</b>                           |              |   |   |         |               |     |       |
| Motion Analysis: Basic Rotation Matrices, Euler Angles, Composite Rotation Matrices. Homogeneous transformations as applicable to rotation and translation – problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                  |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>MANIPULATOR &amp; DIFFERENTIAL KINEMATICS</b> |              |   |   |         |               |     |       |
| Manipulator Kinematics: D-H notation, D-H method of Assignment of frames, D-H Transformation Matrix, joint coordinates and world coordinates, Forward kinematics of 2R, RP, 3R and SCARA manipulators and Inverse kinematics of 2R, RP and 3R manipulators.                                                                                                                                                                                                                                                                                                                                                               |                                                  |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>TRAJECTORY PLANNING &amp; DYANIMCS</b>        |              |   |   |         |               |     |       |
| Trajectory Planning: Definition of Trajectory planning, Path, Trajectory, Knot points, Steps involved in trajectory planning, Trajectory planning techniques-Joint space and Cartesian space techniques, Cubic polynomial trajectory.                                                                                                                                                                                                                                                                                                                                                                                     |                                                  |              |   |   |         |               |     |       |

| UNIT-V | ACTUATORS, FEEDBACK COMPONENTS & APPLICATIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | <p>Actuators: Pneumatic, Hydraulic actuators, electric &amp; stepper motors, comparison of Actuators, Feedback Components: Position sensors – potentiometers, resolvers, optical encoders, Velocity sensor, Contact Sensors-Touch sensors, Tactile and Range sensors, Force and Torque sensors, Proximity sensor, Inductive sensor.</p> <p>Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding &amp; spray painting - Assembly and Inspection.</p> |
|        | <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|        | <ol style="list-style-type: none"> <li>1. Groover M P, Industrial Robotics, McGraw-Hill Education(India) Pvt. Limited, 1986.</li> <li>2. Mittal RK &amp; Nagrath IJ, Robotics and Control, McGraw-Hill Education(India) Pvt. Limited, 1<sup>st</sup> Edition, 2003.</li> <li>3. Reza N. Jazar Theory of Applied Robotics: kinematics, Dynamics and Control, 2<sup>nd</sup> Edition, Springer New York, NY, 2010.</li> </ol>                                                                                                                   |
|        | <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|        | <ol style="list-style-type: none"> <li>1. Fu K S, Gonzalez R C, and Lee C S G., Robotics: Control, Sensing, Vision, and Intelligence, Indian Edition, McGraw Hill, 2017.</li> <li>2. Mark W. Spony and M. Vidyasagar, Robot Dynamics and Control, John Wiley, 1991.</li> <li>3. Asada H, and Slotine J.-J.E., Robot Analysis and control, Wiley, 1991.</li> <li>4. John J Craig, Introduction to Robotics: Mechanics and Control, 3<sup>rd</sup> Edition, Pearson Education, 2004.</li> </ol>                                                 |

## INDUSTRIAL SAFETY AND MAINTENANCE ENGINEERING

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                            |              |   |   |         |               |     |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Category                                                                   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PEC                                                                        | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                            | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>Safety is one of the key dimensions of engineering asset management. Safety by design or prevention through design is in the core for maintaining engineering systems safe. The objective of this course is to impart knowledge on different facets and aspects of engineering systems safety, focusing on tools, techniques and methodologies needed for prevention of occurrences of unsafe operations and accidents under different industrial settings. Upon completion of the course, the students will be equipped with concepts of engineering systems safety, dimensions of engineering systems safety, safety design and analysis mathematics, design for engineering systems safety and control for safety, and integrating safety with other operational goals such as quality and reliability. |                                                                            |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br><ol style="list-style-type: none"><li>1. Understand the need of industrial safety and have knowledge on organizational implementation of safety measures.</li><li>2. Identifies the various hazardous location in the plant and knowledge on equipment with concept of engineering safety systems.</li><li>3. Understand the industrial safety acts as per Government norms.</li><li>4. Analyse the operational principle and maintenance and knowledge on condition monitoring to achieve goal such as qualities and reliability.</li><li>5. Implement techniques and methodology needed for preventive maintenance that occur on unsafe operation.</li></ol>                                                                          |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>INTRODUCTION TO THE DEVELOPMENT OF INDUSTRIAL SAFETY AND MANAGEMENT</b> |              |   |   |         |               |     |       |
| Introduction to the development of industrial safety and management: History and development of Industrial safety: Implementation of factories act, Formation of various councils, Safety and productivity, Safety organizations. Safety committees, safety committee structure, roll of management and roll of Govt. in industrial safety, Safety analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ACCIDENT PREVENTIONS PROTECTIVE EQUIPMENT'S AND THE ACTS</b>            |              |   |   |         |               |     |       |
| Personal protective equipment, Survey the plant for locations and hazards, Part of body to be protected, Education and training in safety, Prevention causes and cost of accident, Housekeeping, First aid, Fire fighting equipment, Accident reporting, Investigations, Industrial psychology in accident prevention, Safety trials.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>SAFETY ACTS</b>                                                         |              |   |   |         |               |     |       |
| Features of Factory Act, Introduction of Explosive Act, Boiler Act, ESI Act, Workman's compensation Act, Industrial hygiene, Occupational safety, Disease prevention, Ergonomics, Occupational diseases, stress, fatigue, health, safety and the physical environment, Engineering methods of controlling chemical hazards, safety and the physical environment, Control of industrial noise and protection against it, Code and regulations for worker safety and health.                                                                                                                                                                                                                                                                                                                                                            |                                                                            |              |   |   |         |               |     |       |

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| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>PRINCIPLES AND PRACTICES OF MAINTENANCE PLANNING:</b> |
| Basic Principles of maintenance planning – Objectives and principles of planned maintenance activity – Importance and benefits of sound Maintenance systems – Reliability and machine availability, Equipment Life cycle, Measures for Maintenance Performance: Equipment’s breakdowns, Mean Time Between Failures, Mean Time To Repair, Factors of availability, Maintenance organization, Maintenance economics.                                                                                                                                                                                             |                                                          |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>MAINTENANCE POLICIES AND PREVENTIVE MAINTENANCE</b>   |
| Maintenance categories – Comparative merits of each category – Preventive maintenance, Maintenance schedules: Repair cycle, Principles and methods of lubrication, Fault Tree Analysis, Total Productive Maintenance: Methodology and Implementation                                                                                                                                                                                                                                                                                                                                                           |                                                          |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                          |
| <ol style="list-style-type: none"> <li>1. Srivastava, S.K., “Industrial Maintenance Management”, S. Chand and Co.</li> <li>2. Bhattacharya, S.N., “Installation, Servicing and Maintenance”, S. Chand and Co.</li> <li>3. Willie Hammer, “Occupational Safety Management and Engineering”, Prentice Hall</li> </ol>                                                                                                                                                                                                                                                                                            |                                                          |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                          |
| <ol style="list-style-type: none"> <li>1. White, E.N., “Maintenance Planning”, Documentation, Gower Press</li> <li>2. Garg, M.R., “Industrial Maintenance”, S. Chand and Co.</li> <li>3. Higgins, L.R., “Maintenance Engineering Hand book”, 5th Edition, McGraw Hill</li> <li>4. Armstrong, “Condition Monitoring”, BSIRSA</li> <li>5. Davies, “Handbook of Condition Monitoring”, Chapman and Hall</li> <li>6. Ray Asfahl, C., “Industrial Safety and Health Management”, 5th Edition, Prentice Hall</li> <li>7. S.C. Mishra, “Reliability and Maintenance Engineering”, New Age Publishing house</li> </ol> |                                                          |

**PROFESSIONAL ELECTIVE – VI**

**PROFESSIONAL ELECTIVE - VI**  
**INDUSTRIAL ENGINEERING AND MANAGEMENT**

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                     |              |   |   |         |               |     |       |
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| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Category                                            | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME76                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PEC                                                 | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                     | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course introduces the basic concepts of Industrial Management which is concerned with the design, improvement and installation of integrated systems of men, materials, equipment and energy. The emphasis of this course is laid on the basic analysis of designing organizational structures, plant layouts, inspection and quality control, work study, Project Management (PERT/CPM) & fundamental management principles and practices. basics  |                                                     |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Describe the most well-known theories and perspectives on management<br>2. Basic understanding of the relationship between organizational structure, technology and the conditions of the organizational environment<br>3. Describe the dynamics of change management<br>4. Pursuit a minor organizational analysis<br>5. Basic understanding of contemporary organizational forms |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTRODUCTION TO MANAGEMENT</b>                   |              |   |   |         |               |     |       |
| Entrepreneurship and organization – Nature and Importance of Management, Functions of Management, Taylor’s Scientific Management Theory, Fayol’s Principles of Management, Maslow’s Theory of Human Needs, Douglas McGregor’s Theory X and Theory Y, Herzberg’s Two-Factor Theory of Motivation, Systems Approach to Management, Leadership Styles, Social responsibilities of Management.                                                                                          |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>DESIGNING ORGANIZATIONAL STRUCTURES</b>          |              |   |   |         |               |     |       |
| Departmentalization and Decentralization, Types of Organization structures – Line organization, Line and staff organization, functional organization, Committee organization, matrix organization, Virtual Organization, Cellular Organization, team structure, boundary less organization, inverted pyramid structure, lean and flat organization structure and their merits, demerits and suitability.                                                                            |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>OPERATIONS MANAGEMENT</b>                        |              |   |   |         |               |     |       |
| Objectives- product design process- Process selection-Types of production system(Job, batch and Mass Production),Plant location-factors- Urban-Rural sites comparison- Types of Plant Layouts-Design of product layout- Line balancing(RPW method) Value analysis-Definition-types of values-Objectives- Phases of value analysis- Fast diagram.                                                                                                                                    |                                                     |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>WORK STUDY &amp; STATISTICAL QUALITY CONTROL</b> |              |   |   |         |               |     |       |

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| <p><b>Work Study:</b> Introduction - definition - objectives - steps in work study - Method study - definition, objectives - steps of method study. Work Measurement - purpose - types of study - stop watch methods - steps - key rating - allowances - standard time calculations - work sampling.</p> <p><b>Statistical Quality Control:</b> variables attributes, Shewhart control charts for variables-chart, R chart - Attributes- Defective - Defect- Charts for attributes - p - chart - c chart (simple Problems), Acceptance Sampling- Single sampling- Double sampling plans-OC curves.</p>                                                                                                                                                                                                                   |                                                |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>JOB EVALUATION &amp; PROJECT MANAGEMENT</b> |
| <p><b>Job Evaluation:</b> Methods of job evaluation — simple routing objective systems—classification method factor comparison method, point method, benefits of job evaluation and limitations.</p> <p><b>Project Management(PERT/CPM):</b> Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing. (simple problems).</p>                                                                                                                                                                                                                                                                                                                         |                                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                |
| <ol style="list-style-type: none"> <li>1. O.P. Khanna, Industrial Engineering and Management, Dhanpat Rai Publications (P) Ltd., 2018.</li> <li>2. T.R. Banga and S.C. Sharma, Industrial Engineering and Management (Including Production Management), Khanna Publishers, 2008.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                |
| <ol style="list-style-type: none"> <li>1. Ralph M Barnes, Motion and Time Study: Design and Measurement of Work, 7<sup>th</sup> Edition, John Willey &amp; Sons, 1991.</li> <li>2. Mark S. Sanders, Ernest J McCormick, Human factors in Engineering and Design, 7<sup>th</sup> Edition, McGraw-Hill, Inc., 1993.</li> <li>3. R. Paneerselvam, Production and Operations Management, 3<sup>rd</sup> Edition, PHI, 2012.</li> <li>4. NVS Raju, Industrial Engineering and Management, 1<sup>st</sup> Edition, Cengage Learning, 2013.</li> <li>5. Kjell B. Zandin, Maynard's Industrial Engineering Hand Book, 5<sup>th</sup> Edition, McGraw Hill Companies, 2001.</li> <li>6. Ravi Shankar, Industrial Engineering and Management, 2<sup>nd</sup> Edition, Galgotia Publications Pvt. Ltd., New Delhi, 2017.</li> </ol> |                                                |

## OPERATIONS RESEARCH

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                      |              |   |   |         |               |     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Category                             | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME77                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | PEC                                  | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                      | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <p><b>COURSE OVERVIEW:</b></p> <p>Operations research (OR) is an analytical method of problem-solving and decision-making that is useful in the management of organizations. In operations research, problems are broken down in to basic components and then solved in defined steps by mathematical analysis. This course gives insight of Linear Programming, Transportation, assignment problems, sequencing etc.</p> <p><b>COURSE OUTCOMES:</b></p> <p><b>At the end of the course, the student should be able to:</b></p> <ol style="list-style-type: none"><li>1. Describe types of models and solve linear programming problem.</li><li>2. Solve transportation and assignment problems.</li><li>3. Analyze sequencing and replacement models and apply them for optimization.</li><li>4. Apply gaming theory for optimal decision making and analyze inventory models to optimize the cost.</li><li>5. Model and solve problems using dynamic programming</li></ol> |                                      |              |   |   |         |               |     |       |
| UNIT-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | INTRODUCTION & ALLOCATION            |              |   |   |         |               |     |       |
| <p>Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications.</p> <p>ALLOCATION: Linear Programming Problem - Formulation – Graphical solution–Simplex method– Artificial variables techniques: Two–phase method, Big-M method; Duality Principle.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                      |              |   |   |         |               |     |       |
| UNIT-II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TRANSPORTATION & ASSIGNMENT PROBLEMS |              |   |   |         |               |     |       |
| <p>TRANSPORTATION PROBLEM: Formulation–Optimal solution, unbalanced transportation problem, Degeneracy.</p> <p>ASSIGNMENT PROBLEM: Formulation – Optimal solution - Variants of Assignment Problem; Traveling Salesman problem.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                      |              |   |   |         |               |     |       |
| UNIT-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SEQUENCING & REPLACEMENT             |              |   |   |         |               |     |       |
| <p>SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines – Job shop sequencing – two jobs through ‘m’ machines REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely- Group Replacement.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |              |   |   |         |               |     |       |



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| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>THEORY OF GAME &amp; INVENTORY</b>          |
| <p>THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points- 2 x 2 games – m x 2 &amp; 2 x n games - graphical method – m x n games – dominance principle.</p> <p>INVENTORY: Introduction – Single item, Deterministic models – Types - Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.</p>                                                                         |                                                |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>WAITING LINES &amp; DYNAMIC PROGRAMMING</b> |
| <p>WAITING LINES: Introduction – Terminology-Single Channel – Poisson arrivals and Exponential Service times – with infinite population and finite population models– Multichannel – Poisson arrivals and exponential service times with infinite population.</p> <p>DYNAMIC PROGRAMMING: Introduction – Terminology- Bellman’s Principle of Optimality – Applications of dynamic programming- shortest path problem – linear programming problem.</p>                                                                                                        |                                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                |
| <ol style="list-style-type: none"> <li>1. J.K.Sharma, Operation Research, MacMilan. India Ltd, 3<sup>rd</sup> Edition, 2008.</li> <li>2. ACS Kumar, Operations Research, Yesdee Publishing pvt Ltd, 1<sup>st</sup> Edition, 2015.</li> </ol>                                                                                                                                                                                                                                                                                                                  |                                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                |
| <ol style="list-style-type: none"> <li>1. Maurice Saseini, Arhur Yaspan and Lawrence Friedman, Operations Research: Methods and Problems, Literary Licensing, LLC, 11-illustrated, 2013.</li> <li>2. M. Natarajan, P. Balasubramaniam, A.Tamilarasi, Operations Research, A Pearson Education, 2<sup>nd</sup> Edition, 2014.</li> <li>3. Wagner, Operations Research, PHI Private Limited; 2<sup>nd</sup> edition, January 1980.</li> <li>4. Hillier &amp; Libermann, Introduction to Operations Research, TMH, 8<sup>th</sup> Edition, July 2017.</li> </ol> |                                                |

## TOTAL QUALITY MANAGEMENT

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                             |              |   |   |         |               |     |       |
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| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Category                                    | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME78                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PEC                                         | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                             | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>Total Quality Management (TQM) is an approach that organizations use to improve their internal processes and increase customer satisfaction. When it is properly implemented, this style of management can lead to decreased costs related to corrective or preventative maintenance, better overall performance, and an increased number of happy and loyal customers. This course is developed to impart knowledge on principles, tools and techniques of TQM, control charts, quality cost and systems. |                                             |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Demonstrate basic concepts of total quality management.<br>2. Explain principles of total quality management.<br>3. Explain seven tools of quality, management tools and bench marking.<br>4. Enumerate and describe various control charts, quality costs and quality management.<br>5. Explain quality systems for TQM.                                                                                                            |                                             |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>INTRODUCTION</b>                         |              |   |   |         |               |     |       |
| Introduction – Need for quality – Evolution of quality – Definitions of quality – Dimensions of product and service quality – Basic concepts of TQM – TQM Framework – Contributions of Deming, Juran and Crosby – Barriers to TQM – Quality statements – Customer focus – Customer orientation, Customer satisfaction, Customer complaints, Customer retention                                                                                                                                                                        |                                             |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>TQM PRINCIPLES</b>                       |              |   |   |         |               |     |       |
| Leadership – Strategic quality planning, Quality Councils – Employee involvement – Motivation, Empowerment, Team and Teamwork, Quality circles Recognition and Reward, Performance appraisal – Continuous process improvement – PDCA cycle, 5S, Kaizen – Supplier partnership – Partnering, Supplier selection, Supplier Rating.                                                                                                                                                                                                      |                                             |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>TQM TOOLS &amp; TECHNIQUES</b>           |              |   |   |         |               |     |       |
| The seven traditional tools of quality – New management tools – Six sigma: Concepts, Methodology, applications to manufacturing, service sector including IT – Bench marking – Reason to bench mark, Bench marking process – FMEA – Stages, Types.                                                                                                                                                                                                                                                                                    |                                             |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CONTROL CHARTS &amp; COST OF QUALITY</b> |              |   |   |         |               |     |       |
| Control Charts – Process Capability – Concepts of Six Sigma – Quality Function Development (QFD) – Taguchi quality loss function – TPM – Concepts, improvement needs – Performance measures.<br>The Cost of Quality: Definition of the Cost of Quality, Quality Costs, Measuring Quality Costs, use of Quality Cost information, Accounting Systems and Quality Management.                                                                                                                                                           |                                             |              |   |   |         |               |     |       |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | QUALITY SYSTEMS |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Need for ISO 9000 – ISO 9001-2008 Quality System – Elements, Documentation, Quality Auditing – QS 9000 – ISO 14000 – Concepts, Requirements and Benefits – TQM Implementation in manufacturing and service sectors.                                                                                                                                                                                                                                                                                                                                                                                          |                 |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |
| <ol style="list-style-type: none"><li>1. Joel E. Ross, Total Quality Management, Taylor and Francis Limited, 3<sup>rd</sup> Edition, 2017.</li><li>2. P. N. Mukherjee, Total Quality Management, PHI, 2<sup>nd</sup> Edition, 2006</li></ol>                                                                                                                                                                                                                                                                                                                                                                 |                 |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |
| <ol style="list-style-type: none"><li>1. Robert L. Flood, Beyond TQM, John Wiley &amp; Sons Ltd, 2<sup>nd</sup> Edition, 1993</li><li>2. E.L. Grant, Statistical Quality Control, McGraw Hill Education; 7<sup>th</sup> Edition, 2017.</li><li>3. H. Lal, Total Quality Management: A Practical Approach, New Delhi: Wiley Eastern, 3<sup>rd</sup> Edition, 1990.</li><li>4. Kanishka Bedi, Quality Management, Oxford University Press, 2<sup>nd</sup> Edition, 2011.</li><li>5. Sunil Sharma, Total Engineering Quality Management, Macmillan, Laxmi Publications, 5<sup>th</sup> Edition, 2003.</li></ol> |                 |

## PRODUCT LIFE CYCLE MANAGEMENT

| IV B. Tech. II Semester: MECHANICAL ENGINEERING                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            |              |   |   |         |               |     |       |
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| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Category                   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME79                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PEC                        | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                            | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course aim at provides the students with knowledge about how a Product Lifecycle Management (PLM) system is used to structure and manage the information which guides the product during its lifecycle. The course identifies different stakeholders which both generates and consumes information related to the product and its manufacturing system over the lifecycle. The course also presents an overview of integration of PLM with other applications. |                            |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br><ol style="list-style-type: none"><li>1. Explain basic concepts of product life cycle management.</li><li>2. Demonstrate product development approaches.</li><li>3. Explain elements of product modelling.</li><li>4. Discuss in detail the concept of product data management.</li><li>5. Discuss about integration of PLM with other applications.</li></ol>                                   |                            |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>INTRODUCTION TO PLM</b> |              |   |   |         |               |     |       |
| Background, Overview, Need, Benefits, Concept of Product Life Cycle. Components / Elements of PLM, Emergence of PLM, Significance of PLM, Customer Involvement. Product Data and Product Workflow, Company’s PLM vision, The PLM Strategy, Principles for PLM strategy, Preparing for the PLM strategy, Developing a PLM strategy, Strategy identification and selection, Change Management for PLM                                                                                            |                            |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>PRODUCT DEVELOPMENT</b> |              |   |   |         |               |     |       |
| Product Development Approaches: Bottom-up design, Top-down design, Front-loading design workflow, Design in context, Modular design. Concurrent engineering, partnership with supplier, collaborative and Internet based design, work structuring and team deployment, Product and process systemization, problem, identification and solving methodologies, improving product development solutions                                                                                           |                            |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PRODCUT MODELING</b>    |              |   |   |         |               |     |       |
| Product Modelling - Definition of concepts - Fundamental issues – Role of Process chains and product models -Types of product models – model standardization efforts-types of process chains - Industrial demands. Foundation technologies and standards (e.g. visualization, collaboration and enterprise application integration)                                                                                                                                                            |                            |              |   |   |         |               |     |       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>PRODUCT DATA MANAGEMENT</b>                                   |
| Product Data Management (PDM) – Benefits and Terminology, PDM functions, definition and architectures of PDM systems, product data interchange, portal integration, PDM acquisition and implementation. Information authoring tools (e.g., MCAD, ECAD, and technical publishing), Core functions (e.g., data vaults, document and content management, workflow and program management), Functional applications (e.g., configuration Management). |                                                                  |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INTEGRATION OF PLM WITH OTHER APPLICATIONS, PLM SOFTWARES</b> |
| Different ways to integrate PLM systems, Transfer file, Database integration, System roles, ERP, Optimization of ERP for PLM and CAD.<br>Different ways to integrate PLM systems, Transfer file, Database integration, System roles, ERP, Optimization of ERP for PLM and CAD.<br>PLM Softwares - Basic features and modules of ENOVIA and Windchill.                                                                                             |                                                                  |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                  |
| 1. Grieves, Michael. Product Lifecycle Management, McGraw-Hill, 2006. ISBN 0071452303<br>2. Saaksvuori Antti / Immonen Anselmie, product Life Cycle Management Springer, Dreamtech, 3-540- 25731-4.                                                                                                                                                                                                                                               |                                                                  |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                  |
| 1. Stark, John. Product Lifecycle Management: Paradigm for 21st Century Product Realisation, Springer-Verlag, 2004. ISBN 1852338105.<br>2. Fabio Giudice, Guido La Rosa, Product Design for the environment -A life cycle approach, Taylor & Francis 2006.                                                                                                                                                                                        |                                                                  |

**OPEN ELECTIVES**  
**OFFERED BY**  
**DEPARTMENT OF MECHANICAL ENGINEERING**

|                                                     |                    |                                          |
|-----------------------------------------------------|--------------------|------------------------------------------|
| <b>III B. Tech. II Semester: OPEN ELECTIVE - I</b>  |                    |                                          |
| <b>Sl. No.</b>                                      | <b>Course Code</b> | <b>Subject Name</b>                      |
| 1                                                   | A6ME55             | Basics of Mechanical Engineering         |
| 2                                                   | A6ME60             | Fundamentals of Engineering Materials    |
| <b>IV B. Tech. I Semester: OPEN ELECTIVE - II</b>   |                    |                                          |
| <b>Sl. No.</b>                                      | <b>Course Code</b> | <b>Subject Name</b>                      |
| 1                                                   | A6ME65             | Fundamentals of Operations Research      |
| 2                                                   | A6ME70             | Introduction to Industrial Engineering   |
| <b>IV B. Tech. II Semester: OPEN ELECTIVE - III</b> |                    |                                          |
| <b>Sl. No.</b>                                      | <b>Course Code</b> | <b>Subject Name</b>                      |
| 1                                                   | A6ME75             | Renewable Energy Sources                 |
| 2                                                   | A6ME80             | Elements of Electric and Hybrid Vehicles |

**OPEN ELECTIVE – I**

## BASICS OF MECHANICAL ENGINEERING

### III B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-I)

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME55      | OEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course deals with fundamental concepts of design, manufacturing and thermal engineering.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. Explain fundamentals of metallurgy and material science.
3. Discuss basic concepts and principles of thermodynamics.
4. Demonstrate the elements of fabrication processes and equipment.
5. Explain basic concepts of CAD/CAM.

#### UNIT-I

#### MECHANICS

MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law – derivation and problems, polygon law, resolution of forces in coplanar concurrent force system – problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, Definition of centroid and centre of gravity, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia.

MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strains, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.

#### UNIT-II

#### METALLURGY & MATERIAL SCIENCE

Bonds in Solids, crystallization of metals, Classification of steels, Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide equilibrium diagram.

Heat treatment – Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.

#### UNIT-III

#### THERMO DYNAMICS

System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic view points, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry.

First law of Thermodynamics – Corollaries, First law applied to a Closed System, applied to a flow system, Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>MANUFACTURING</b> |
| <p>Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies.</p> <p>Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.</p> |                      |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>CAD/CAM</b>       |
| <p>Fundamentals of CAD/CAM – Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, disadvantages, wire frame entities – analytic entities and synthetic entities.</p> <p>Surface modeling: Definition, advantages, disadvantages, surface entities – analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.</p>                                                                                                         |                      |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |
| <ol style="list-style-type: none"> <li>1. K. Vijay Kumar Reddy and J. Suresh Kumar, Singer's Engineering Mechanics, B.S Publications</li> <li>2. Avner, Introduction to physical metallurgy, Tata McGraw Hill Edition.</li> <li>3. P.K Nag, Thermodynamics, McGraw Hill Education.</li> <li>4. R.K Jain, Production Technology, Khanna Publishers.</li> <li>5. Ibrahim Zeid &amp; R. Sivasubramanian, CAD/CAM Theory &amp; Practice, McGraw Hill Education.</li> </ol>                                                                                                                |                      |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |
| <ol style="list-style-type: none"> <li>1. R.K. Bansal, Engineering Mechanics, Laxmi Publications</li> <li>2. O.P. Khanna, Material science and metallurgy, Dhanpat Ray Publications</li> <li>3. Pakirappa, V.Naveen Kumar, V.Naresh, Engineering Thermodynamics, Durga Publishing House</li> <li>4. P.C Sharma, Production Technology, S. CHAND.</li> <li>5. P. Radha Krishnan, S. Subramanyan and V. Raju, CAD/CAM, New Age International Publishers.</li> <li>6. Sadhu Singh, Strength of Materials, Khanna Publishers.</li> </ol>                                                  |                      |

## FUNDAMENTALS OF ENGINEERING MATERIALS

III B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-I)

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME60      | OEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

COURSE OVERVIEW:

This course will enable students to understand basic structure and crystal arrangement of materials, the phase diagrams, advantages of heat treatment, various heat treatment processes, the need and application of composite materials.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
2. Demonstrate the concept of alloying and formation of different types of phases in alloys.
3. Differentiate ferrous and non-ferrous alloys.
4. Explain various heat treatment processes.
5. Classify and explain polymers, ceramics and composites.

| UNIT-I                                                                                                                                                                                                                                                                                                              | CRYSTAL STRUCTURE                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.                                                                   |                                   |
| UNIT-II                                                                                                                                                                                                                                                                                                             | ALLOYS & PHASE DIAGRAMS           |
| Alloys- substitutional and interstitial solid solutions.<br>Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite. |                                   |
| UNIT-III                                                                                                                                                                                                                                                                                                            | FERROUS AND NON FERROUS ALLOYS    |
| Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons.<br>Copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.                                                                        |                                   |
| UNIT-IV                                                                                                                                                                                                                                                                                                             | HEAT TREATMENT OF STEEL           |
| Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.                                                                                                    |                                   |
| UNIT-V                                                                                                                                                                                                                                                                                                              | POLYMERS, CERAMICS AND COMPOSITES |
| Classification, properties and applications of polymers, ceramics, composites and nano materials.                                                                                                                                                                                                                   |                                   |

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**Text Books:**

1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999.
2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
3. Sidney H. Avener, Introduction to Physical Metallurgy, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2007.

**Reference Books:**

1. W. D. Callister, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India, 2006.
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. D. Kodgire, Material Science and Metallurgy for engineers, 1st Edition, Everest, Pune, India, 2006.

**OPEN ELECTIVE – II**

## FUNDAMENTALS OF OPERATIONS RESEARCH

IV B. Tech. I Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-II)

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME65      | OEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

COURSE OVERVIEW:

This course aims to introduce the basic concepts of Operational Research such as linear programming, transportation and assignment problem, sequencing, game theory and inventory to the students.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1.

Describe types of models and solve linear programming problem.

2.

Solve transportation and assignment problems.

3.

Analyze sequencing and replacement models and apply them for optimization.

4.

Apply gaming theory for optimal decision making.

5.

Analyze inventory models to optimize the cost

|                                                                                                                                                                                                                                                                                                          |                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| UNIT-I                                                                                                                                                                                                                                                                                                   | INTRODUCTION & ALLOCATION            |
| Development – Definition – Characteristics and Phases – Types of models – Operations Research models – applications.<br>ALLOCATION: Linear Programming Problem – Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two-phase method, Big-M method; Duality Principle. |                                      |
| UNIT-II                                                                                                                                                                                                                                                                                                  | TRANSPORTATION & ASSIGNMENT PROBLEMS |
| TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy.<br>ASSIGNMENT PROBLEM: Formulation – Optimal solution – Variants of Assignment Problem; Traveling Salesman problem.                                                                              |                                      |
| UNIT-III                                                                                                                                                                                                                                                                                                 | SEQUENCING & REPLACEMENT             |
| SEQUENCING: Introduction – Flow –Shop sequencing – $n$ jobs through two machines – $n$ jobs through three machines.<br>REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.            |                                      |
| UNIT-IV                                                                                                                                                                                                                                                                                                  | THEORY OF GAMES                      |
| THEORY OF GAMES: Introduction –Terminology – Solution of games with saddle points and without saddle points – $2 \times 2$ games – $m \times 2$ & $2 \times n$ games – graphical method – $m \times n$ games - dominance principle.                                                                      |                                      |
| UNIT-V                                                                                                                                                                                                                                                                                                   | INVENTORY                            |
| INVENTORY: Introduction – Single item, Deterministic models – Types – Purchase inventory models with one price break and multiple price breaks – Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.                                           |                                      |

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**Text Books:**

1. J.K. Sharma, Operation Research: Theory and Applications, 5<sup>th</sup> Edition, MacMillan India Ltd., 2012.
2. A.C.S. Kumar, Operations Research: Quantitative Analysis for Business Decisions, 1<sup>st</sup> Edition, Yes Dee Publications, 2015.

**Reference Books:**

1. A.M. Natarajan, P. Balasubramaniam, A. Tamilarasi, Operations Research, 2<sup>nd</sup> Edition, Pearson Education India, 2014.
2. Harvey M. Wagner, Principles of Operations Research, 2<sup>nd</sup> Edition, Prentice Hall India Learning Pvt. Ltd., 1980.
3. Frederick S. Hillier, Gerald J. Lieberman, Bodhibrata Nag, Preetam Basu, Introduction to Operations Research, 11<sup>th</sup> Edition, Tata McGraw Hill India, 2021.

## INTRODUCTION TO INDUSTRIAL ENGINEERING

### IV B. Tech. I Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-II)

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME70      | OEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

#### COURSE OVERVIEW:

This course focuses on production systems that produce goods or provide services for customers. Industrial engineers define, design, build, operate and improve production processes that convert resources to high quality products or services effectively, efficiently and safely.

#### COURSE OUTCOMES:

**At the end of the course, the student should be able to:**

1. Describe the tools and concepts of industrial engineering principles.
2. Differentiate various types of plant layouts and maintenance.
3. Apply the principles of method study and time study to analyze and optimize work processes for increased effectiveness.
4. Apply Statistical Methods for Quality Control and explain Total Quality Management principles.
5. Use the techniques like human resource management, industrial relations, wage incentive plans, Value engineering for the effective enterprise resource planning.

#### UNIT-I INTRODUCTION

Definition of industrial engineering (I.E), development, applications, role of an industrial engineer, differences between production management and industrial engineering, quantitative tools of IE and productivity measurement. concepts of management, importance, functions of management, scientific management, Taylor's principles, theory X and theory Y, Fayol's principles of management.

#### UNIT-II PLANT LAYOUT

Factors governing plant location, types of production layouts, advantages and disadvantages of process layout and product layout, applications, quantitative techniques for optimal design of layouts, plant maintenance, preventive and breakdown maintenance.

#### UNIT-III WORK STUDY

Importance, types of production, applications, workstudy, method study and time study, work sampling, PMTS.  
Micro-motion study, rating techniques, MTM, work factor system, principles of Ergonomics, flow process charts, string diagrams and Therbligs.

#### UNIT-IV STATISTICAL QUALITY CONTROL

Quality control, Queuing assurance and its importance, SQC, attribute sampling inspection with single and double sampling, Control charts – X and R – charts X and S charts and their applications, numerical examples.  
TOTAL QUALITY MANAGEMENT: zero defect concept, quality circles, implementation, applications, ISO quality systems. six sigma – definition, basic concepts.

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                               | RESOURCE MANAGEMENT |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <p>Concept of human resource management, personnel management and industrial relations, functions of personnel management, Job-evaluation, its importance and types, merit rating, quantitative methods, wage incentive plans, types.</p> <p>VALUE ANALYSIS: Value engineering, implementation procedure, enterprise resource planning and supply chain management.</p>                              |                     |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                   |                     |
| <ol style="list-style-type: none"><li>1. O.P Khanna, Industrial Engineering and management, 2<sup>nd</sup> Edition, Dhanpat Rai publications, 2018.</li><li>2. Martand Telsang, Industrial Engineering and Production Management, 3<sup>rd</sup> Edition, S.Chand &amp; Company Ltd. New Delhi, 2013.</li></ol>                                                                                      |                     |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                              |                     |
| <ol style="list-style-type: none"><li>1. Bhattacharya D. K., Industrial Management, 2<sup>nd</sup> Edition, Vikas publishers, 2011.</li><li>2. J.G Monks, Operations Management, 3<sup>rd</sup> Edition, McGraw Hill Publishers, 2007.</li><li>3. T.R. Banga, S.C. Sharma, N. K. Agarwal, Industrial Engineering and Management Science, 12<sup>th</sup> Edition, Khanna Publishers, 2008.</li></ol> |                     |



**OPEN ELECTIVE – III**

## RENEWABLE ENERGY SOURCES

IV B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-III)

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6ME75      | OEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 40            | 60  | 100   |

COURSE OVERVIEW:

The energy has become an important and one of the basic infrastructures for the economic development of the country. It is imperative for the sustained growth of the economy. This course envisages the new and renewable source of energy, available in nature and to expose the students on sources of energy crisis and the alternates available, also stress up on the application of non-conventional energy technologies.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Demonstrate various energy sources and their availability.
2. Demonstrate the knowledge of solar radiation and its applications.
3. Analyse the performance of wind energy conversion systems.
4. Explain Biomass conversion techniques and Geothermal sources and resources
5. Explain various methods for harvesting the ocean energy sources.

|                                                                                                                                                                                                                                                                                                                                              |                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| UNIT-I                                                                                                                                                                                                                                                                                                                                       | ENERGY SOURCES & THEIR AVAILABILITY                              |
| Energy Sources and their Availability - Importance of Renewable Energy Resources – Present energy scenario – World energy status – Energy scenario in India – Classification of RES.<br>Introduction to Solar, Wind, Geothermal, Bio-mass, Ocean Energy Sources, Comparison of these Energy Sources, Advantages and Limitations of RES.      |                                                                  |
| UNIT-II                                                                                                                                                                                                                                                                                                                                      | INTRODUCTION TO SOLAR RADIATION & ITS APPLICATIONS               |
| Introduction to Solar radiation - beam and diffuse radiation, solar constant, measurement of solar radiation. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, Focusing and non-focusing type.<br>Applications of Solar Energy: Solar water heater – Solar dryer – Solar distillation – Solar Photo-Voltaic Cell. |                                                                  |
| UNIT-III                                                                                                                                                                                                                                                                                                                                     | PRINCIPLE OF WIND ENERGY CONVERSION                              |
| Principle of Wind Energy Conversion – Basic components of Wind Energy Conversion Systems, Wind mill Components, Various Types and their Constructional Features.<br>Effect of Density, Frequency Variances, Angle of attack, and Wind Speed – Design Considerations of Horizontal and Vertical Axis Wind Machines.                           |                                                                  |
| UNIT-IV                                                                                                                                                                                                                                                                                                                                      | BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES |
| Biomass conversion techniques – Biogas Generation – Types of biogas plants – Advantages and disadvantages of biogas plants – Types of Bio-digester – Advantages and disadvantages.<br>Geothermal Sources like hydrothermal, geo-pressured hot dry rock, magma – Simple geothermal Power plant.                                               |                                                                  |

| UNIT-V                                                                                                                                                                                                                                                                                                                                                                                                       | TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Tidal Energy – Principle of working, performance and limitations. Wave Energy – Principle of working.<br>Ocean Thermal Energy Conversion – OTEC power plants – working principle and Limitations.                                                                                                                                                                                                            |                                                |
| <b>Text Books:</b>                                                                                                                                                                                                                                                                                                                                                                                           |                                                |
| <ol style="list-style-type: none"><li>1. G. D. Rai, Non-Conventional Energy Sources, 4th Edition, Khanna Publishers, 2000.</li><li>2. B H Khan, Non-Conventional Energy Resources, 2nd Edition, Tata McGraw Hill Education Pvt. Ltd, 2011</li></ol>                                                                                                                                                          |                                                |
| <b>Reference Books:</b>                                                                                                                                                                                                                                                                                                                                                                                      |                                                |
| <ol style="list-style-type: none"><li>1. S. Hasan Saeed and D.K. Sharma, Non-Conventional Energy Resources, 3rd Edition, S.K. Kataria &amp; Sons, 2012.</li><li>2. N.K. Bansal, M. Kleemann, M. Heliss, Renewable energy sources and conversion technology, Tata McGraw Hill, 1990.</li><li>3. S. P. Sukhatme, Solar Energy Principles and Application, Tata McGraw Hill Education Pvt. Ltd, 2009.</li></ol> |                                                |

## ELEMENTS OF ELECTRIC AND HYBRID VEHICLES

| IV B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-III)                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                            |              |   |   |         |               |     |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------|---|---|---------|---------------|-----|-------|
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Category                                                                   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
| A6ME80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | OEC                                                                        | L            | T | P | C       | CIE           | SEE | Total |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                            | 3            | 0 | 0 | 3       | 40            | 60  | 100   |
| <b>COURSE OVERVIEW:</b><br>This course introduces the fundamental concepts, principles, analysis, design, and control systems of electric and hybrid vehicles.                                                                                                                                                                                                                                                                                                                                              |                                                                            |              |   |   |         |               |     |       |
| <b>COURSE OUTCOMES:</b><br><b>At the end of the course, the student should be able to:</b><br>1. Analyze the impact of conventional vehicles on the society and different types of drive train topologies<br>2. Evaluate the performance and energy consumption in electric vehicles<br>3. Assess the configuration and architecture in hybrid vehicles<br>4. Choose the type of motor used in electric and hybrid electric vehicles<br>5. Design the drive train suitable for electric and hybrid vehicles |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-I</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES</b>                        |              |   |   |         |               |     |       |
| History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies                                                                                                                                                                                                                                                                                                                                              |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>ELECTRIC VEHICLES</b>                                                   |              |   |   |         |               |     |       |
| Configurations of EVs, Performance of EVs, Tractive Effort in Normal Driving, Energy Consumption.                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-III</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>HYBRID ELECTRIC VEHICLES</b>                                            |              |   |   |         |               |     |       |
| Concept of Hybrid Electric Drive Trains.<br>Architectures of Hybrid Electric Drive Trains.                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>ELECTRIC PROPULSION SYSTEMS</b>                                         |              |   |   |         |               |     |       |
| Introduction, DC Motor Drives, Induction Motor Drives, Permanent Magnetic BLDC Motor Drives, SRM Drives                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                            |              |   |   |         |               |     |       |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>DESIGN PRINCIPLE OF SERIES AND PARALLEL HYBRID ELECTRIC DRIVE TRAIN</b> |              |   |   |         |               |     |       |
| Design Principles of a Series (Electrical Coupling) Hybrid Drive Train: Operation Patterns, Control Strategies, Design Example.<br>Design Principles of Parallel (Mechanically Coupled) Hybrid Electric Drive Train: Drive Train Configuration and Design Objectives, Control Strategies, Parametric Design of a Drive Train.                                                                                                                                                                               |                                                                            |              |   |   |         |               |     |       |
| <b>Text Books:</b><br>1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004.                                                                                                                                                                                                                                                                                                                  |                                                                            |              |   |   |         |               |     |       |

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| 2. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003. |
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| <b>Reference Books:</b> |
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| <ol style="list-style-type: none"><li>1. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.</li><li>2. Allen Fuhs, Hybrid Vehicles and the future of personal transportation, CRC Press, 2011.</li></ol> |
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## VISION STATEMENT

To create and nurture competent Engineers and managers who would be enterprise leaders in all parts of the world with aims of reaching the skies and touching the stars and yet feet firmly planted on the ground – good human beings steeped in ethical and moral values.

## MISSION STATEMENT

MLR Institute of Technology is committed to providing a positive, professional and conducive learning environment where all students are inspired to achieve their potential and strive for excellence in a global society as dignified professionals with the cooperation of all stakeholders.

## QUALITY POLICY

We, at MLRIT, are committed to Educate, Enrich and Excel, in imparting Professional Education, by top-quality faculty; who endeavor to mentor the students as turn-key solution providers, while striving continually to improve through team work, innovation and research.

## GOALS OF MLRIT

Goals of Engineering education at undergraduate / graduate level:

- Equip students with industry – accepted career and life skills
- To create a knowledge warehouse for students
- To disseminate information on skills and competencies that are in use and in demand by the industry.
- To create learning environment where the campus culture acts as a catalyst to student fraternity to understand their core competencies, enhance their competencies and improve their career prospects.
- To provide base for lifelong learning and professional development in support of evolving career objectives, which include being informed, effective, and responsible participants within the engineering profession and in society.
- To prepare students for graduate study in Engineering and Technology.
- To prepare graduates to engineering practice by learning from professional engineering assignments.